

What kind of drugs should we develop

*Mathematical and Computational Biology in Drug Discovery
(MCBDD) Module III*

*Dr. Jitao David Zhang
April 2026*

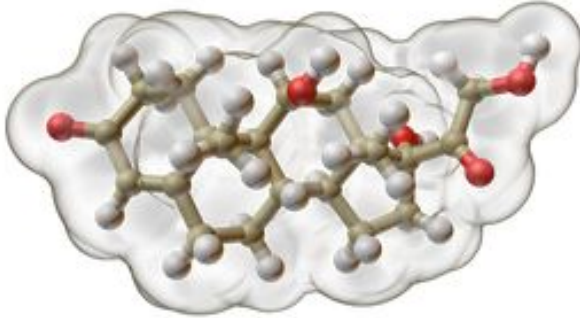
Objectives of the module

- Essentials of modalities: small molecules, large molecules (i.e. antibodies), antisense oligonucleotides, and gene and cell therapy.
- Multiple modalities can be designed for the same disease: the example of spinal muscle atrophy (SMA)
- Therapeutic antibodies: from design to development

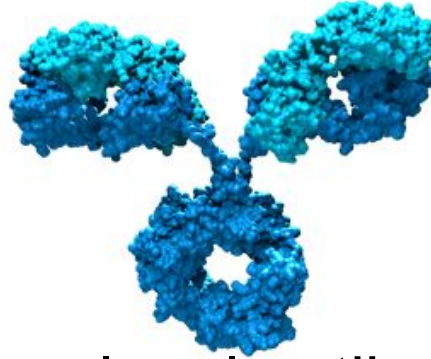
Overview

- Essentials of modalities
 - Small molecules: classical, protein degrader, RNA modulator
 - Large molecules: classical, DUTA-Fabs, protein design
 - Antisense oligonucleotides: siRNA, shRNA, ASO
 - Gene and cell therapy
- Three case studies:
 - Success stories:
 - [Small molecules] SMA (Evrysdi/Risdiplam and Nusinersen)
 - [Antisense] patisiran ([KEGG DRUG](#)) and givosiran ([DrugBank](#), [structure available at EMA](#))
 - [Offline read] mRNA vaccine (MIT Technology Review)
 - Turning failure into successes: [Multispecific drugs] Thalidomide, PROTAC, degraders
 - [Antibody] Cancer immunotherapy (CTLA4, PD1)
 - [Gene and Cell therapy] CAR-T
 - Challenges
 - [Antisense] HTT (Tominersen)
 - Difference between genetic and enzymatic inhibition

A zoo of modalities



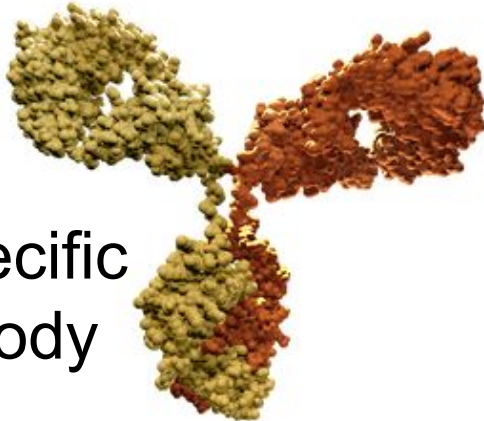
Small molecule



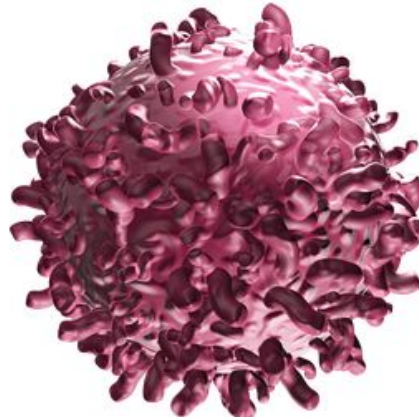
Monoclonal antibody



Oligonucleotides



Bispecific
antibody



Chimeric
Antigen
Receptor
(CAR)
T-cells



mRNA vaccines

Criteria to choose a modality

Disease relevance

Target characteristics

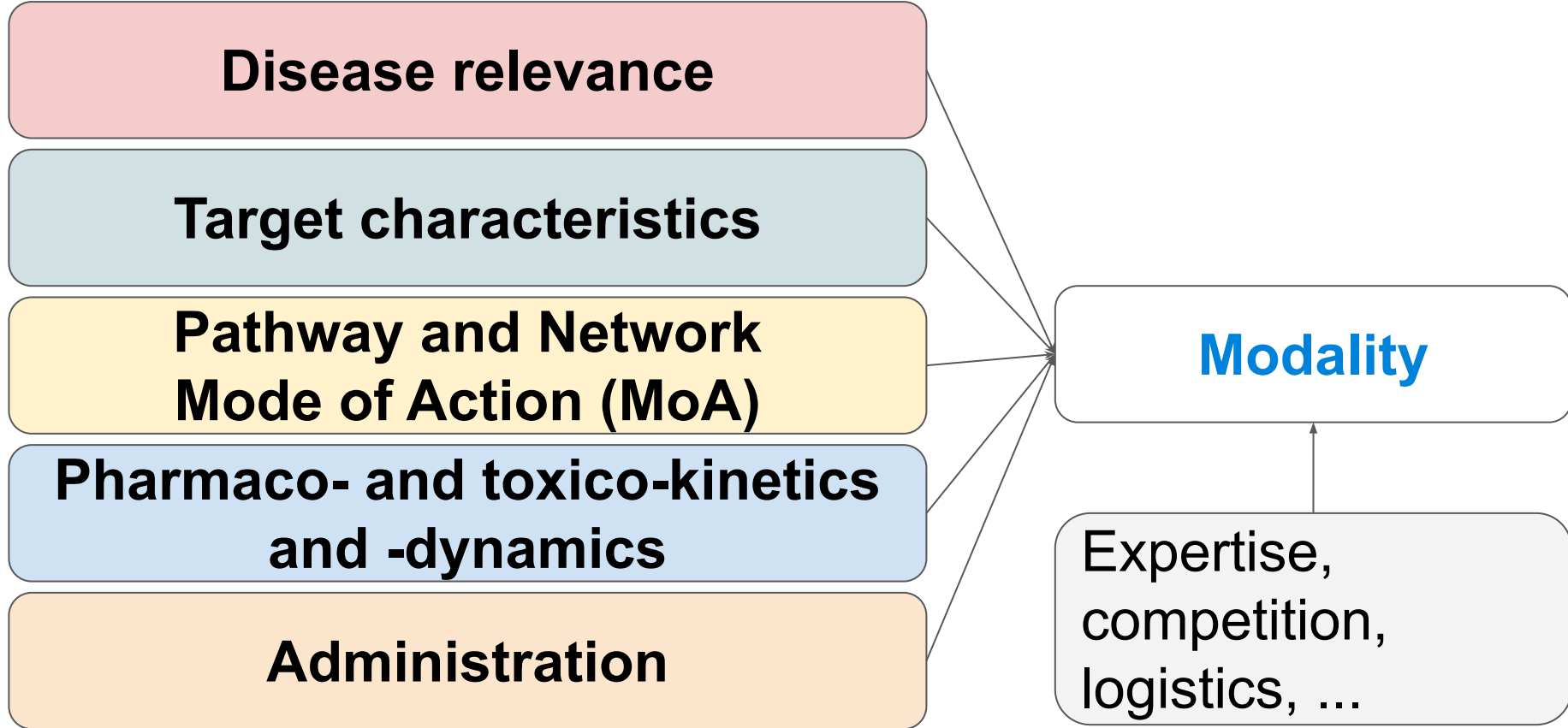
**Pathway and Network
Mode of Action (MoA)**

**Pharmaco- and toxico-kinetics
and -dynamics**

Administration

Modality

Expertise,
competition,
logistics, ...

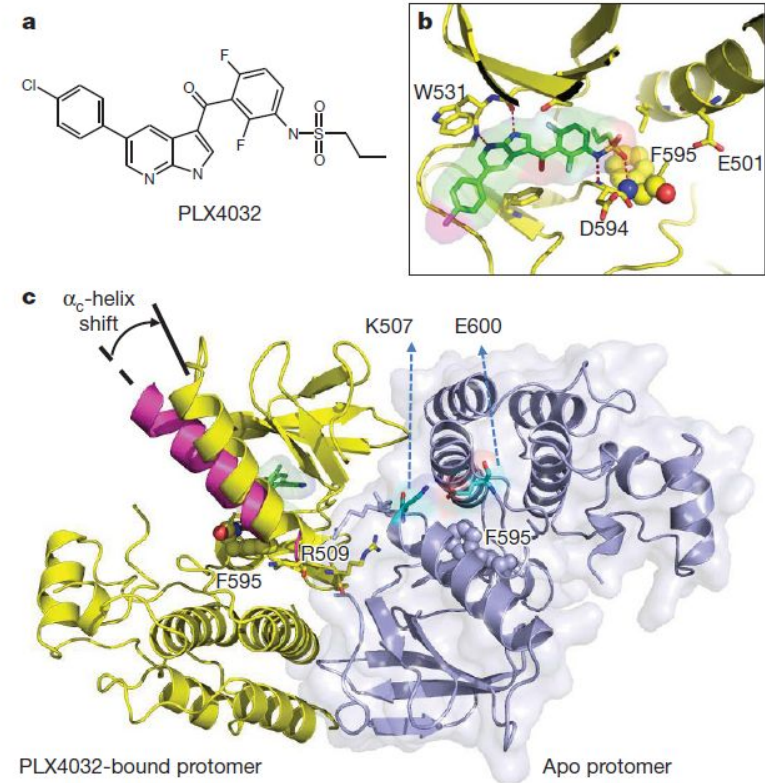


Characteristics of therapeutic modalities

Modality	Cause of disease at the protein level		Molecular target			Protein target localization			Delivery		
	Reduction or loss of function	Excessive or detrimental function	DNA	RNA	Protein	Extracellular	Plasma membrane	Intracellular	Oral	Injection	Inhaled
Small molecule											
Protein replacement											
Antibody											
Oligonucleotide therapy											
Cell and gene therapy*											

Classical small molecules: an example from AMIDD

- Vemurafenib (Zelboraf, PLX4032)
V600**E** mutated **BRAF**
inhibition
- Lock and key: an oversimplified yet powerful metaphor, first proposed by Emil Fischer

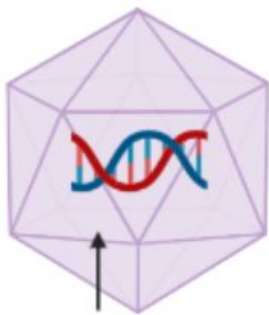


Facts about Spinal Muscular Atrophy (SMA)

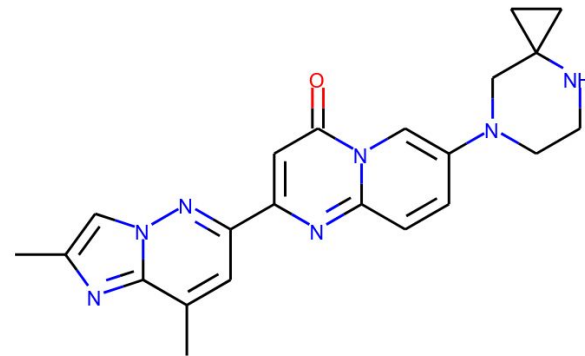
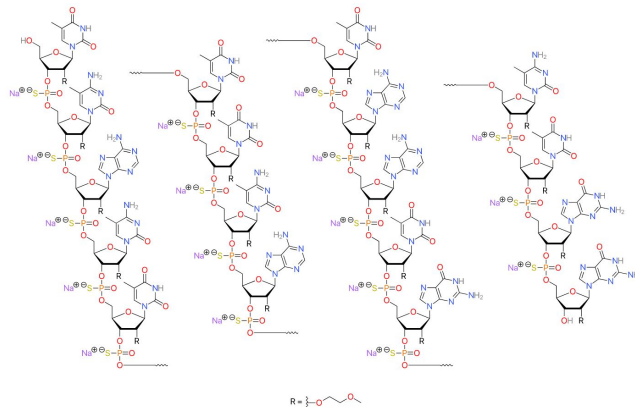
- SMA is caused by a defect in a gene called *SMN1*. People with SMA have reduced levels of the SMN protein.
- When SMN protein levels are reduced, motor neurons are unable to send signals to the muscles, causing them to become smaller and weaker over time.
- Depending on the severity, or type of SMA, people with the disease will have difficulties moving, eating, and in some cases breathing, making them increasingly dependent on parents and caregivers.
- A short movie: <https://www.nejm.org/doi/full/10.1056/NEJMoa2009965>

One Disease, Three Drugs

AAV9 capsid



SMN1 gene

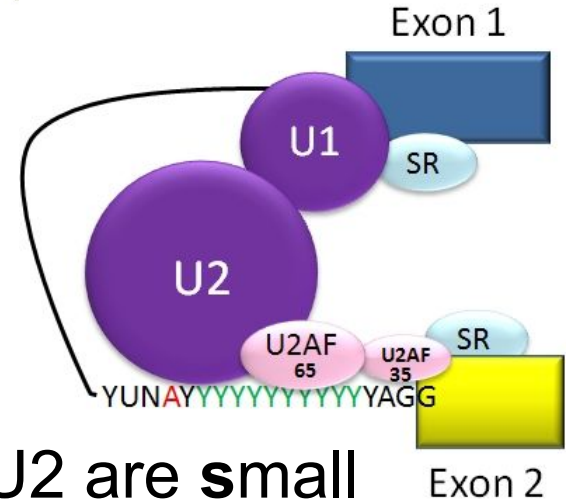
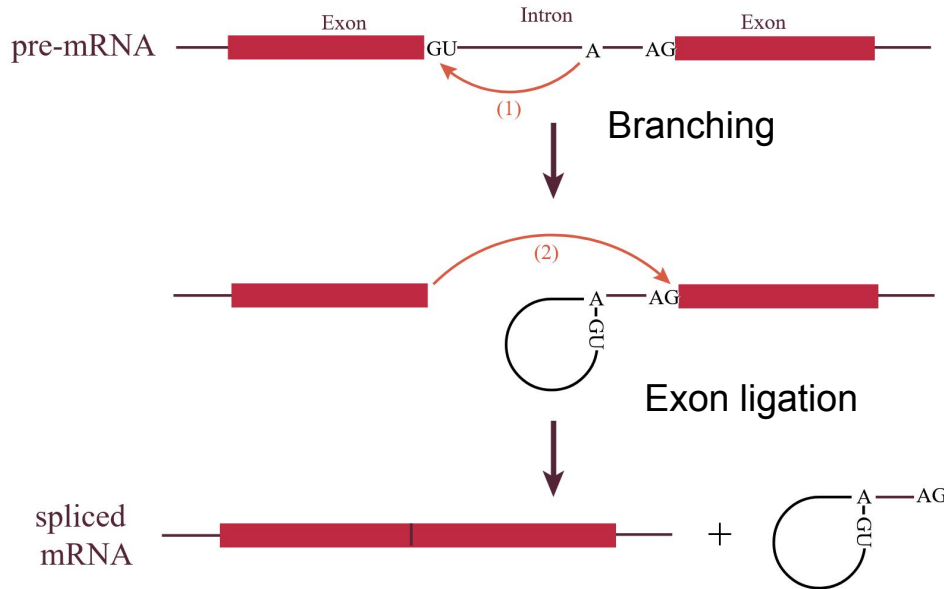
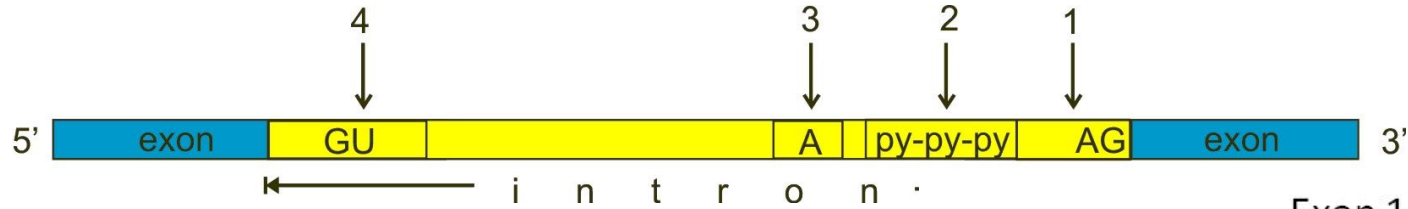


Onasemnogene
Abepravovec/
Zolgensma

Nusinersen sodium/ Spinraza
([CHEMBL3833342](#))

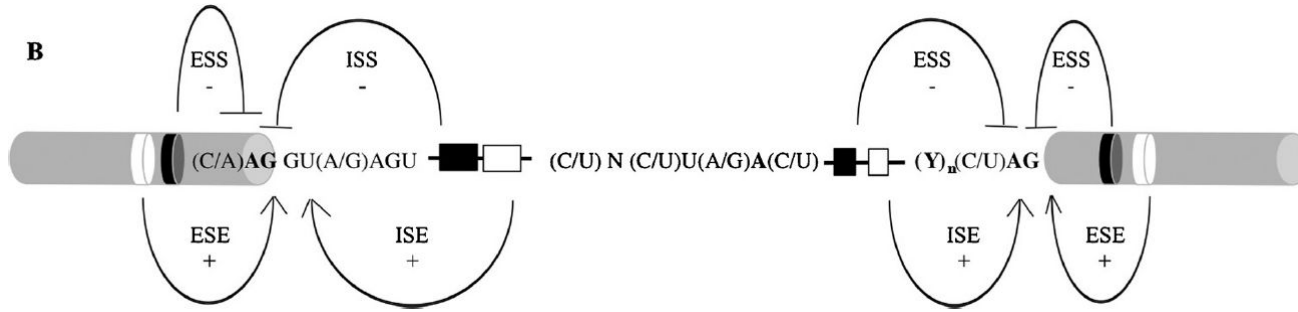
Risdiplam/ *Evrysdi*
([CHEMBL4297528](#))

Spliceosome: the splicing machinery



U1/U2 are **small nuclear ribonucleoproteins** (snRNPs = RNA+proteins)

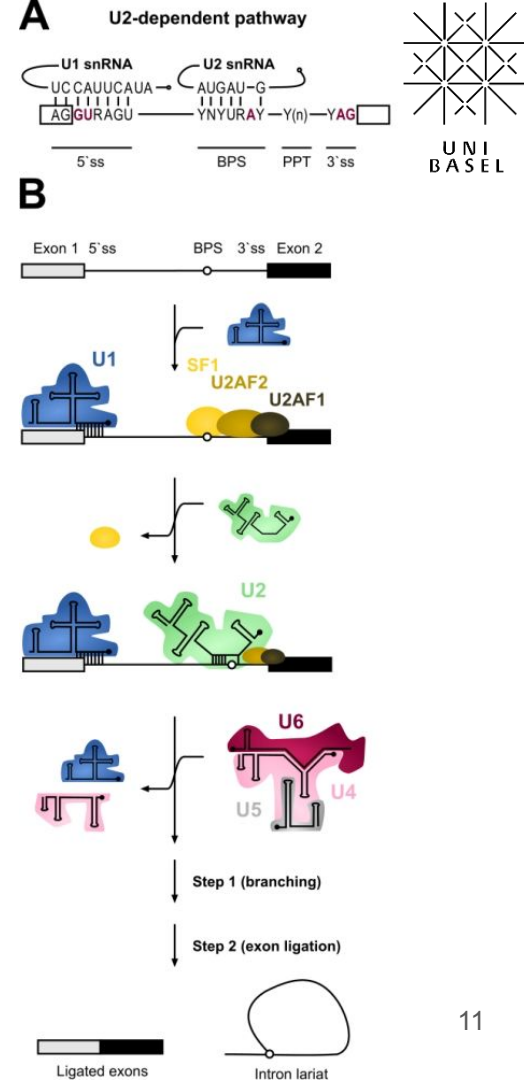
Splicing in action and under regulation



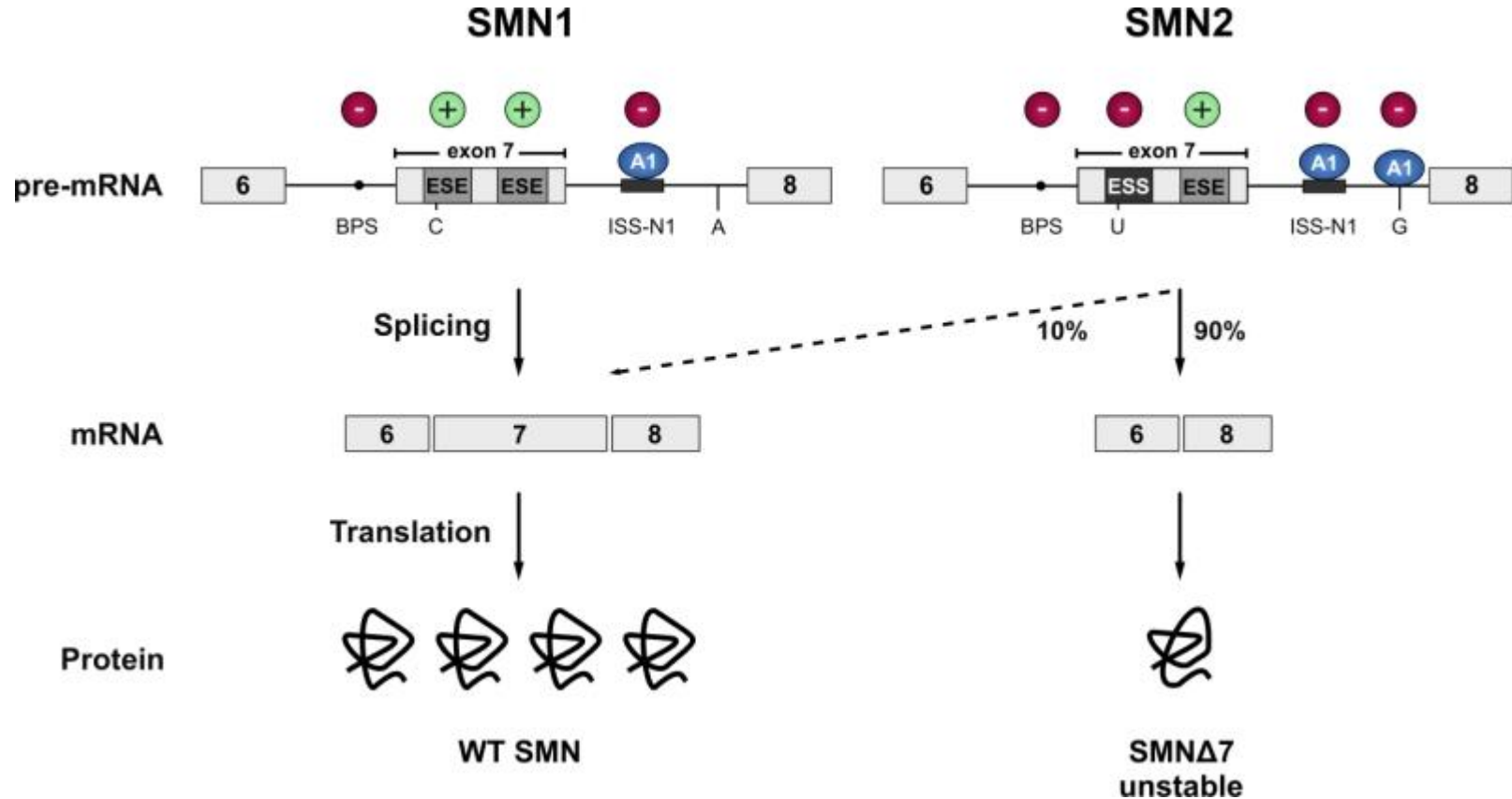
ESS=exon splicing silencer; ESE=exon splicing enhancer;

ISS=intron splicing silencer; ISE=intron splicing enhancer.

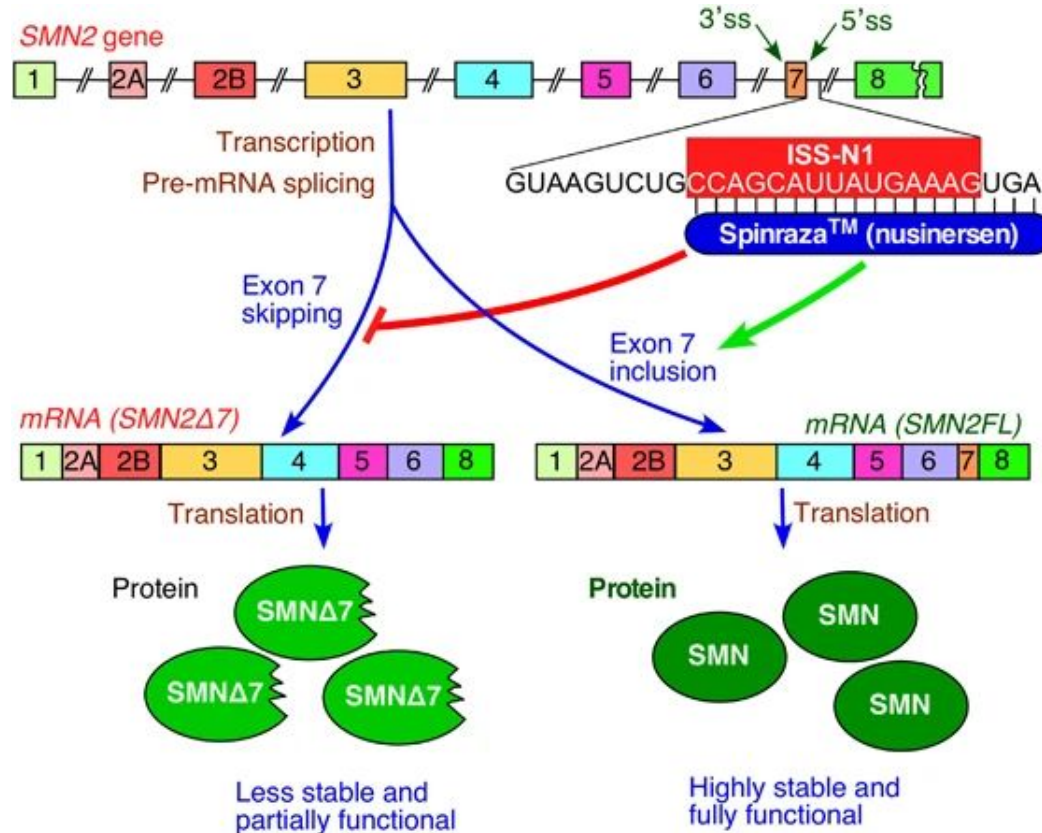
BPS=branch point sequence; PPT=polypyrimidine tract (C/U);



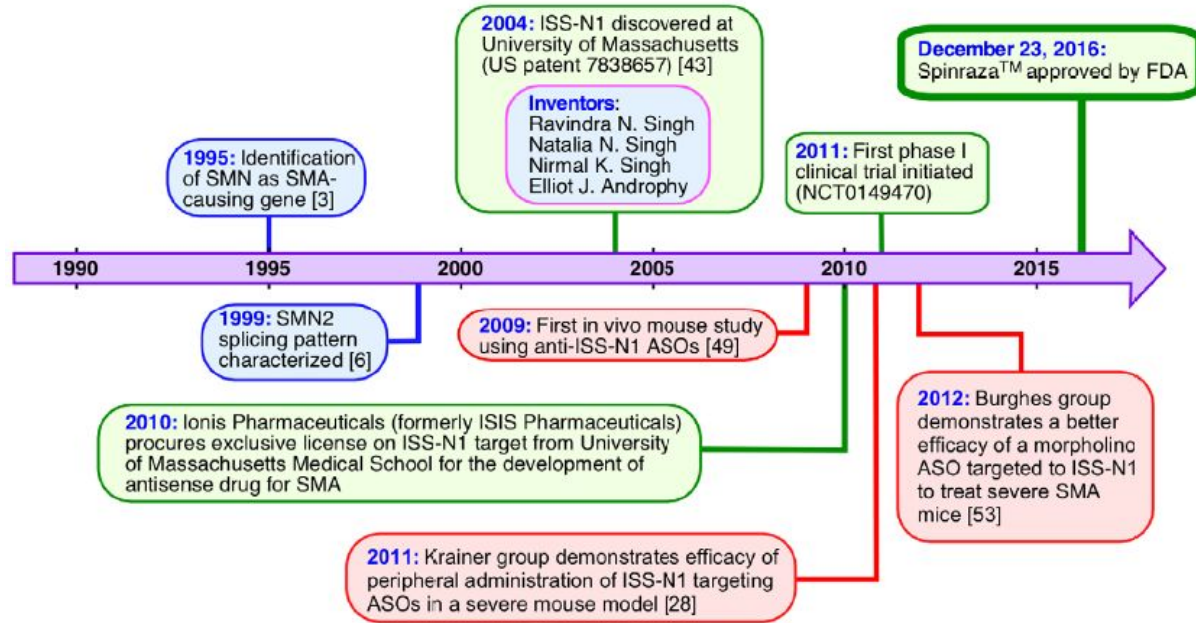
Different splicing of SMN1 and SMN2



How Spinraza (nusinersen) works



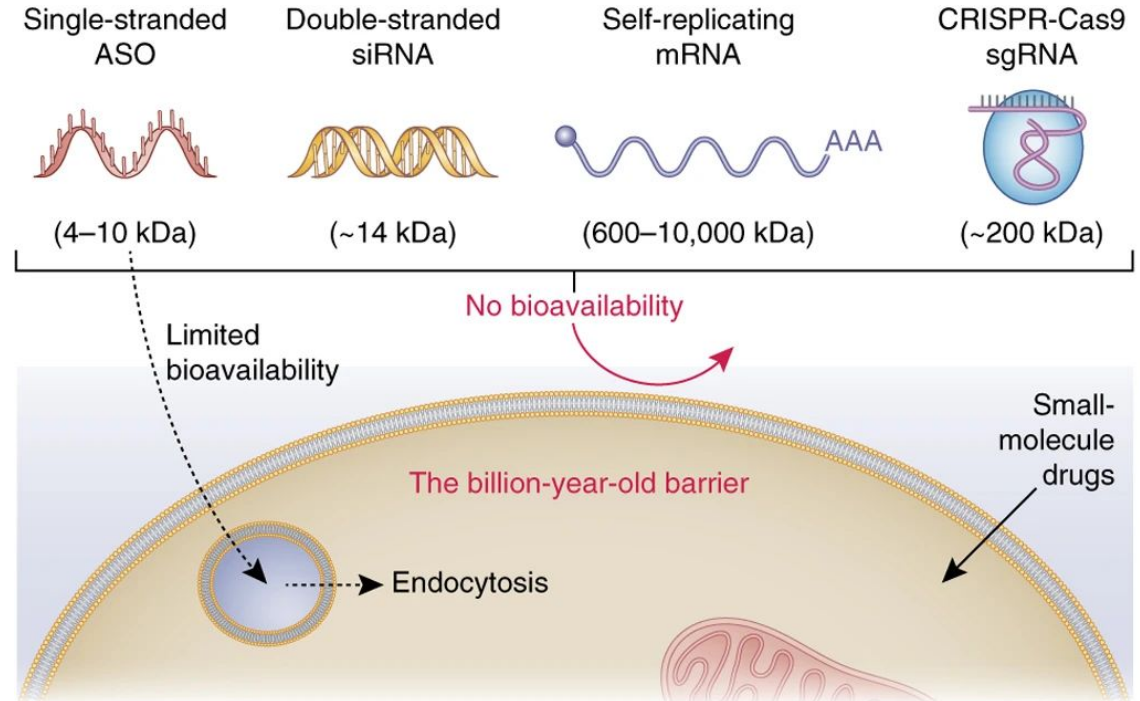
It took 21 years to turn biological insights into a drug



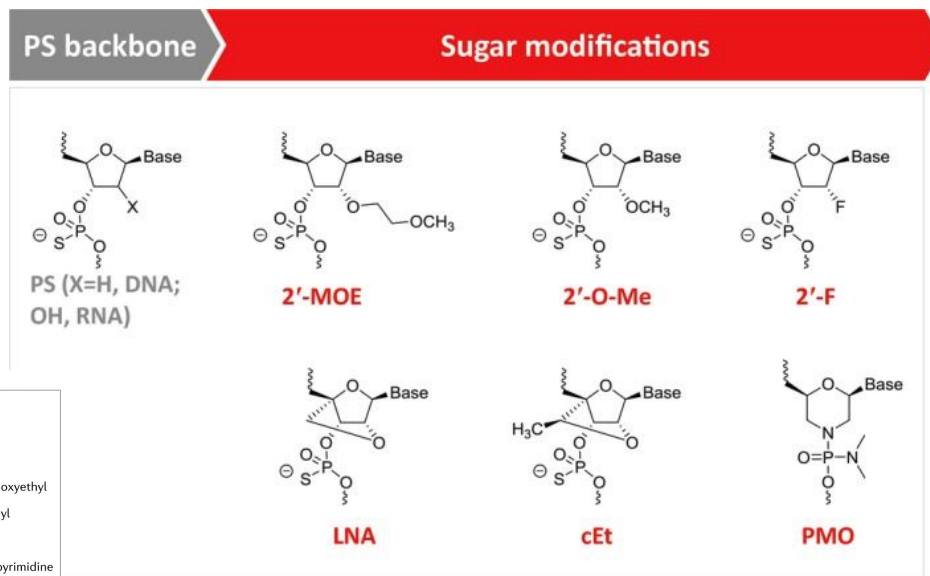
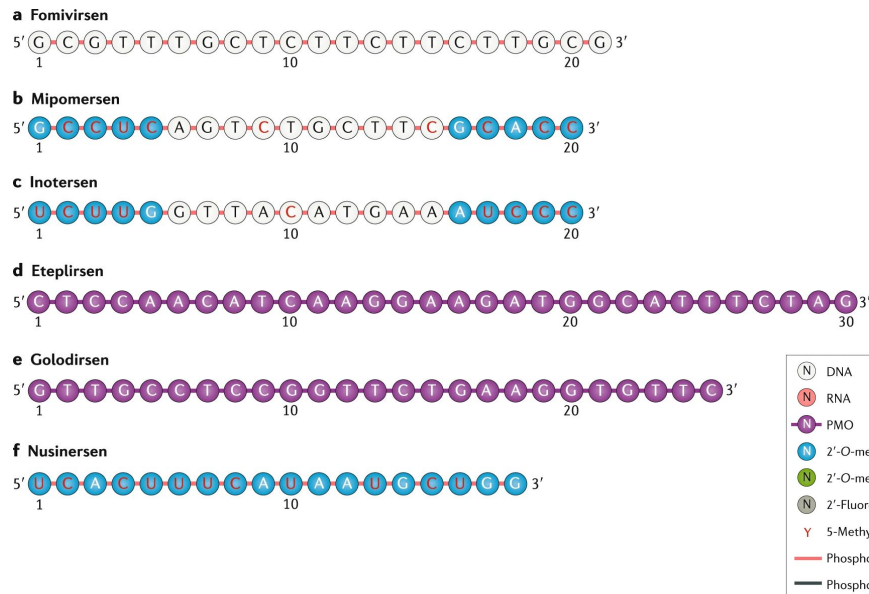
Besides biology understanding, chemistry need to be optimized to achieve good efficacy and favorable ADME, PK, and safety profiles.

The four-billion-year-old barrier to RNA therapeutic

- Too large and charged to pass lipid bilayers
- Degradable by RNases
- Rapid clearance from liver and kidney
- Immunogenicity
- Endocytosis
- Delivery into organs other than liver and eye



A variety of modifications make oligonucleotides better drug-like molecules



[Rin and Rogge, 2019](#)

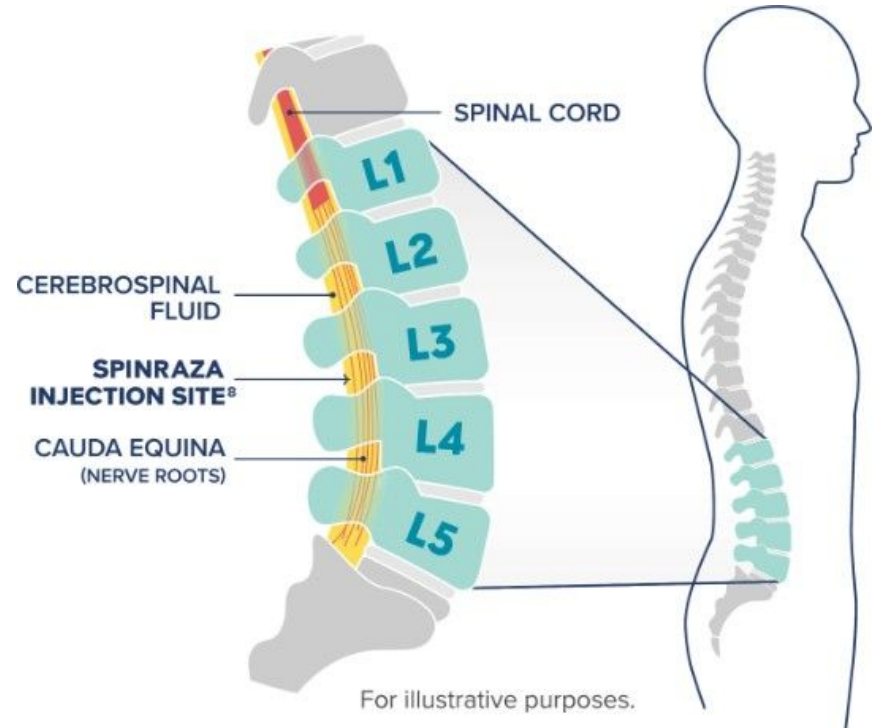
Nusinersen (Spinused 2'-O-methoxyethyl (2'-MOE) modified RNAs to make the molecule resist nuclease degradation so that they can be injected directly. Furthermore, the modifications increase the binding affinity, improve the specificity, and increase the lipophilicity to gain better PK properties. [See details here.](#)

Nusinersin was in 2016 approved as the first drug to treat SMA

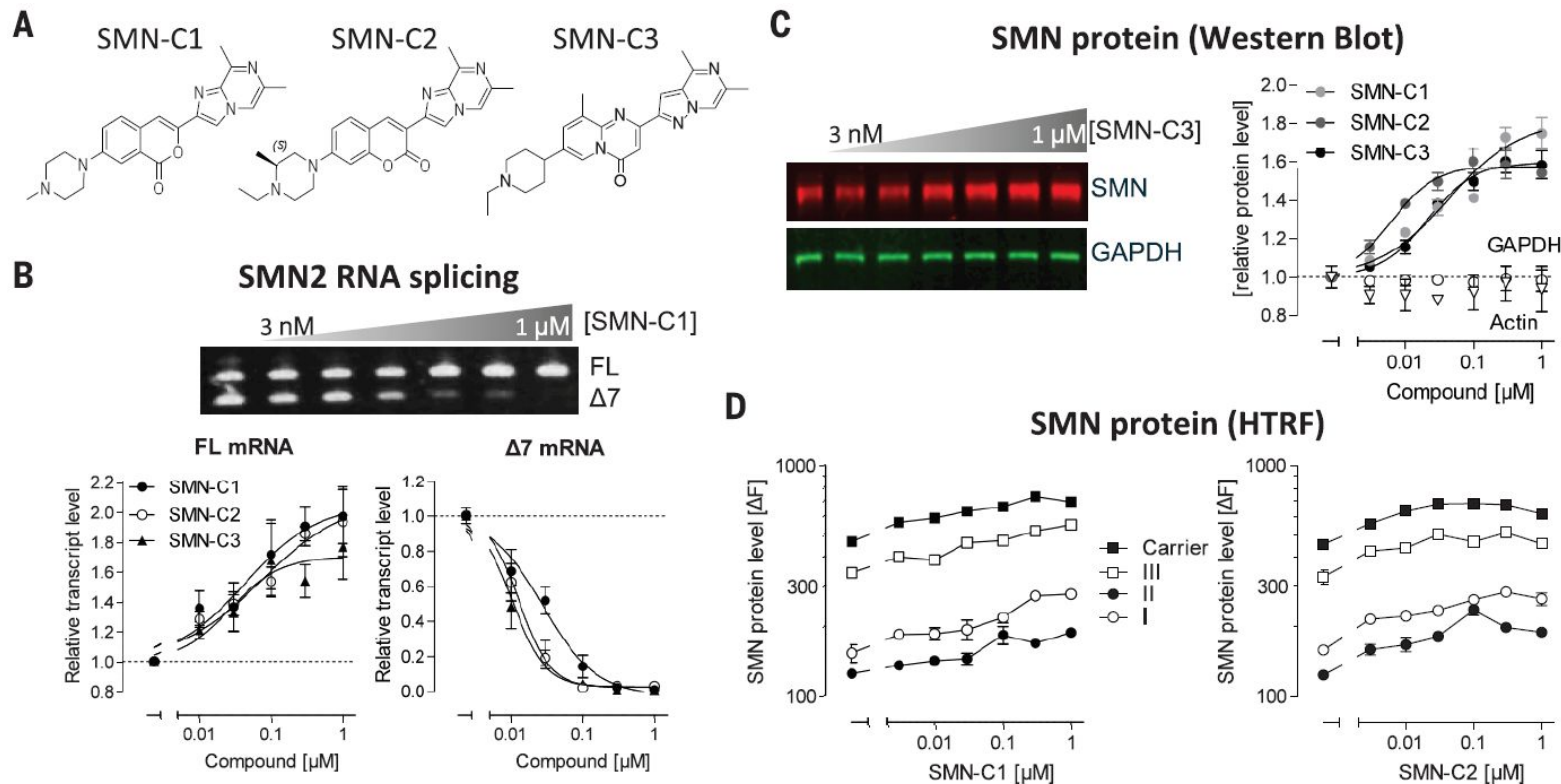
SPINRAZA (nusinersen) is administered intrathecally by healthcare professionals, with a dosing schedule that includes

- 4 initial loading doses, followed by
- maintenance doses every 4 months.

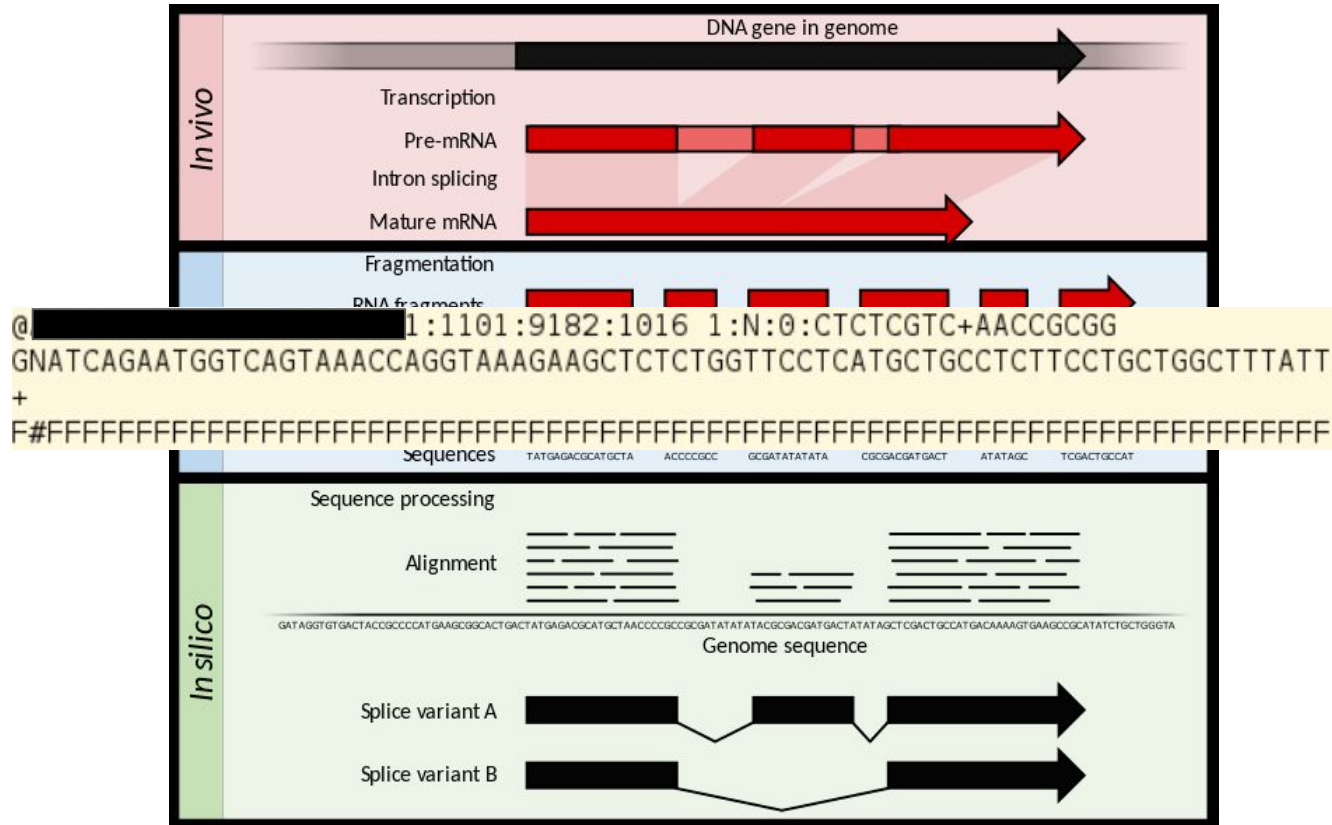
The High Dose Regimen involves two 50 mg loading doses 14 days apart, followed by 28 mg maintenance doses every 4 months.



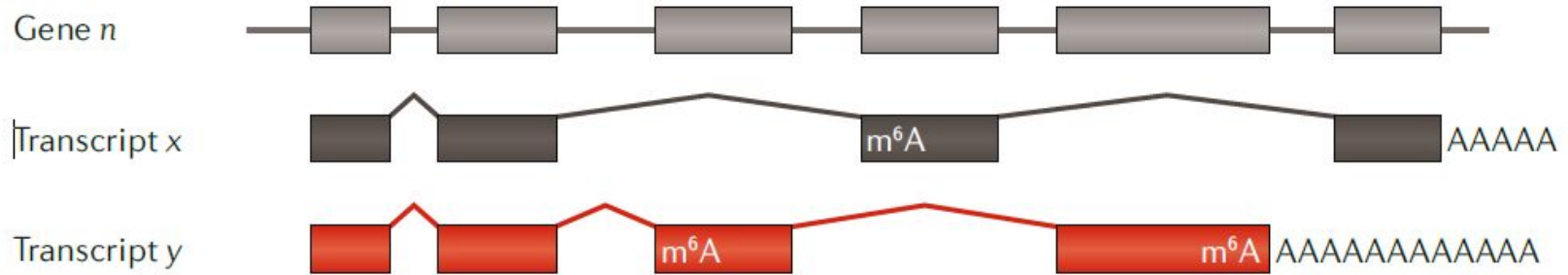
Small molecules as RNA splicing modifiers



Transcriptome profiling by RNA sequencing



Transcriptome profiling by RNA sequencing



Ambiguous
to exon



Unambiguous
to exon



Ambiguous
to isoform



Unambiguous
to isoform

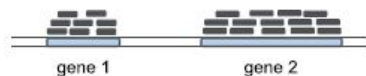


Read Mapping

Count collection

Normalization by library size

Differential Gene Expression Analysis



	sample A1	sample A2	sample B1	sample B2
gene 1	8	10	100	200
gene 2	14	15	15	40
gene 3	33	40	35	70
...	...	...	...	...
gene N	100	120	105	220

	sample A1	sample A2	sample B1	sample B2
gene 1	8	10	100	200
gene 2	14	15	115	40
gene 3	33	40	35	70
...	...	...	...	...
gene N	100	120	105	220

Tot. reads:
5 millions

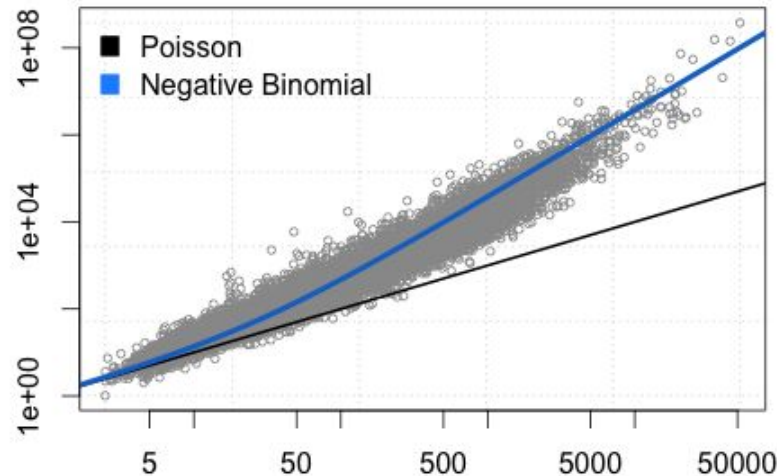
Tot. reads:
10 millions

	sample A1	sample A2	sample B1	sample B2
gene 1	0.16	0.20	2.00	2.00
gene 2	0.28	0.30	0.30	0.40
gene 3	0.66	0.80	0.70	0.70
...	...	...	...	...
gene N	2.00	2.40	2.10	2.20

Differential gene expression



Pooled gene-level variance (log10 scale)



Mean gene expression level (log10 scale)

Tools: *edgeR* and *DESeq2*

Interpret differential gene expression data with gene-set enrichment analysis

Reactome pathways

Gene Ontology

UniProt Keywords

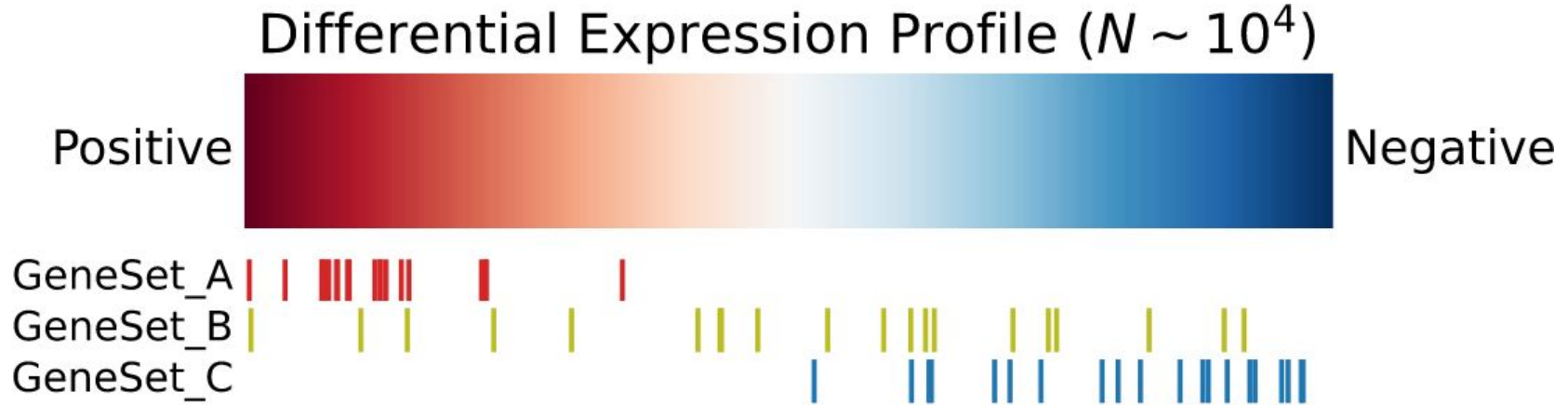
Literature

Gene (N~10 ⁴)	G ₁	G ₂	G ₃	G ₄	G ₅	...	G _{N-3}	G _{N-2}	G _{N-1}	G _N
Change (log2FC)	3.0	2.8	2.5	1.5	1.2	...	-0.8	-1.2	-1.5	-2.2

Differential gene expression results

Gene-set Enrichment Analysis Methods

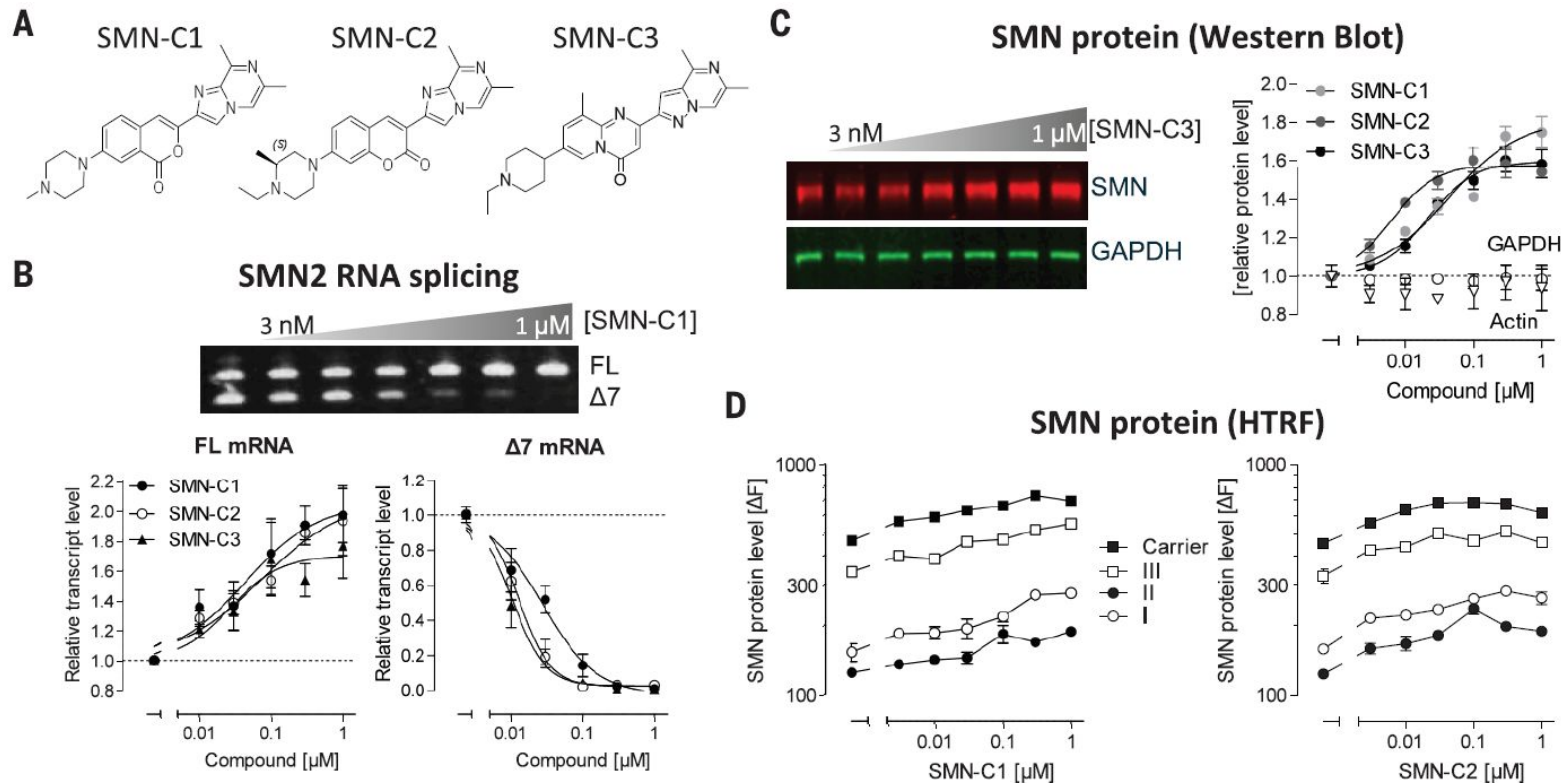
Gene-set enrichment analysis



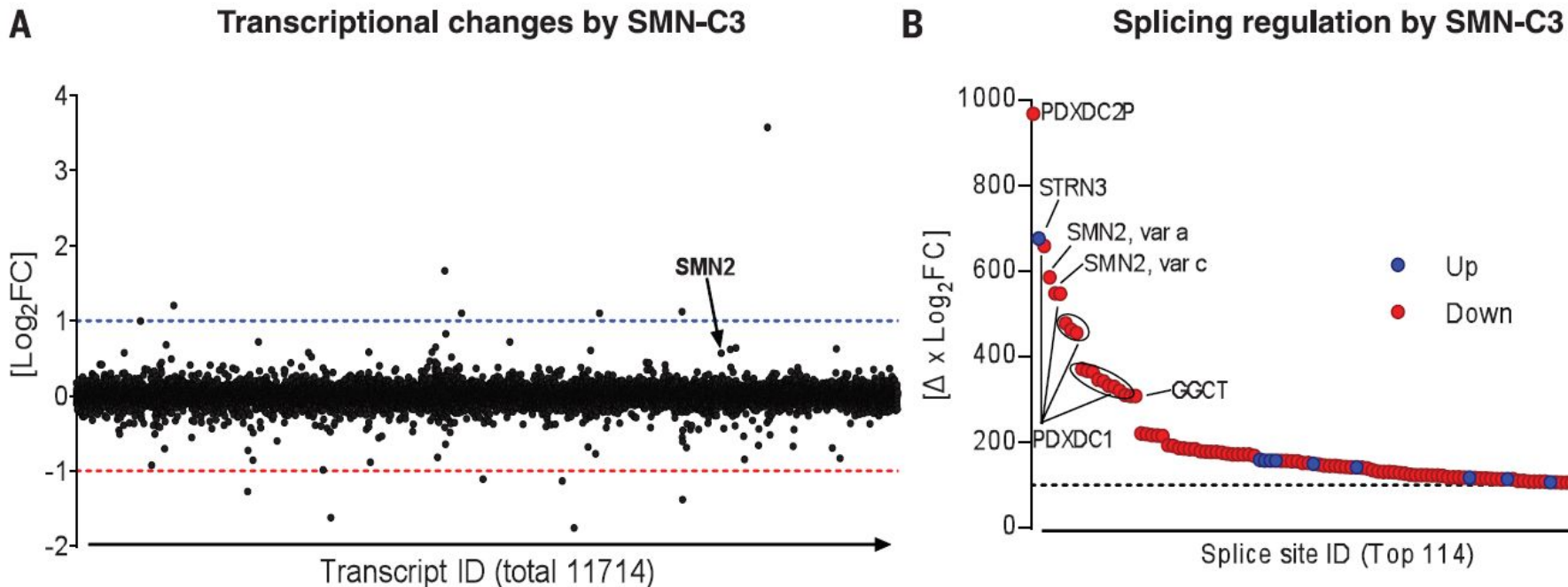
Input: (1) a differential gene expression profile; (2) a set of gene-sets $\{G\}$, each a set of genes.

Output: a ranked list of the input gene-sets by *enrichment*.

Small molecules as RNA splicing modifiers

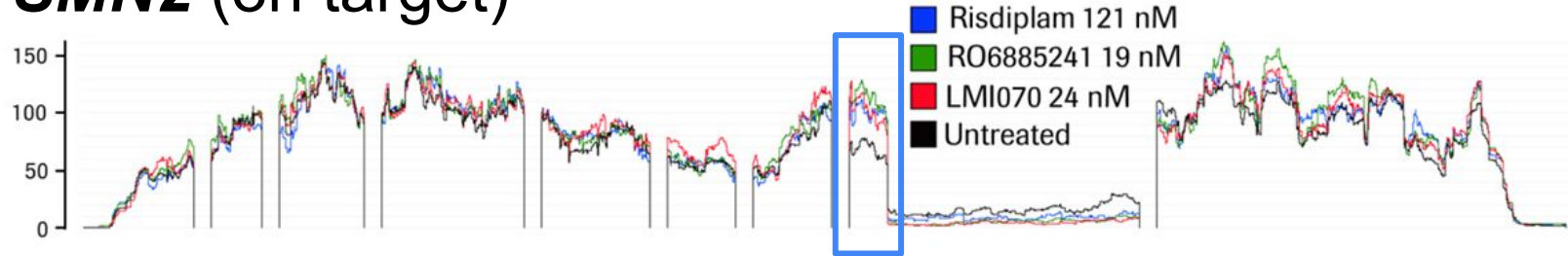


RNA sequencing confirms the specificity of SMN-C3

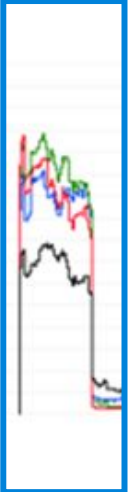
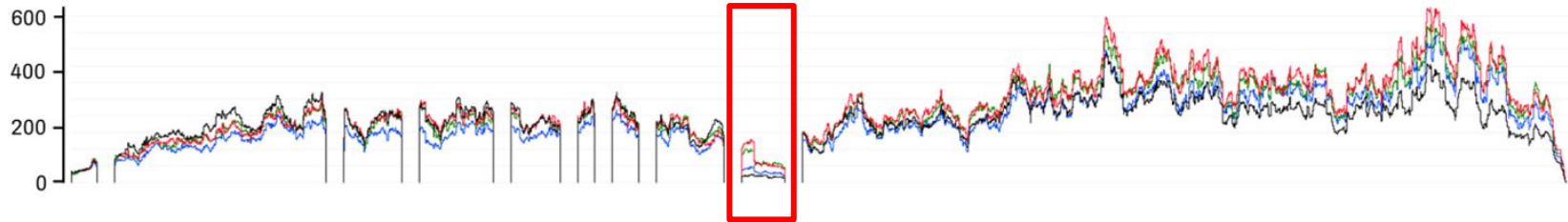


RNA sequencing confirms the specificity of SMN-C3

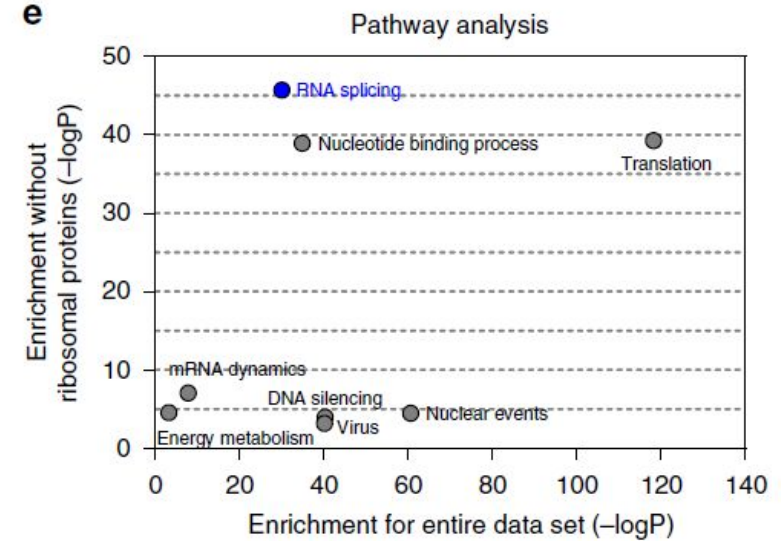
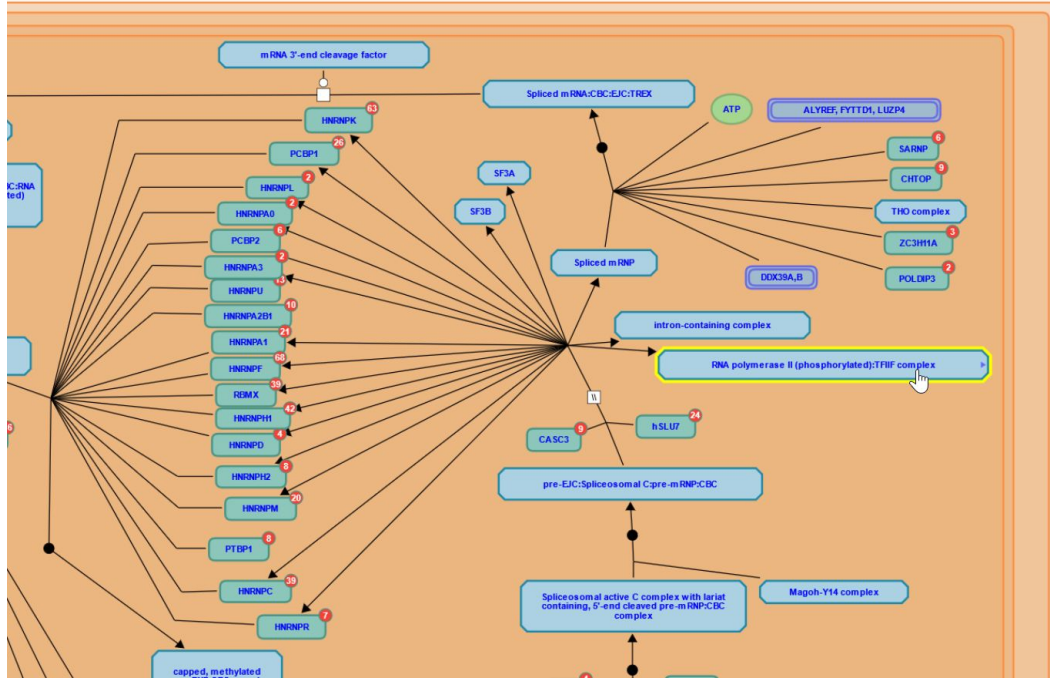
SMN2 (on target)



FOXM2 (off target)

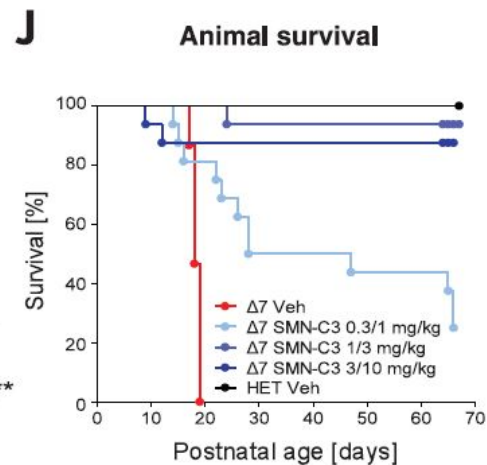
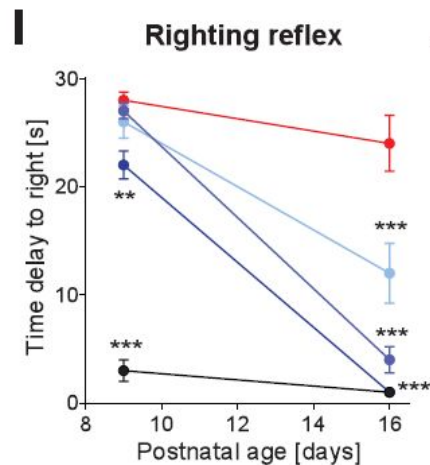
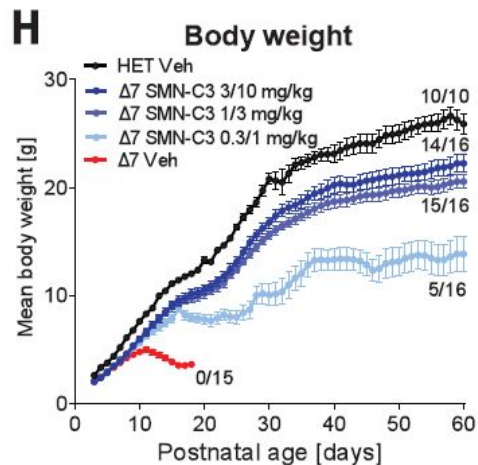
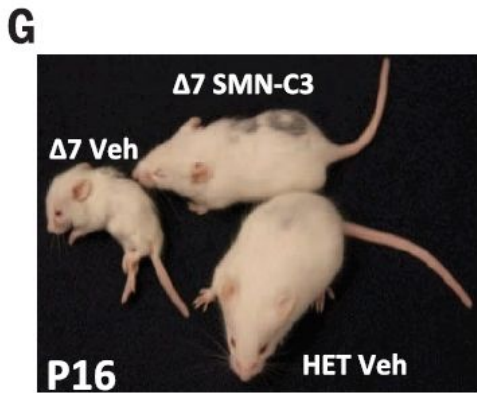
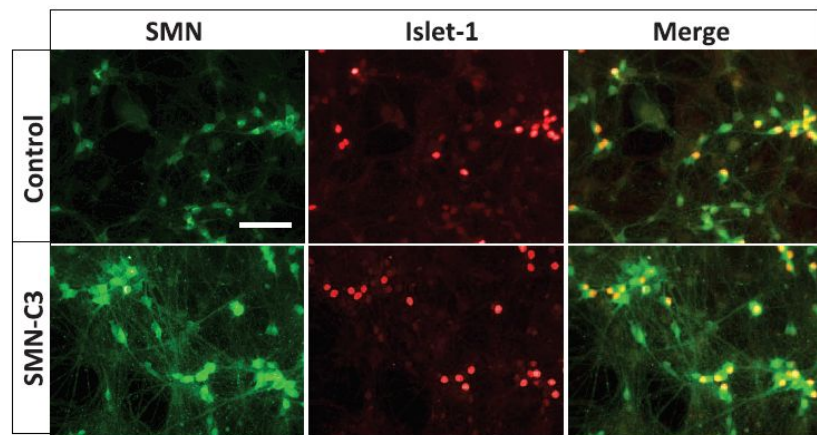


Gene-enrichment analysis confirms specific regulation of RNA splicing

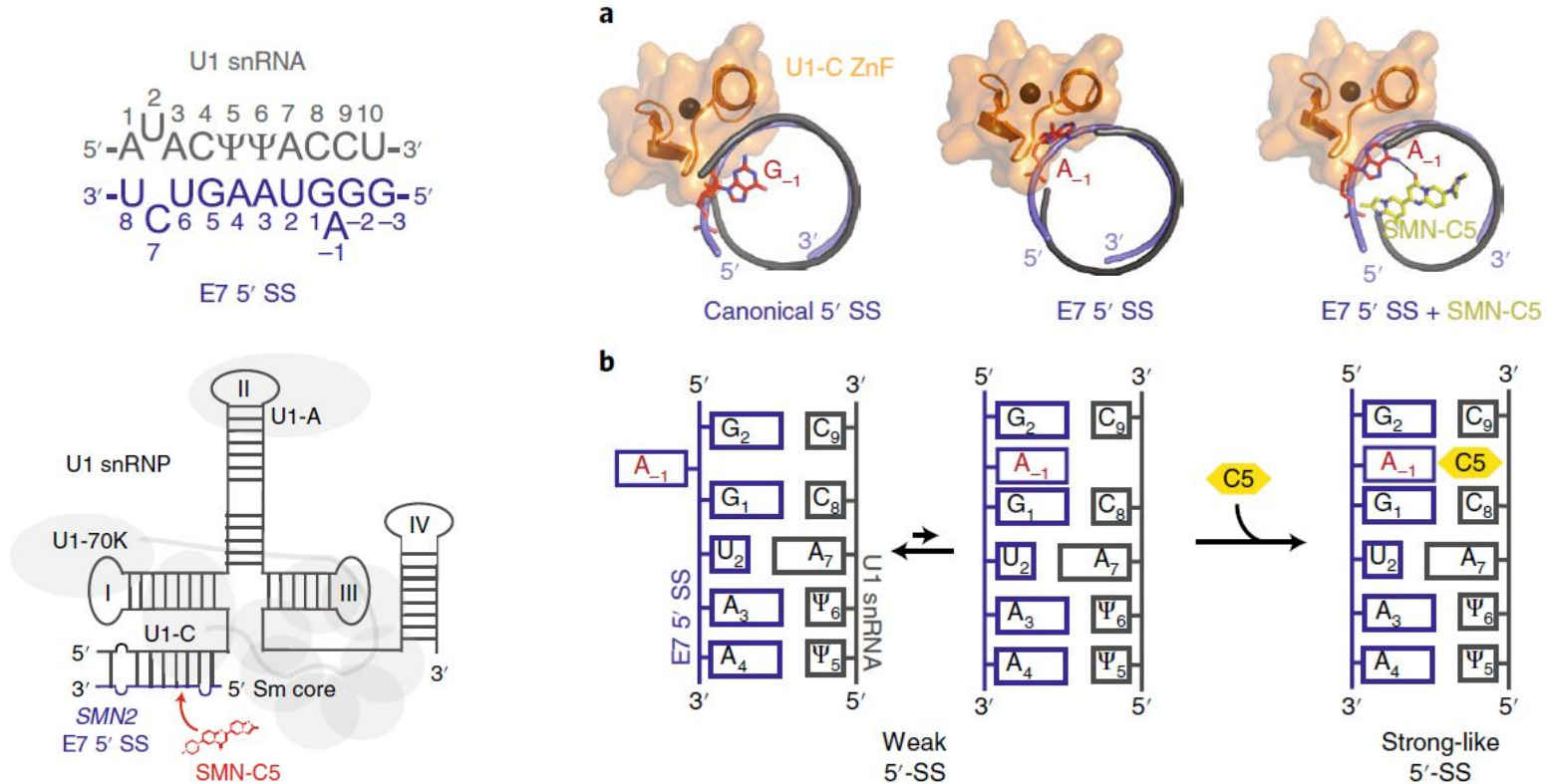


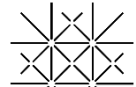
Part of the mRNA splicing pathway in Reacome

Experiments *in vitro* and *in vivo* support efficacy profiles of SMN-C3



Structural basis of specific splicing correction





Clinical trial ([FIREFISH](#) [Part 1](#)) Results

Table 1. Demographic and Clinical Characteristics of the Patients at Baseline.*

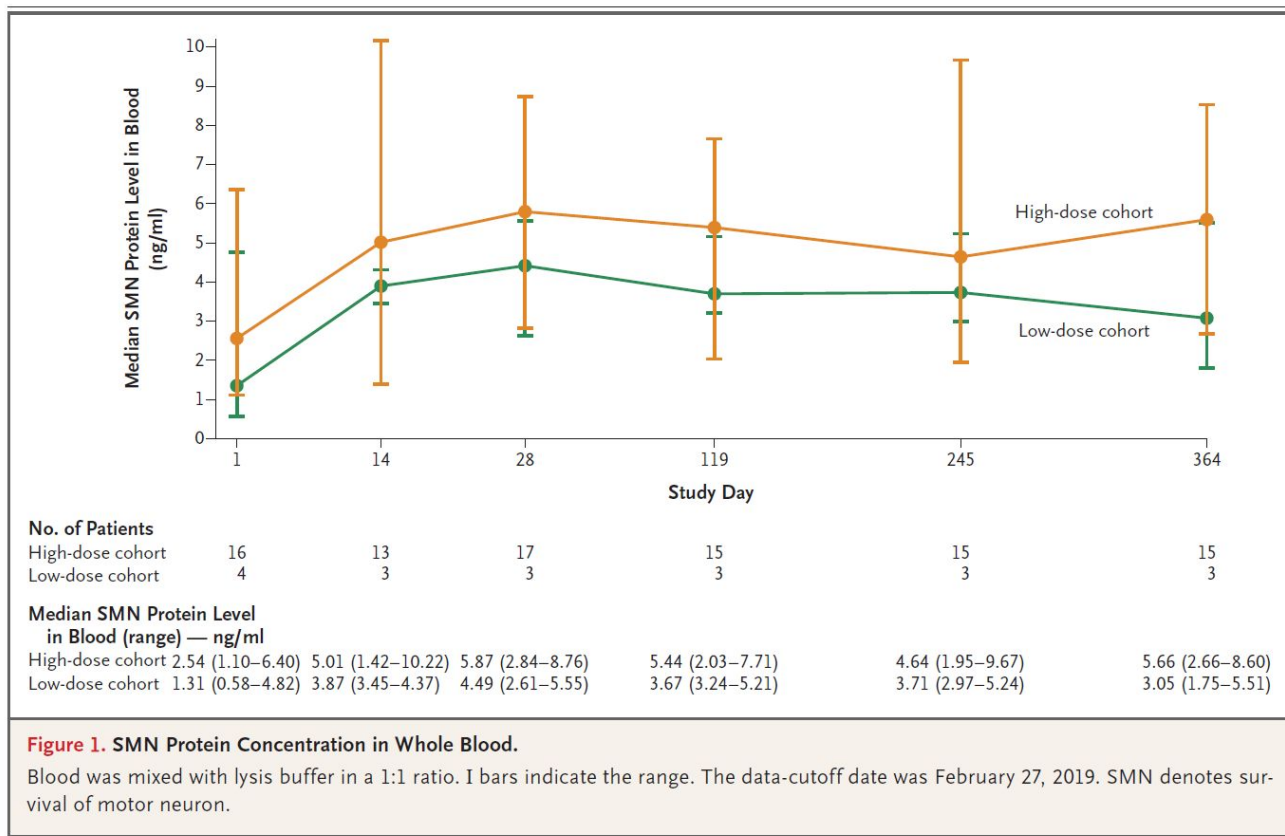
Characteristic	Low-Dose Cohort (N=4)	High-Dose Cohort (N=17)	All Infants (N=21)
Sex — no. (%)			
Female	4 (100)	11 (65)	15 (71)
Male	0	6 (35)	6 (29)
Median age (range) — mo			
At onset of symptoms	2.7 (2.0–3.0)	1.5 (0.9–3.0)	2.0 (0.9–3.0)
At diagnosis	3.3 (2.5–5.1)	3.0 (0.9–5.4)	3.0 (0.9–5.4)
At enrollment	6.9 (6.7–6.9)	6.3 (3.3–6.9)	6.7 (3.3–6.9)
Motor measures†			
Median CHOP-INTEND score (range)	23.5 (10–25)	24 (16–34)	24 (10–34)
Median HINE-2 score (range)	1 (0–3)	1 (0–2)	1 (0–3)
Respiratory support — no. (%)	0	5 (29)‡	5 (24)‡

Note: [Table 2](#) is not complete

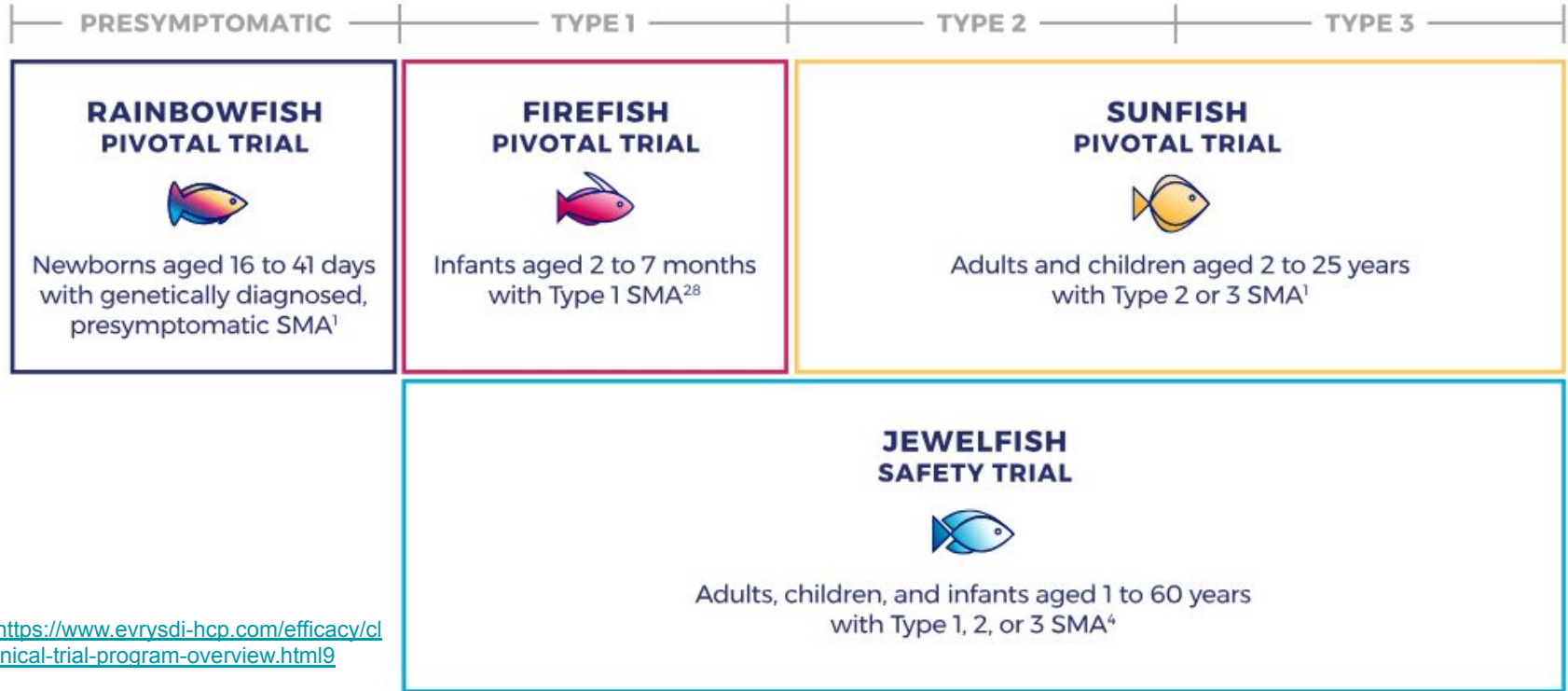
Table 2. Adverse Events.*

Event	Infants (N= 21)
Total no. of adverse events	202
≥1 Adverse event — no. (%)	21 (100)
Total no. of serious adverse events	24
≥1 Serious adverse event — no. (%)	10 (48)
≥1 Adverse event of grade 3–5 — no. (%)	9 (43)
Serious adverse event with fatal outcome — no. (%)†	3 (14)
Most common adverse events — no. (%)‡	
Pyrexia	11 (52)
Upper respiratory tract infection	9 (43)
Diarrhea	6 (29)
Cough	5 (24)

Clinical trial (FIREFISH Part 1) Results



Overview of clinical trials of Evrysdi



<https://www.evrysdi-hcp.com/efficacy/clinical-trial-program-overview.html#9>

Natural History Study:

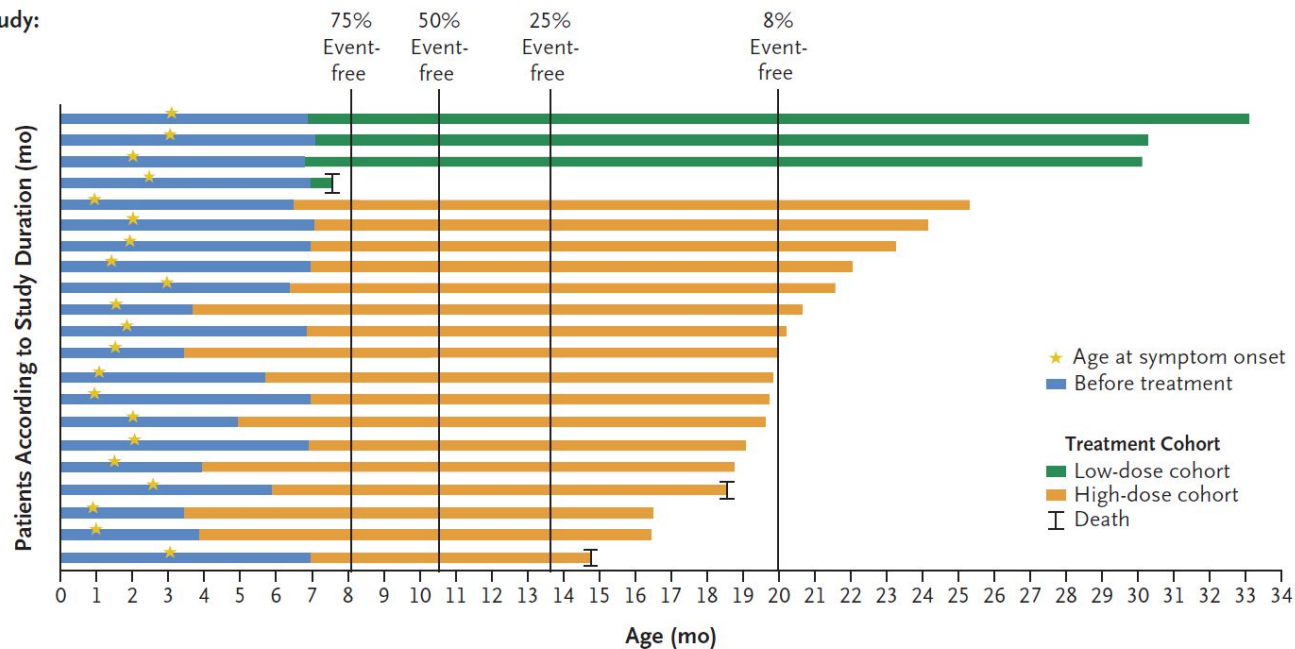
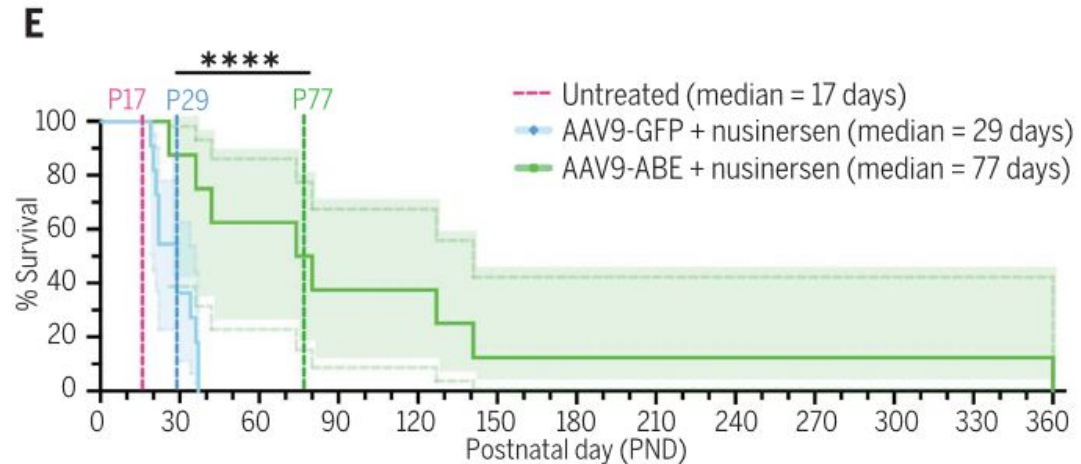
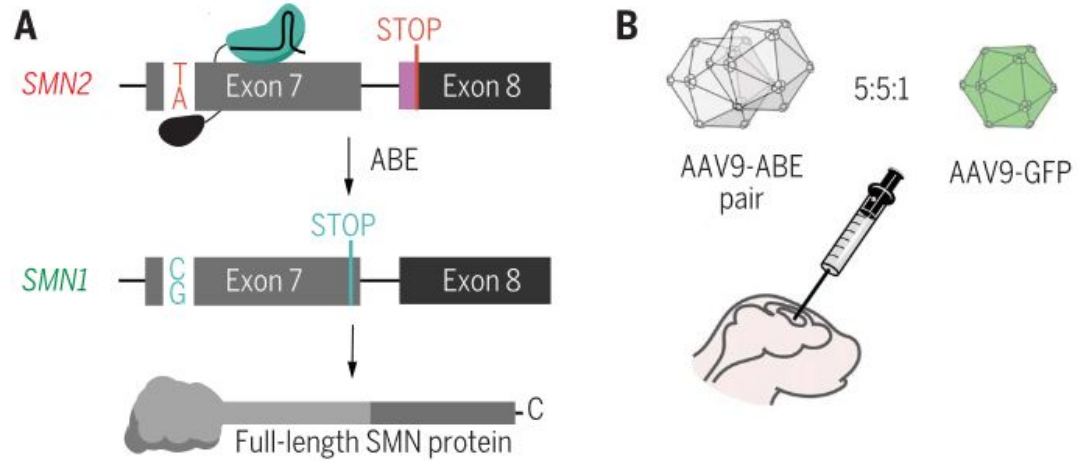


Figure 2. Event-free Survival.

Event-free survival was defined as being alive and not receiving permanent ventilation (tracheostomy or ventilation [bilevel positive airway pressure] for ≥ 16 hours per day continuously for >3 weeks or continuous intubation for >3 weeks, in the absence of, or after the resolution of, an acute reversible event). The percentages of patients who were event-free in a previous natural history study of spinal muscular atrophy⁷ are shown at the top of the graph for comparison. The median age at the combined outcome among patients in the previous study who had two copies of *SMN2* was 10.5 months (interquartile range, 8.1 to 13.6); event-free survival in that study was defined as being alive and not receiving noninvasive ventilation for 16 hours or more per day continuously for 2 or more weeks. The duration of our study was measured from the date of enrollment to the data-cutoff date. As of the data-cutoff date, three infants (one in the low-dose cohort and two in the high-dose cohort) had died; one additional infant in the high-dose cohort died after that date (Table S5).

Base editing rescue of spinal muscular atrophy in cells and in mice

Arbab *et al.*, Science, April 2023



Conclusions

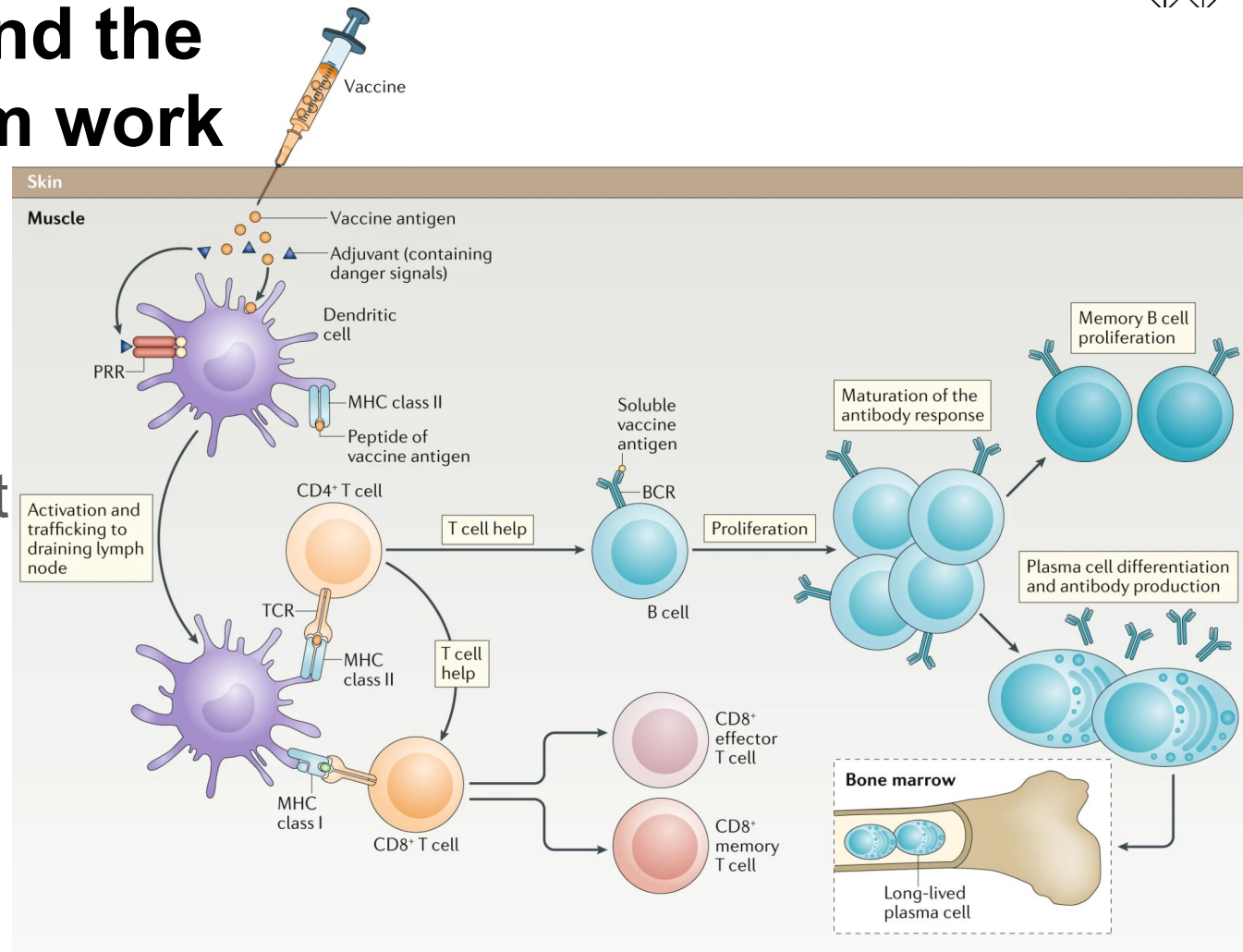
- Drugs of multiple modalities can be developed for the same disease
- Key considerations of modality selection includes efficacy, ADME/PK, and safety.
- Next-generation sequencing allows a comprehensive evaluation of drug's effect on the transcriptome with regard to efficacy and safety
 - Statistical thinking is critical to work with such methods effectively.

Supplementary Information

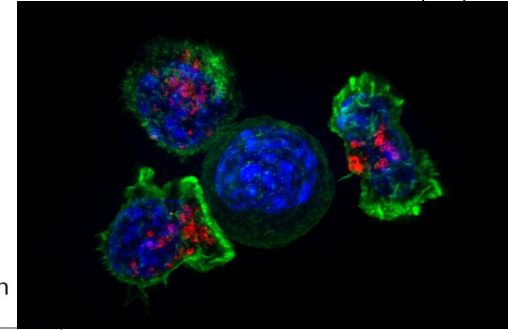
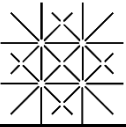
How vaccine and the immune system work

Key players:

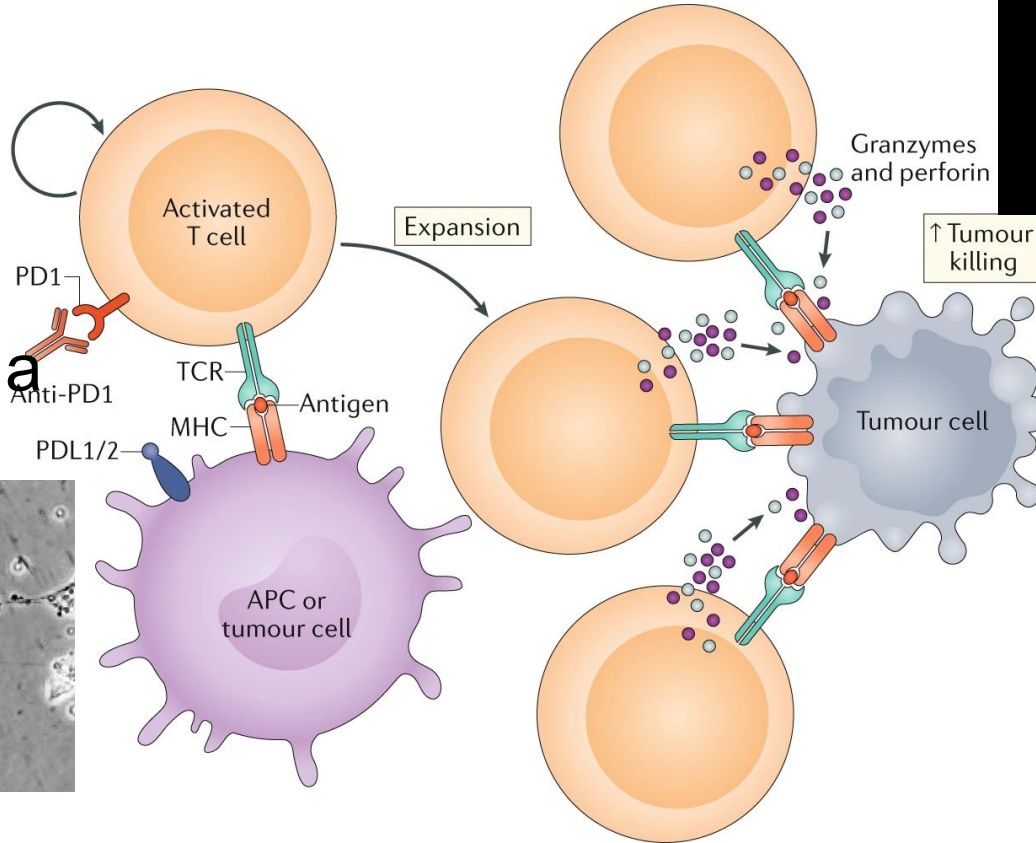
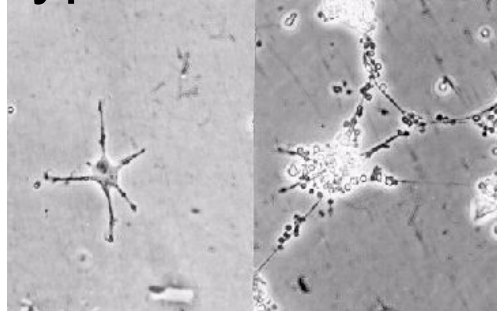
1. Antigen-presenting cells (e.g. dendritic cells)
2. T cells
3. B cells



Antigen-presenting cells (APC) and T cells work together to kill tumour cells

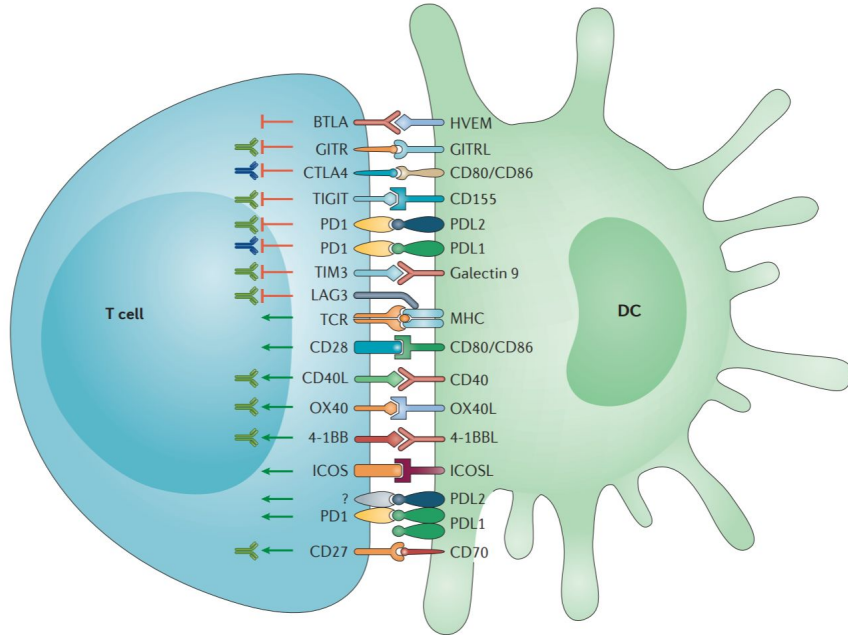


Dendritic cell, a type of APC

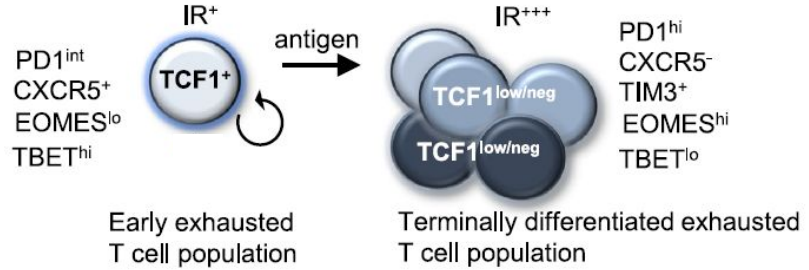


Killer T cells attacking a cancer cell

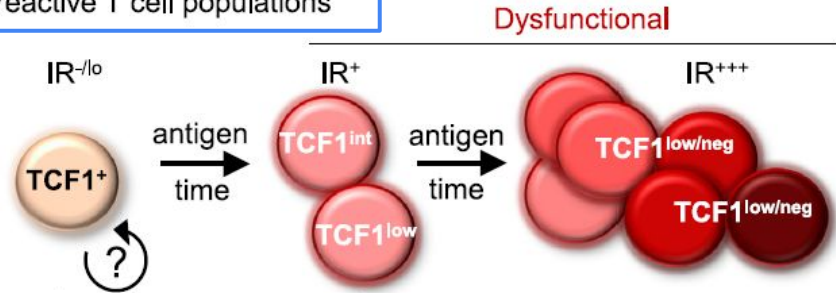
Exhausted T cells reduces immune system's capacity to clear pathogenic cells



Virus-specific exhausted T cell population

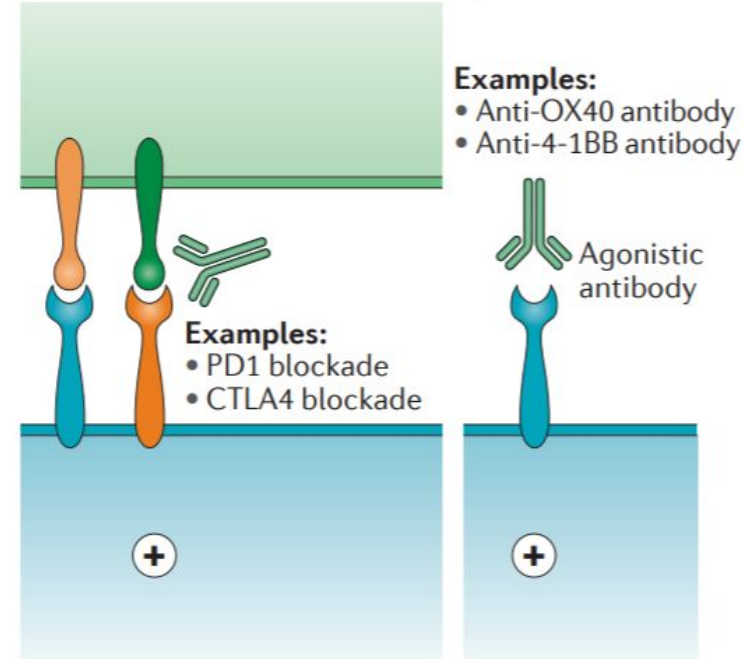
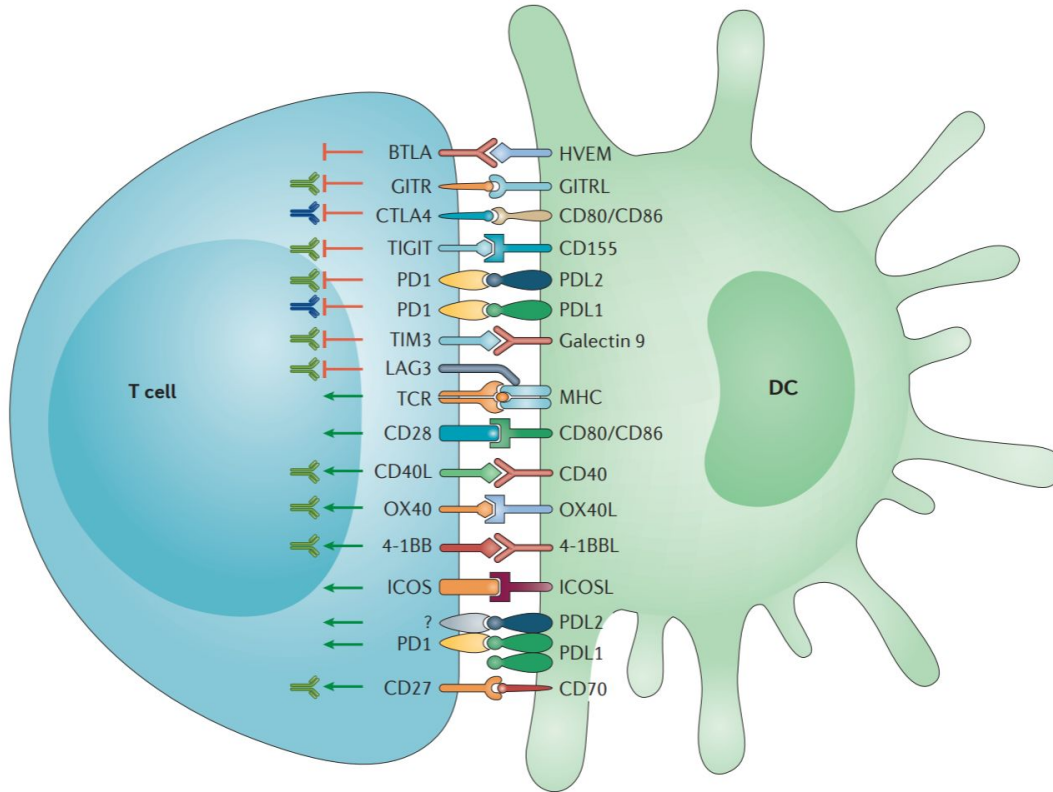


Tumor-reactive T cell populations

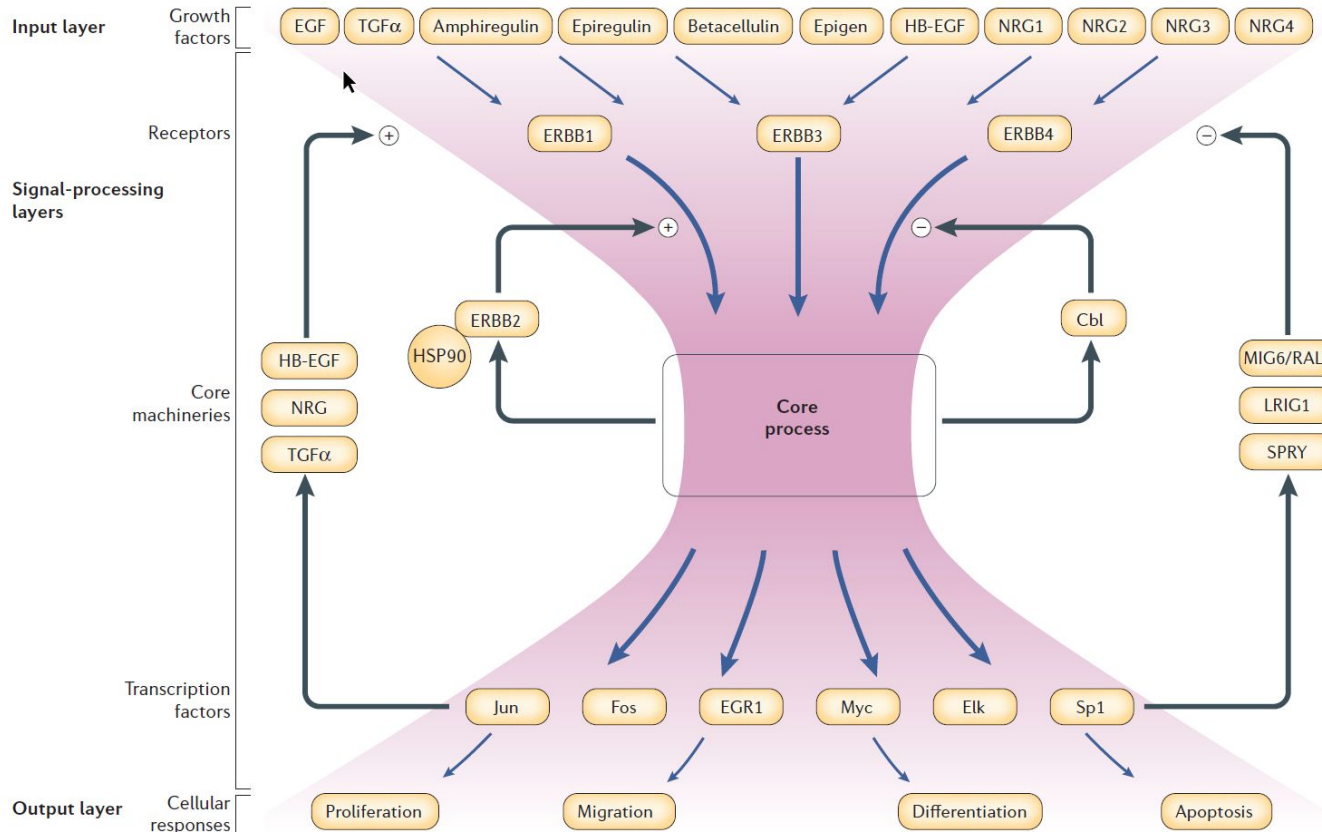


IR=inhibitory receptors (left panel).
They are like 'breaks' controlled
by dendritic cells.

Cancer Immunotherapy with immune checkpoints as drug targets



Why antibodies work like a wonder? The Bow-Tie model of signaling transduction



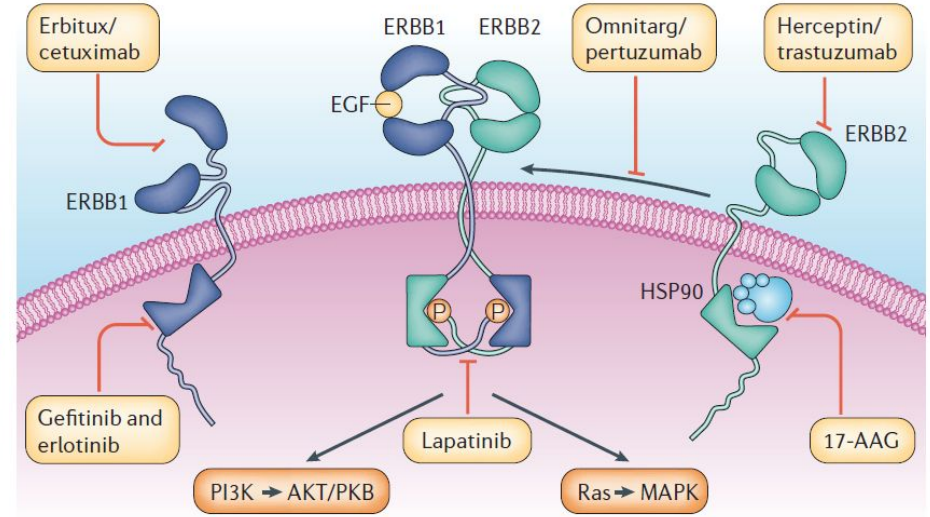
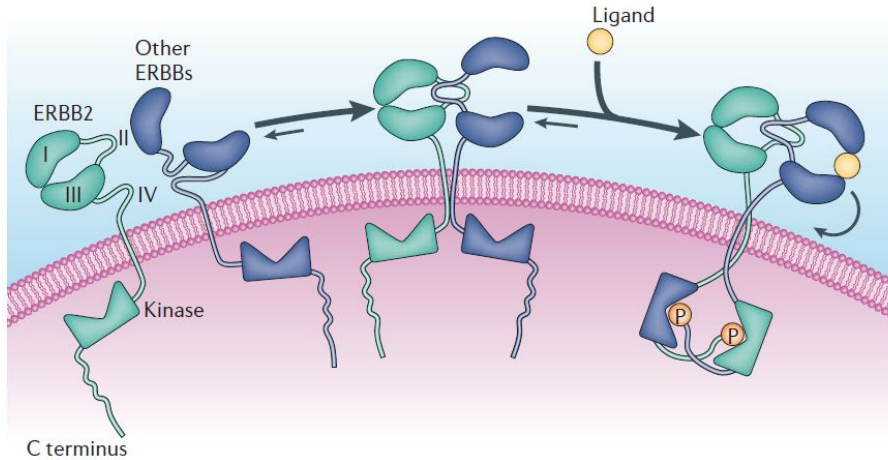
Extracellular
Cell
Membrane

Cytoplasm

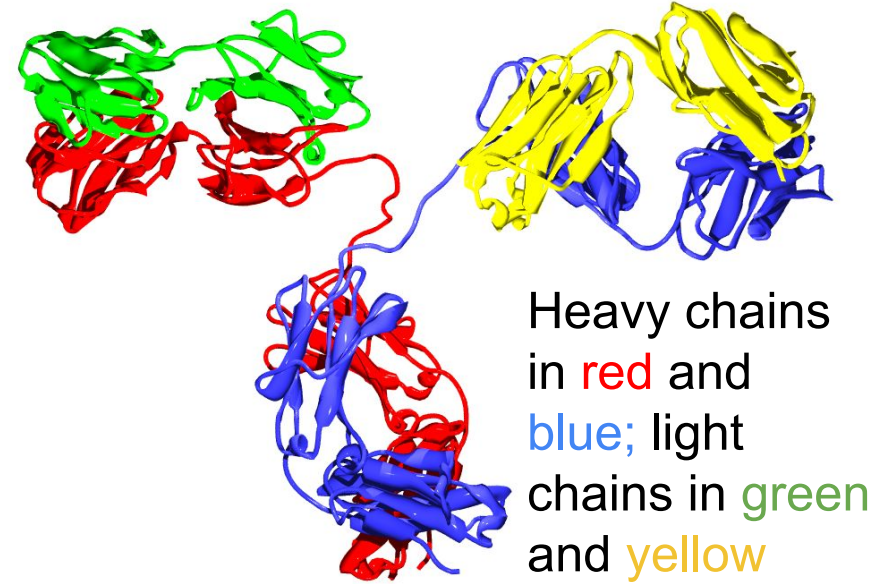
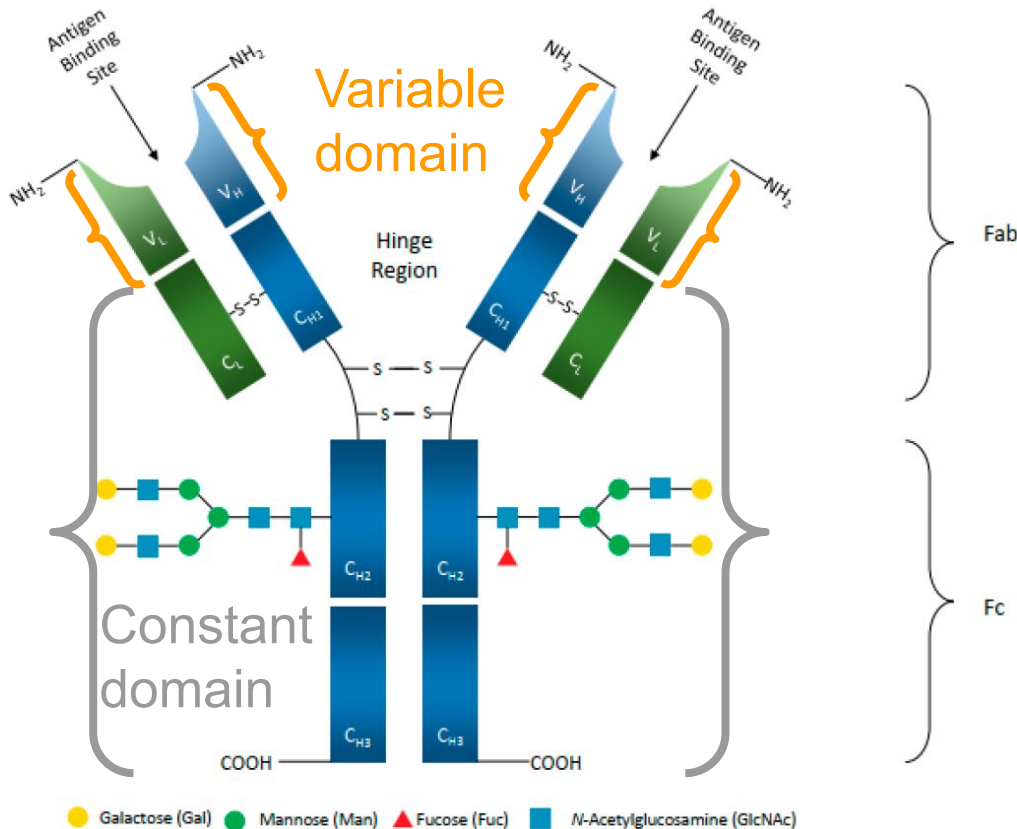
Nucleus

Everywhere 41

ERBB signaling system and antibody drugs



Structure of antibodies



Fc=fragment crystallizable region
Fab=fragment antigen binding

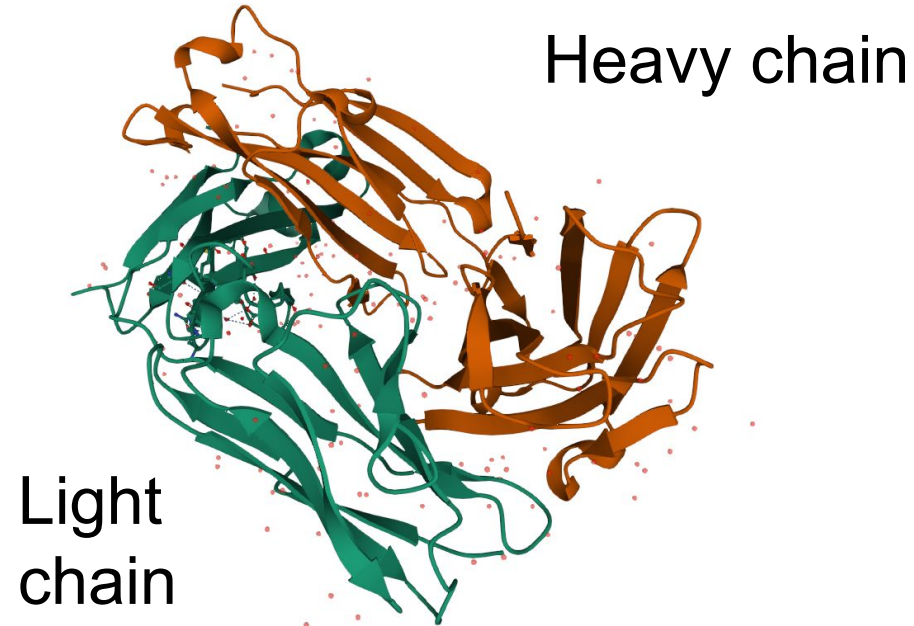
Cetuximab as an example

Variable heavy chain

QVQLKQSGPGLVQPSQSLSITCTVSGF
SLTNYGVHWRQSPGKGLEWLGVIWSG
GNTDYNTPFTSRLSINKDNSKSQVFFK
MNSLQSNDTAIYYCARALTYDYEFAY
WGQGTLLVTVSA

Variable light chain

DILLTQSPVILSVSPGERVSFSCRASQ
SIGTNIHWYQQRNGSPRLLIKAYASES
ISGIPSRFSGSGSGTDFTLSINSVESE
DIADYYCQQNNNWPTTFGAGTKLELK

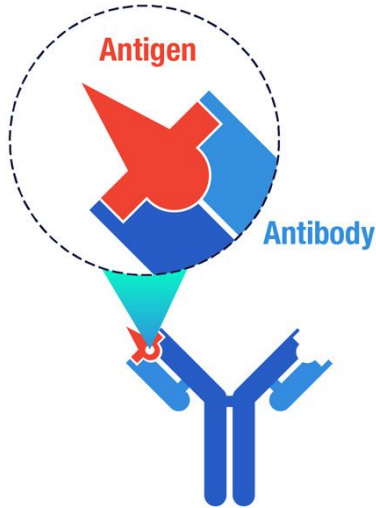


[PDB 1YY8](#)

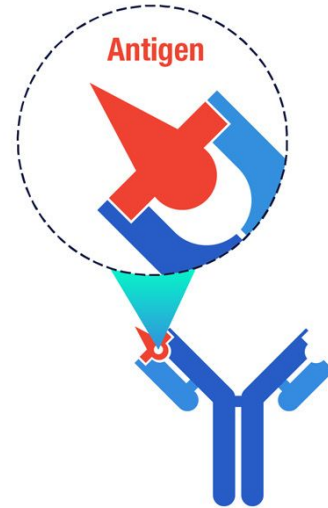
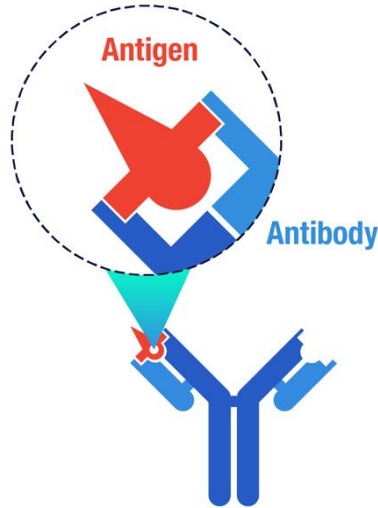
Antibodies work by shape complementarity

Affinity of antibodies for antigens can vary

High-Affinity Binding



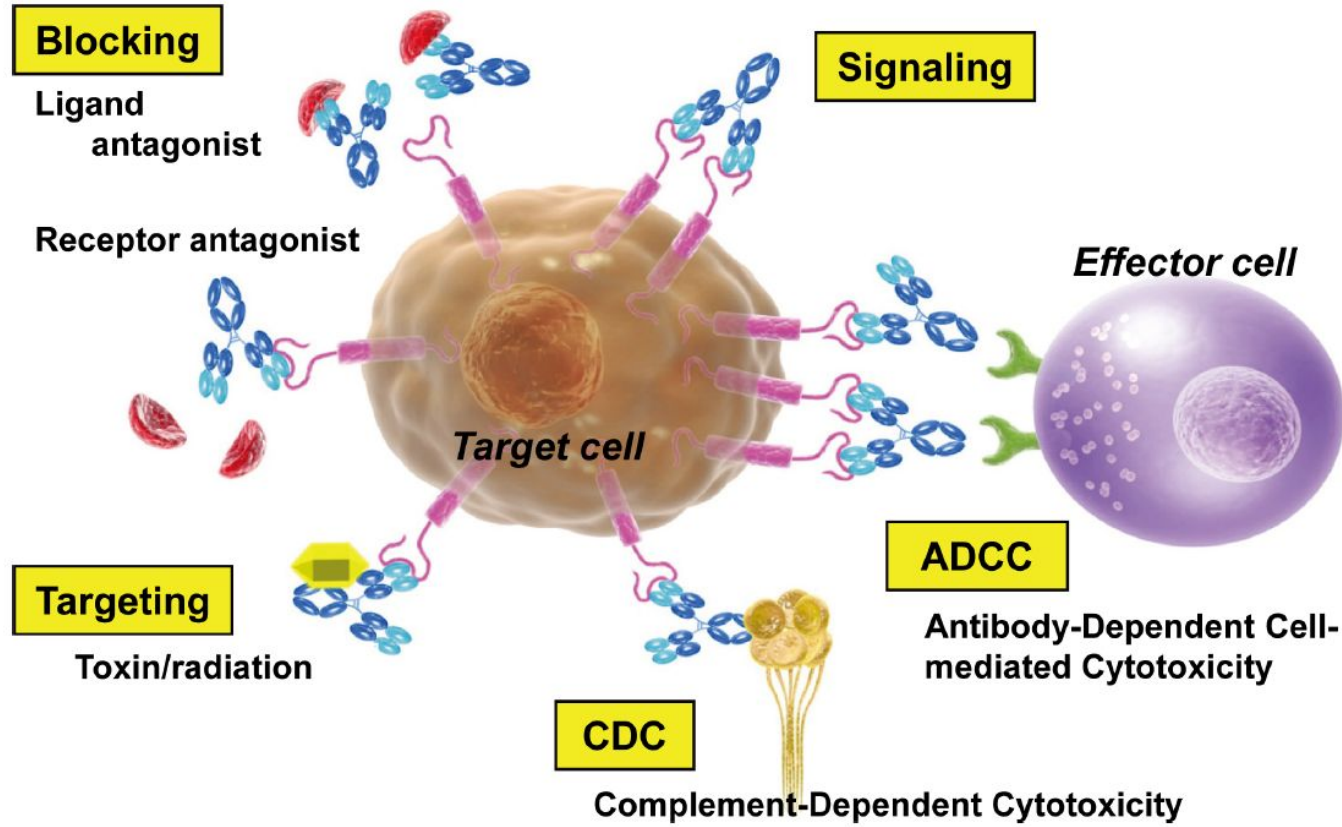
Low-Affinity Binding



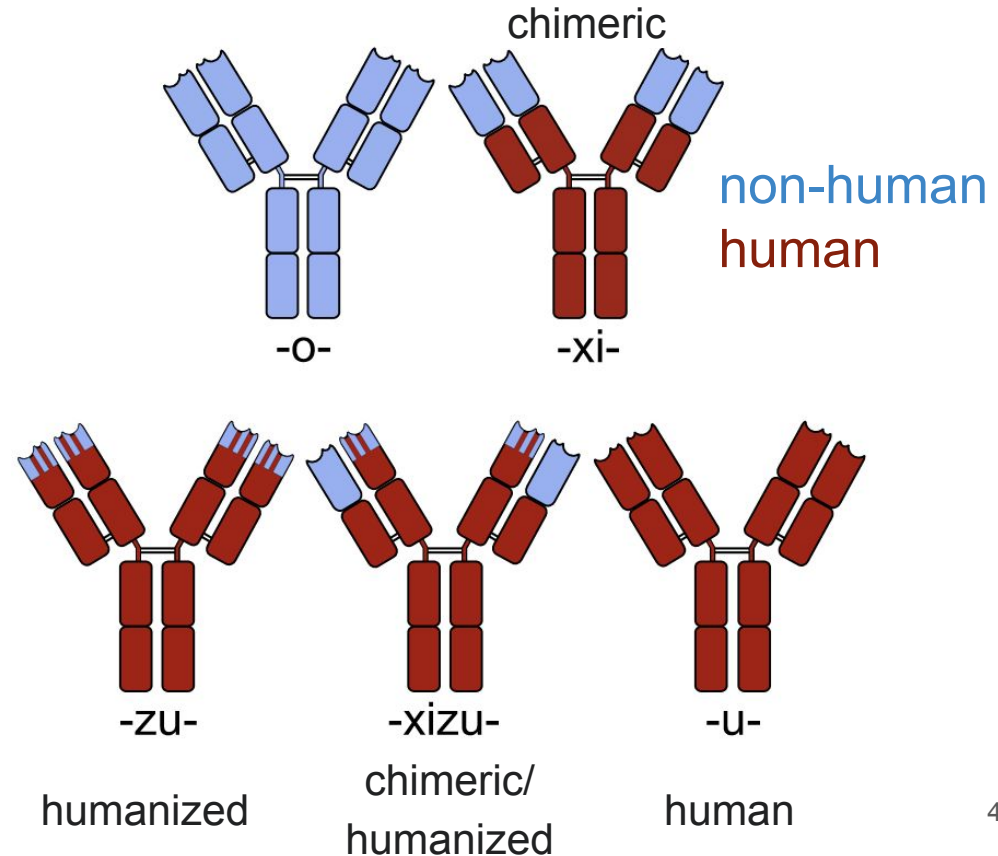
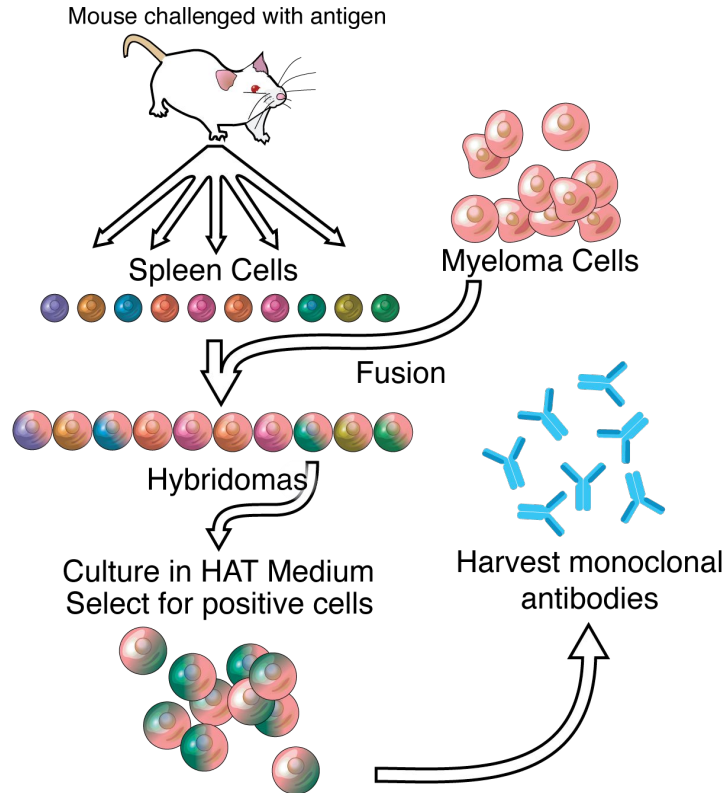
The responsible peptides are known as the Complementarity Determining Region (CDR).

Recommended reading: [10 things to know about antibodies](#) by Amgen

Mechanisms of action of therapeutic antibodies

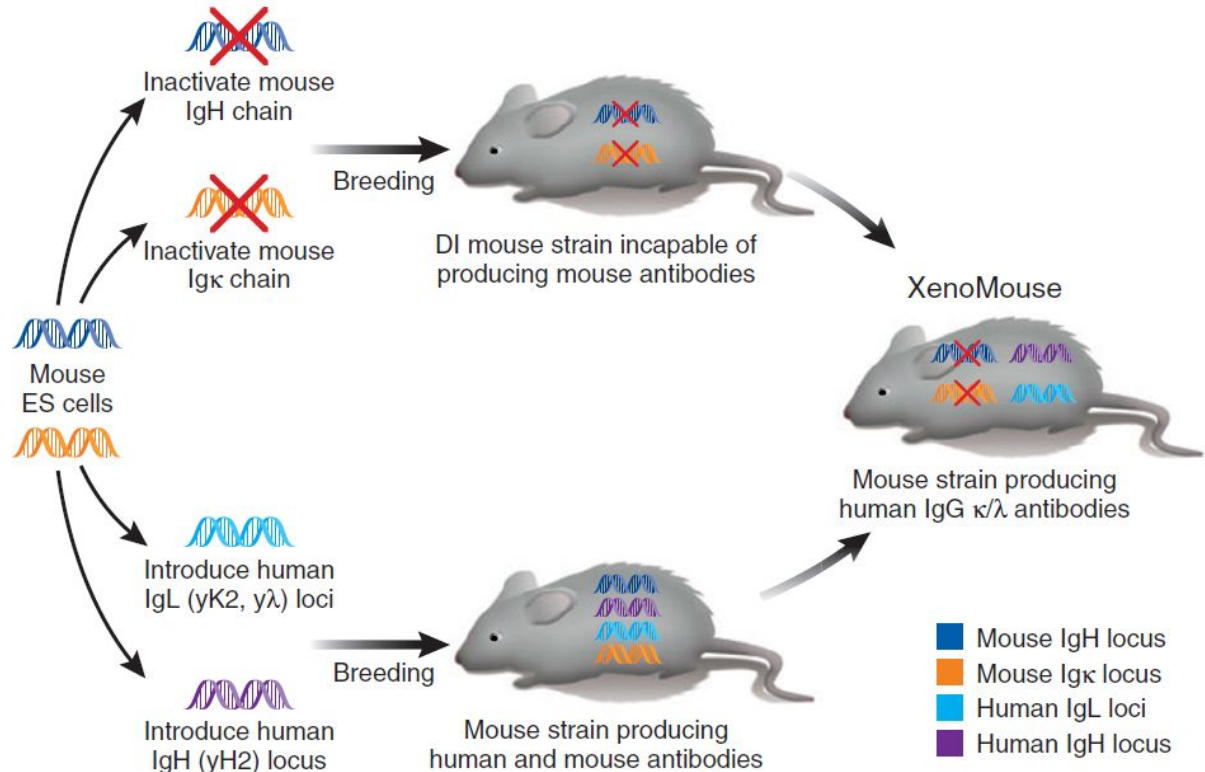


Therapeutic antibody discovery with hybridoma and humanization



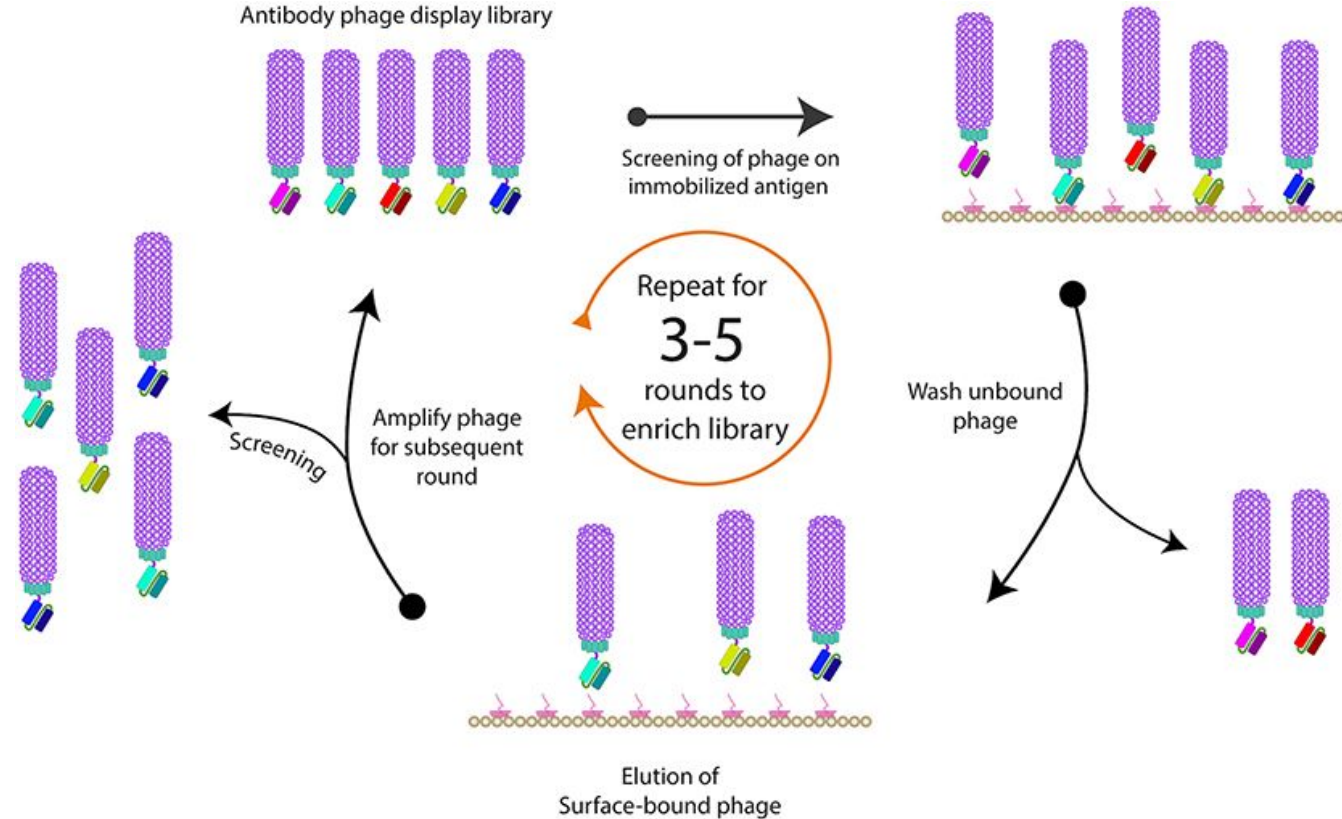
Therapeutic antibody discovery with transgenic animals

The XenoMouse model led to the discovery of panitumumab (Vectibix). Panitumumab targets EGFR for advanced colorectal cancer.



The principle of phage display

A protein-encoding gene is inserted into the phage coat protein gene, causing the phage to **display** the protein, which can be screened *in vitro* iteratively.



Antibody discovery with phage display

Convalescent or patient blood plasma

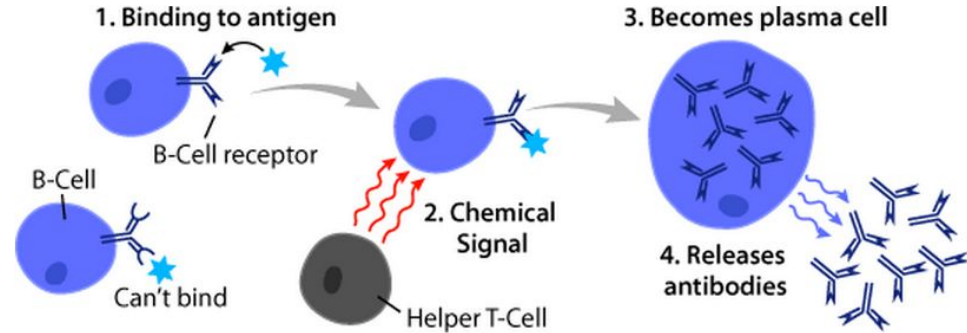
RNA → cDNA

Copy antibody-coding sequences

Paste sequences into phage coats

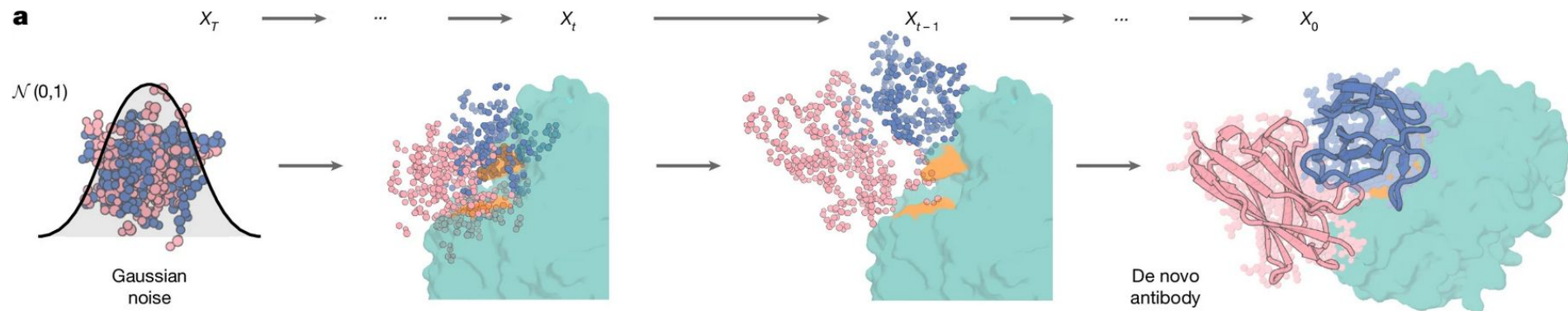
Antibody selection

Copy/Paste: PCR and cloning



Aim: Reverse-engineering existing antibodies

Atomically accurate de novo design of antibodies with *RFdiffusion*



Antibody discovery currently relies on immunization, random library screening, or the isolation of antibodies directly from patients. The study demonstrates that combining computational protein design using a diffusion model with yeast display screening enables the de novo generation of antibodies that bind to user-specified epitopes.

Bennett, Nathaniel R., Joseph L. Watson, Robert J. Ragotte, et al. Baker, David "Atomically Accurate de Novo Design of Antibodies with RFdiffusion." Nature 649, no. 8095 (2026): 183–93. <https://doi.org/10.1038/s41586-025-09721-5>.

Computational design pipeline and study design

Epitope-specific antibody design pipeline

RFdiffusion
backbone design



ProteinMPNN
CDR sequence
design



Fine-tuned
RoseTTAFold2
In silico filtering

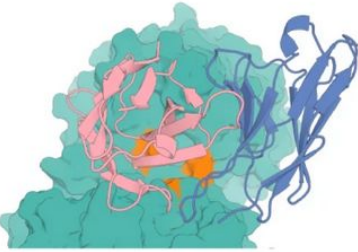
Epitope-specific antibody design



Experimental characterization

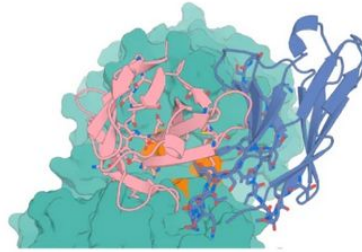


Antibody lead optimization



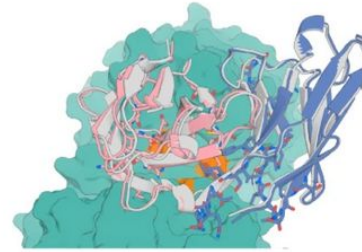
Inputs:

- Target structure
- Target sequence
- Hotspots to bind
- Internal framework structure and sequence



Inputs:

- Target structure
- Target sequence
- Designed antibody structure
- Framework sequence



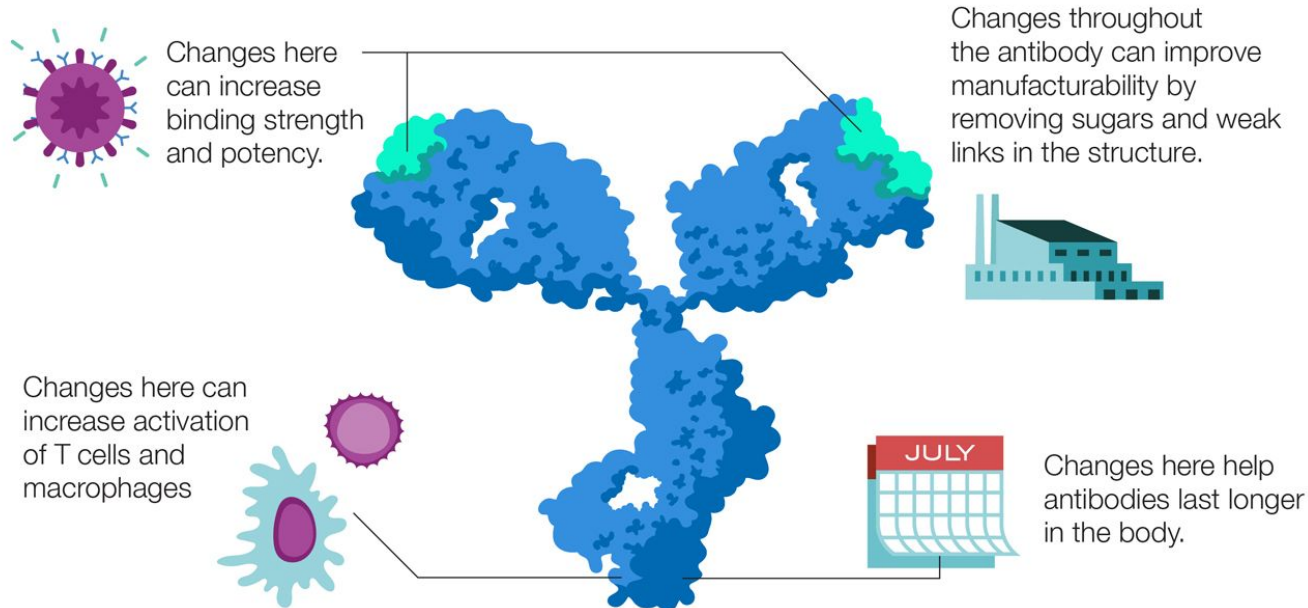
Inputs:

- Target structure
- Target sequence
- Antibody sequence

Source code: <https://github.com/RosettaCommons/RFantibody> (free for academic, personal and commercial use)

Discovered antibodies need further development

Engineering antibodies with strong attributes



Major challenges of antibody discovery and development

- Disease relevance of the target (e.g. [the story of TIGIT](#))
- *Lack of quantitative rules of developability*
- Immunogenicity of therapeutic proteins (see backup)

Biophysical properties of clinical-stage antibodies (N=137 by ~2017)

Name	Light chain class	Type	Original mAb Isotype or Format	Clinical Status	Phage ^c	Year Name Proposed			
abrituzumab	kappa	2L1	IgG2	Phase 2	No	2013			
abrilumab	kappa	t	Name	VH		VL	LC Class	Source	Source Detailed ^a
adalimumab	kappa	t	abrituzumab	QVQLQQSGGELAKPGASVKVSKASG		DIQMTQS	kappa	WHO-INN	PL109
			abrilumab	QVQLVQSGAEVKKPGASVKVSKVSG		DIQMTQS	kappa	WHO-INN	PL111
			adalimumab	EVQLVESGGGLVQPGRSLRLSCAASGF		DIQMTQS	kappa	PDB	4NYL
alemtuzumab	kappa	2	alemtuzumab	QVQLQESGPGLVPRPSQTLSTCTVSGF		DIQMTQS	kappa	PDB	1BEY
alirocumab	kappa	t	alirocumab	EVQLVESGGGLVOPGGSLRLSCAASGF		DIVMTQS	kappa	WHO-INN	PI 107

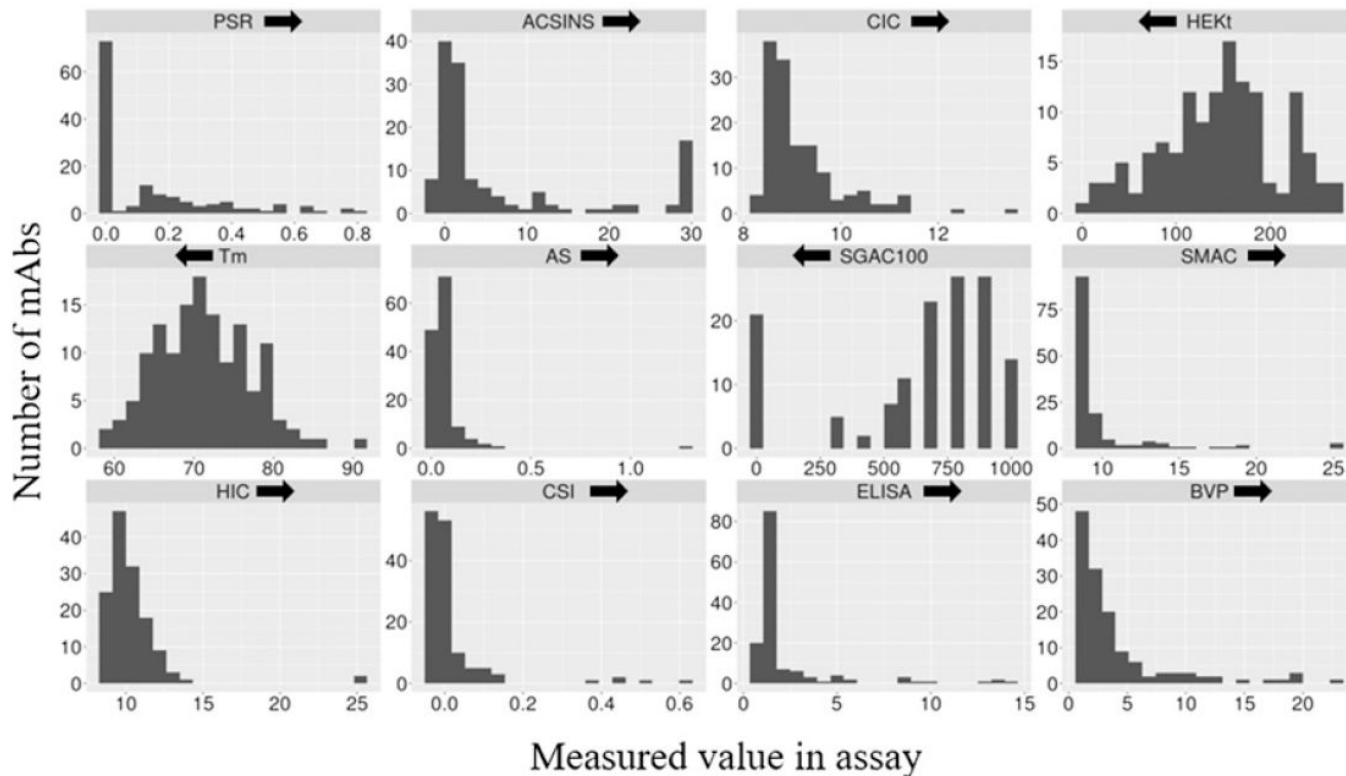
Name	HEK Titer (mg/L)	Fab Tm by DSF (°C)	SGAC-SINS AS100 ((NH4)2SO4 mM)	HIC Retention Time (Min) ^a	SMAC Retention Time (Min) ^a	Slope for Accelerated Stability	Poly-Specificity Reagent (PSR) SMP Score (0-1)	Affinity-Capture Self-Interaction Nanoparticle Spectroscopy (AC-SINS) $\Delta\lambda_{max}$ (nm) Average	CIC Retention Time (Min)	CSI-BLI Delta Response (nm)	ELISA	BVP ELISA
abrituzumab	89.6	75.5	900.0	9.2	8.7	0.06	0.17	1.5	8.6	0.00	1.14	2.72
abrilumab	100.2	71.0	900.0	9.4	8.7	0.03	0.00	-0.9	8.4	-0.02	1.12	1.82
adalimumab	134.9	71.0	900.0	8.8	8.7	0.05	0.00	1.1	8.9	-0.01	1.08	1.49
alemtuzumab	144.7	74.5	1000.0	8.8	8.7	0.06	0.00	-0.8	8.5	-0.02	1.16	1.46
alirocumab	69.2	71.5	900.0	9.0	8.7	0.03	0.00	1.2	8.8	-0.01	1.20	2.18
anifrolumab	82.0	62.5	700.0	8.8	8.6	0.07	0.00	-0.6	8.5	-0.02	1.16	1.62
atezolizumab	164.1	73.5	300.0	13.4	19.3	0.06	0.07	15.0	10.8	0.06	1.29	6.20
bpineuzumab	151.1	73.0	1000.0	8.9	8.7	0.07	0.00	-0.7	8.6	0.06	1.21	3.55
basiliximab	107.5	60.5	0.0	9.6	8.6	0.05	0.40	28.8	9.4	0.00	1.20	2.14
bavituximab	45.1	59.5	0.0	11.5	12.7	0.04	0.56	29.9	11.4	-0.01	1.32	1.69
belimumab	10.5	60.0	800.0	10.5	9.3	0.13	0.00	0.8	8.6	-0.03	3.61	12.23

Twelve different biophysical assays

Code	Name	Purpose
AC-SINS	Affinity-capture self-interaction nanoparticle spectroscopy	Self-interaction
CSI	Clone self-interaction by biolayer interferometry	Self-interaction
PSR	Poly-specificity reagent	Cross-interaction
BVP	Baculovirus particle	Cross-interaction
CIC	Cross-interaction chromatography	Cross-interaction
ELISA	Enzyme-linked immunosorbent assay with commonly used antigens	Cross-interaction

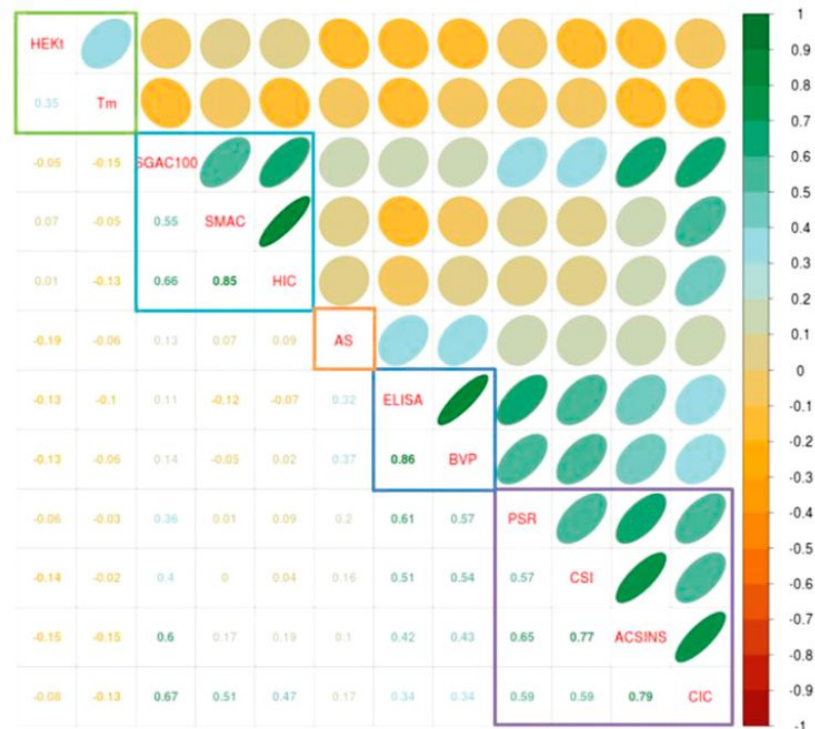
Code	Name	Purpose
HEK	Expression titer in HEK cells	Expression
Tm	Melting temperature	Thermostability
HIC	Hydrophobic interaction chromatography	Species separation and analysis
SAGC-SINS	salt-gradient affinity-capture self-interaction nanoparticle spectroscopy	Species separation and analysis
SMAC	standup monolayer adsorption chromatography	Developability
AS	Size-exclusion chromatography in accelerated stability	Stability

Distribution of results from biophysical assays for 137 monoclonal antibodies

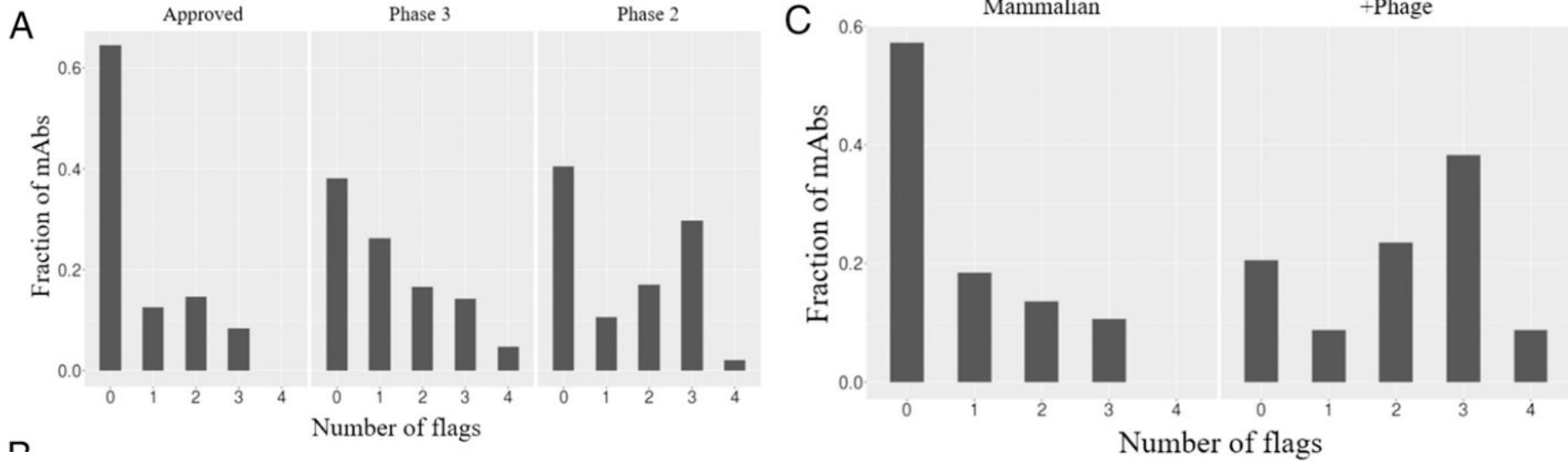


Unsupervised clustering analysis reveals related assays

Group	Assay	Worst 10% threshold
Group 1	PSR	0.27 ± 0.06
	ACSINS	11.8 ± 6.2
	CSI	0.01 ± 0.02
	CIC	10.1 ± 0.5
Group 2	HIC	11.7 ± 0.6
	SMAC	12.8 ± 1.2
	SGAC-SINS	370 ± 133
Group 3	BVP	4.3 ± 2.2
	ELISA	1.9 ± 1.0
Group 4	AS	0.08 ± 0.03



Approved antibodies and antibodies discovery not via phage display tend to have fewer flags



Conclusions

- Given mechanistic understanding of biological processes underlying diseases, we can develop different modalities as therapeutics.
- Mathematical and computational biology
 1. reveals how drug candidate work and ranks them
 2. helps with molecule design
 3. contributes to modality selection

Offline activities

Reading the review draft on leveraging protein turnover for drug discovery, and sharing questions, criticism, and feedback.

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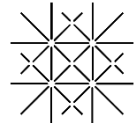
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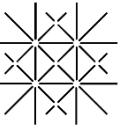


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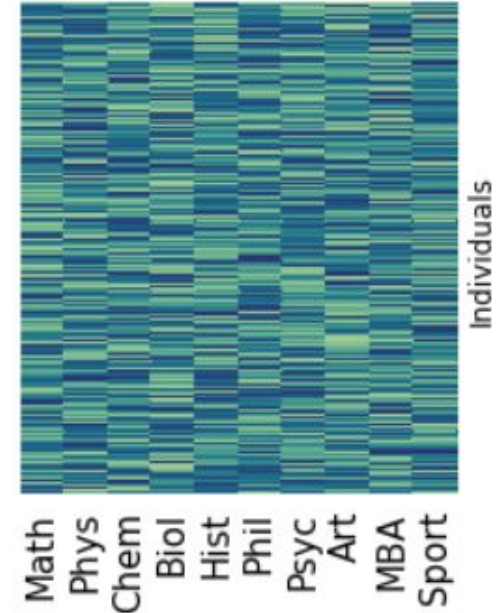
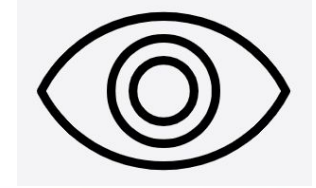
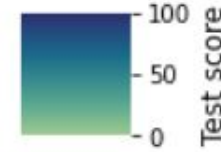
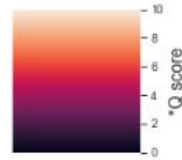


References (continued)

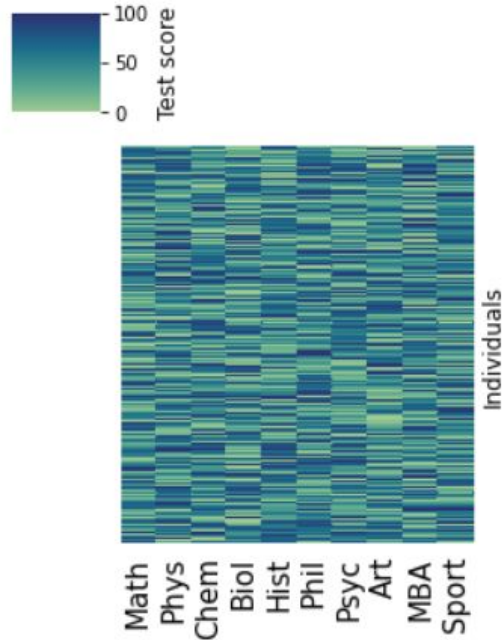
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A fallible fairy tale about intelligence

- IQ** Intellectual intelligence
- EQ** Emotional intelligence
- SQ** Spiritual intelligence



We may infer *Q if *Qs are independent and test scores are largely determined by individual *Qs

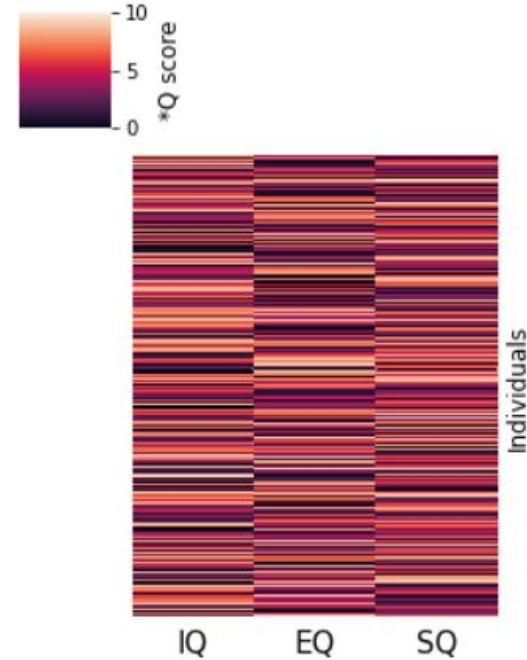


$$\begin{aligned} \text{Math} &= 8 \times \text{IQ} + 2 \times \text{EQ} + 2 \times \text{SQ} + \varepsilon \\ \text{Philosophy} &= 6 \times \text{IQ} + 2 \times \text{EQ} + 8 \times \text{SQ} + \varepsilon \\ &\dots \\ \text{MBA} &= 4 \times \text{IQ} + 7 \times \text{EQ} + 5 \times \text{SQ} + \varepsilon \\ \text{Sport} &= 5 \times \text{IQ} + 6 \times \text{EQ} + 5 \times \text{SQ} + \varepsilon \end{aligned}$$

Numbers in blue: loadings

IQ/EQ/SQ: factors

Factor analysis

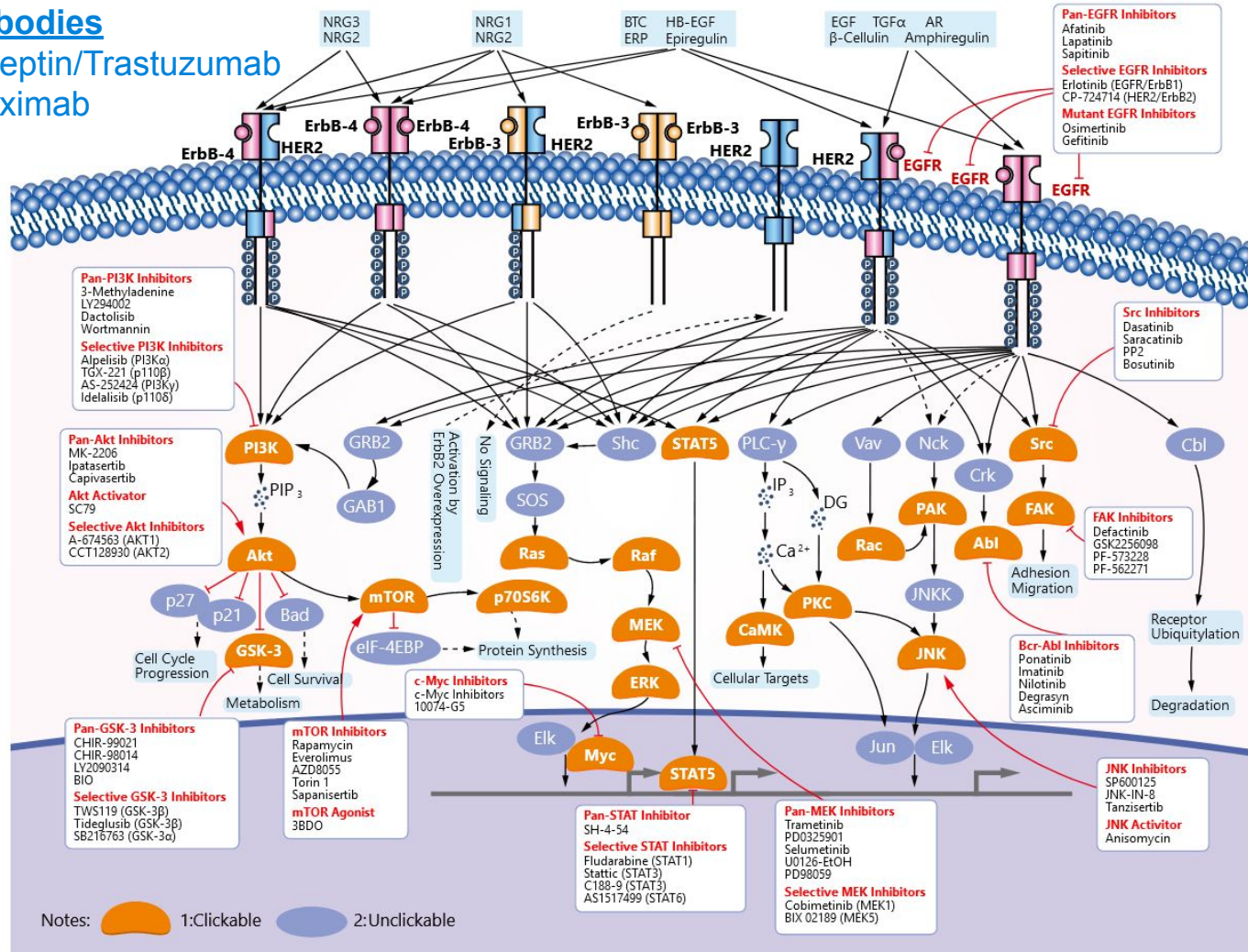


Multiple modalities can target the same biological process

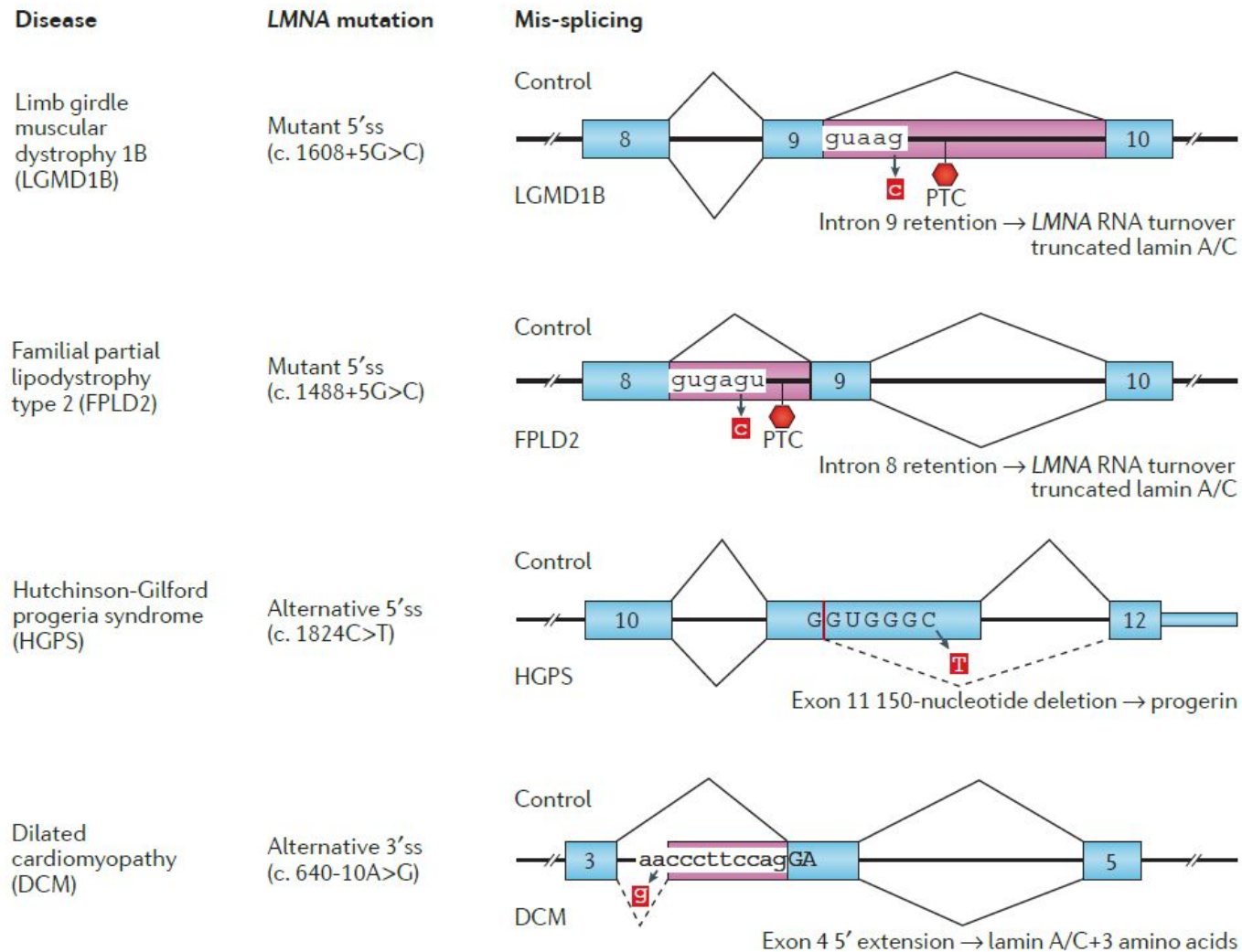
An example: the epidermal growth factor receptor (EGFR) pathway

Antibodies

Herceptin/Trastuzumab
Cetuximab

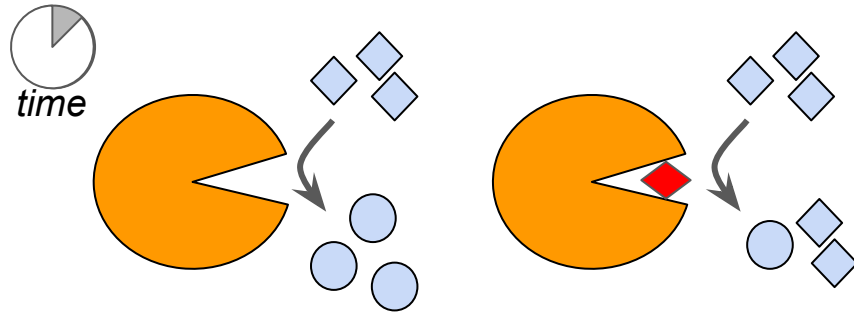


Splicing modifying oligo-nucleotide and other RNA therapeutics

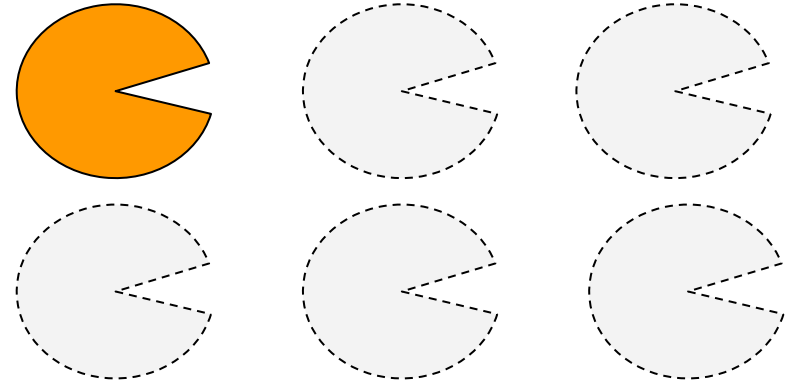


Difference between small molecule and antisense oligonucleotides (ASOs)

Competitive inhibitors reduce reaction rate; antisense oligonucleotides modulate protein abundance

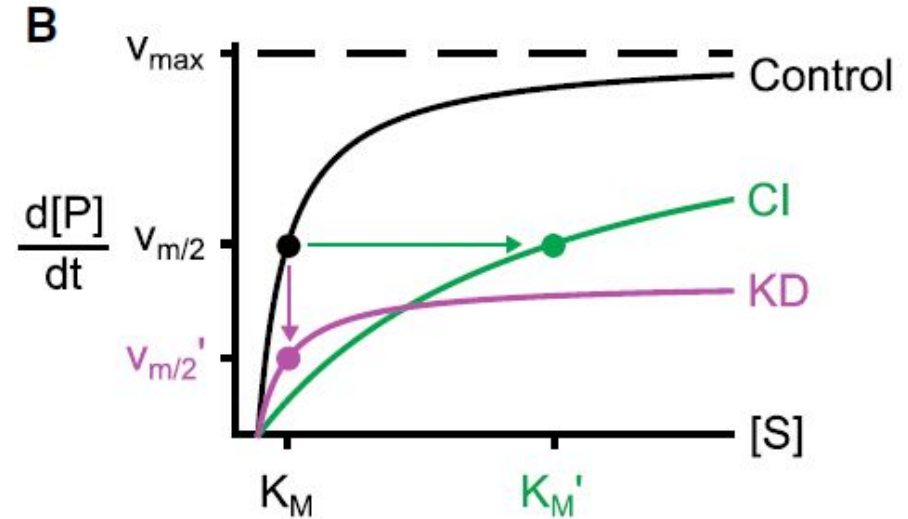
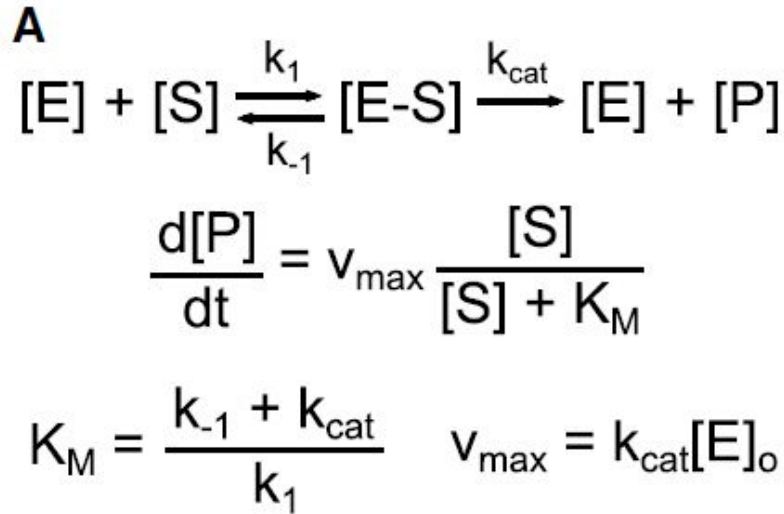


A competitive inhibitor (red diamond) reduces the rate of product generation in an enzymatic reaction.



Antisense oligonucleotides reduce the abundance of the enzyme protein.

Enzymic and genetic inhibition have distinct impact on reaction dynamics

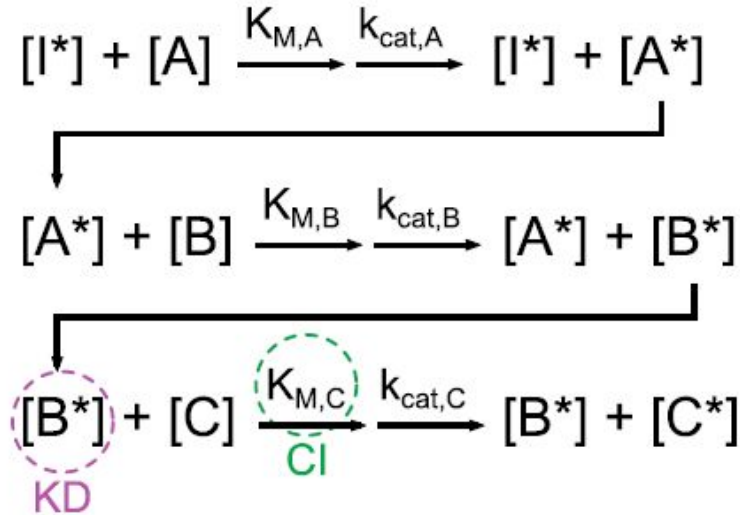


Competitive inhibition (CI)
versus knockdown (KD)

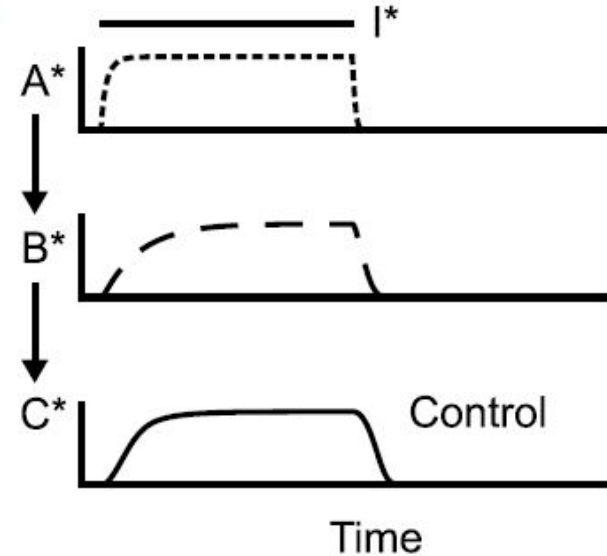
The Michaelis-Menten Equation

A linear system simulating enzymatic reactions

C



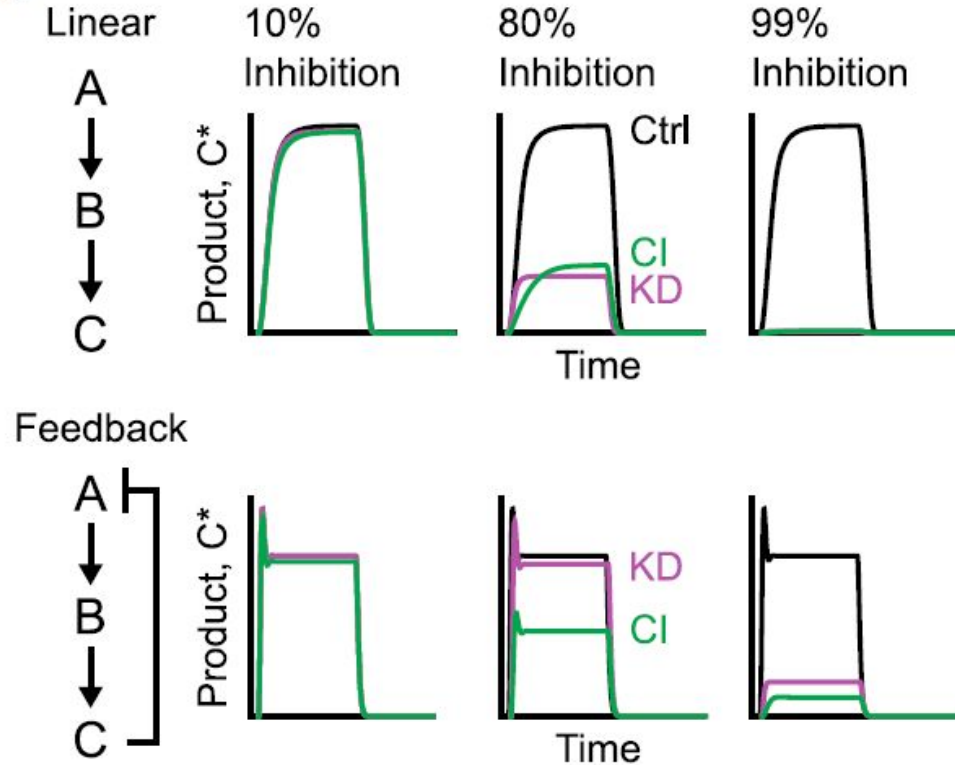
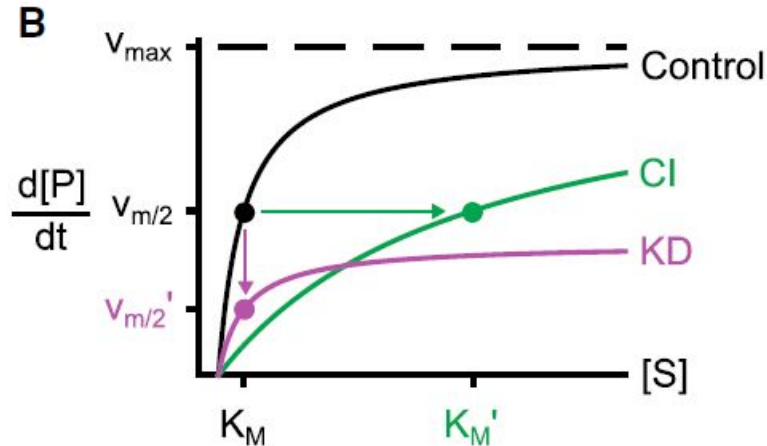
D



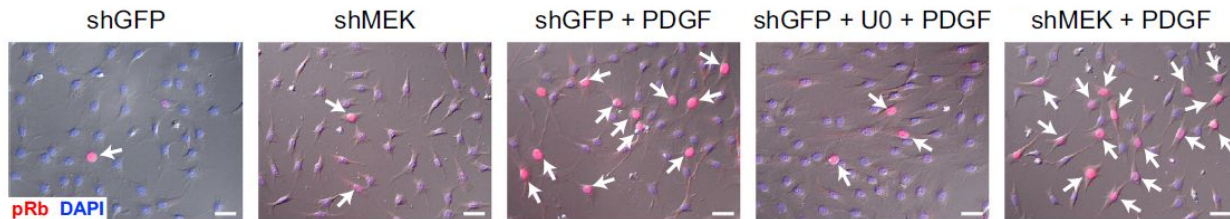
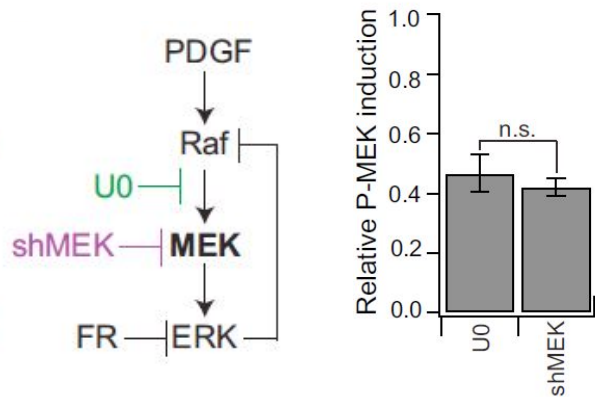
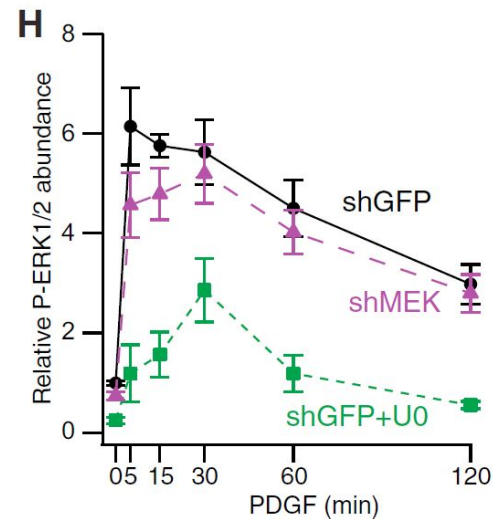
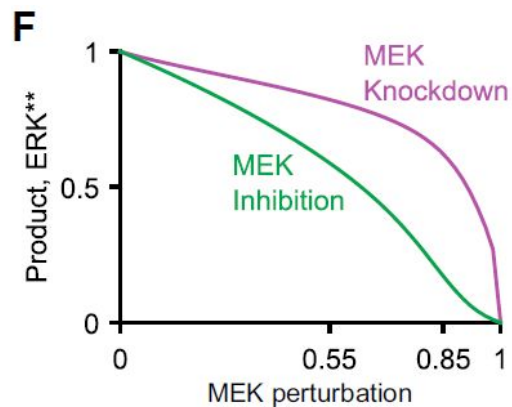
I^* : upstream input; A/A^* and B/B^* : inactivated and activated enzyme; C^* : product

Adding a negative feedback may differentiate effects of enzymatic and genetic inhibition

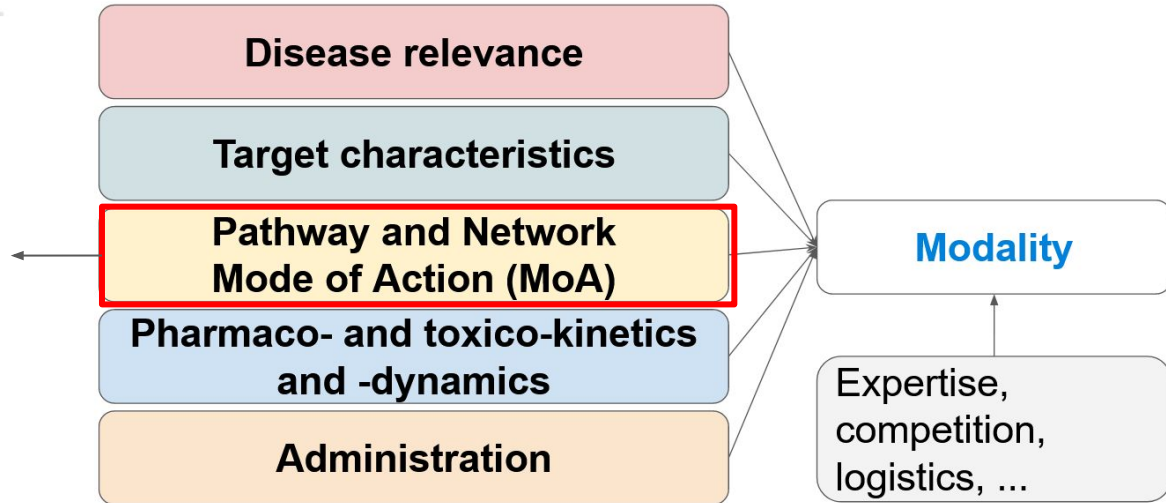
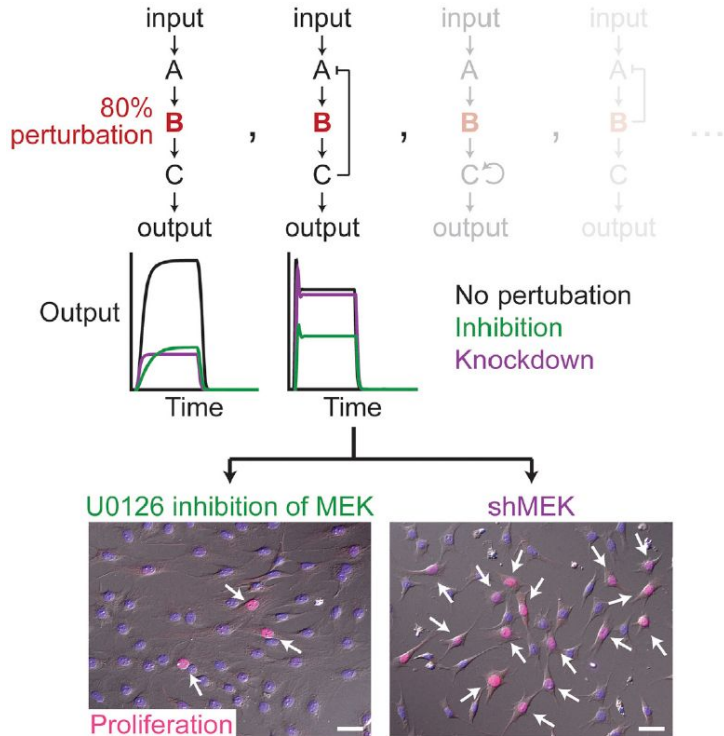
Intuition: when $[B^*]$ stays low, CI leads to **slower** accumulation of C^* than KD.



Confirmation of predicted difference of KD and CI



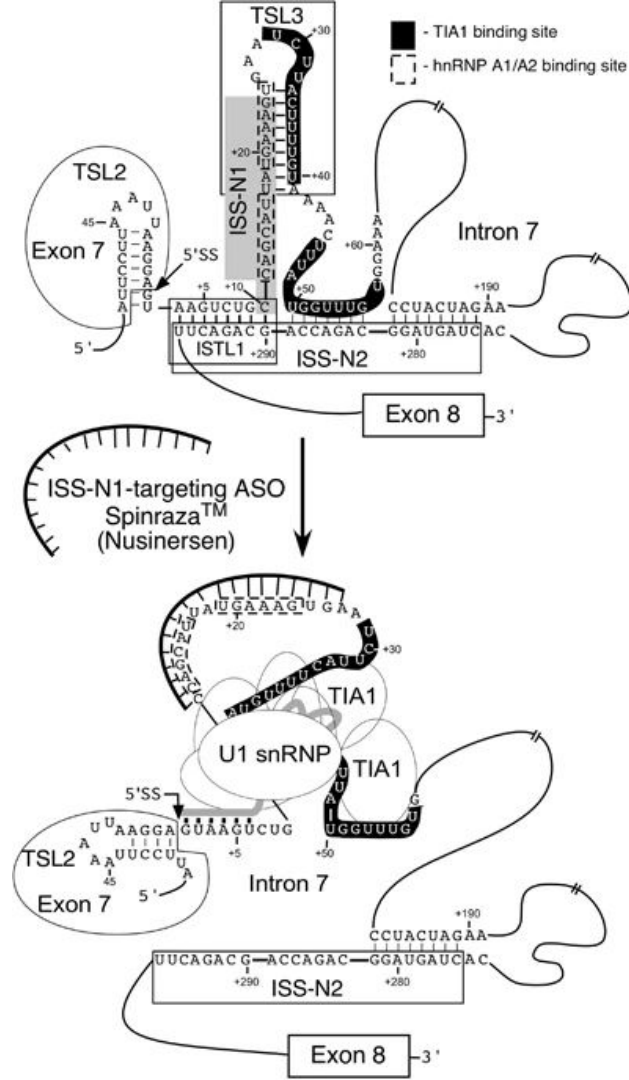
Computational biology may empower our choice of modality



How Spinaraza (nusinersen) works, base by base

Nusinersen binds to ISS-N1, causing structural rearrangement and recruitment of U1 snRNP by TIA1.

- ISS-N1: Intronic splicing silencer N1;
- [TIA1](#): TIA1 cytotoxic granule associated RNA binding protein;
- TSLs: (inhibitory) terminal stem-loop structures;
- ISTL1: internal stem formed by a long-distance interaction



Clinical-stage siRNAs

The growth of siRNA-based therapeutics

FDA-approved
siRNA drugs

Patisiran

Givosiran

Lumasiran

siRNA drugs
in clinical trials

Vutrisiran

Nedosiran

Inclisiran

Fitusiran

Tepasiran

Cosdosiran

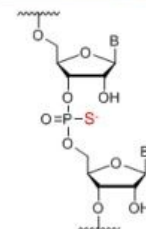
Tivanisiran

Drug	Alternative name	Company	Disease	Updated status
Patisiran	ONPATTRO	Alnylam	Hereditary transthyretin mediated amyloidosis	FDA approval in 10/08/2018 210922Orig1s000*
Givosiran	GIVLAARI	Alnylam	Acute hepatic porphyria	FDA approval in 11/20/2019 212194Orig1s000
Lumasiran	ALN-GO1	Alnylam	Primary hyperoxaluria type 1 (PH1)	FDA approval on 11/23/2020 214103Orig1s000
Vutrisiran	ALN-TTRsc02	Alnylam	Hereditary transthyretin mediated amyloidosis	Phase 3 trials ELIOS-A (NCT03759379)** HELIOS-B (NCT04153149)
Nedosiran	DCR-PHXC	Dicerna Alnylam	Primary hyperoxaluria	Phase 3 trial PHYOX 3 (NCT04042402)
Inclisiran	ALN-PCSSC	Alnylam Novartis	Hypercholesterolemia	Phase 3 trials ORION-9 (NCT03397121) ORION-10 (NCT03399370) ORION-11 (NCT03400800)
Fitusiran	ALN-AT3sc ALN-APC SAR439774	Alnylam Sanofi Genzyme	Hemophilia A and B	Phase 3 trials ATLAS-A/B (NCT03417245) ATLAS-INH (NCT03417102) ATLAS-PPX (NCT03549871) ATLAS-PEDS (NCT03974113) ATLAS-OLE (NCT03754790)
Tepasiran	AKI-5, DGFI, I-5NP, QPI-1002	Quark Novartis	Acute kidney injury Delayed graft function	Phase 3 trial ReGIFT (NCT02610296)
Cosdosiran	QPI-1007	Quark	Non-arteritic anterior ischemic optic neuropathy (NAION)	Phase 2/3 trial NCT02341560
Tivanisiran	SYL-1001	Sylentis	Dry eyes Ocular pain	Phase 3 trial HELIX (NCT03108664)

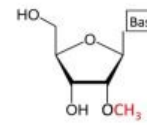
* FDA application number.

** ClinicalTrials.gov identifier number at <https://clinicaltrials.gov/ct2/>

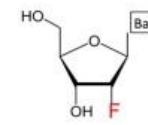
Drug	Chemical modifications				Delivery platform
	Backbone	2'-OMe	2'-F	2'-MOE	
Patisiran	-	+	-	-	LNP
Givosiran	+	+	+	-	GalNAc
Lumasiran	+	+	+	-	GalNAc
Vutrisiran	+	+	+	-	GalNAc
Nedosiran	+	+	+	-	GalNAc
Inclisiran	+	+	+	+	GalNAc
Fitusiran	+	+	+	-	GalNAc
Tepasiran	-	+	-	-	None
Cosdosiran	-	+	-	-	None
Tivanisiran	-	-	-	-	None



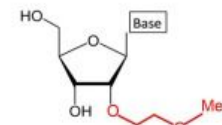
Phosphorothioate (PS)



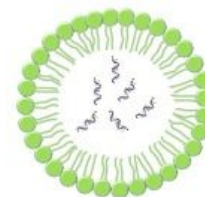
2-O-methyl (2'-OMe)



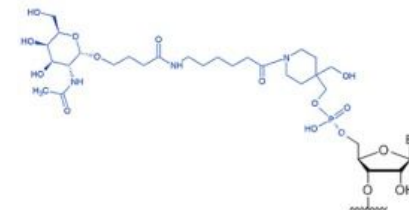
2'-fluoro (2'-F)



2'-O-methoxyethyl (2'-MOE)



Lipid nanoparticle (LNP)

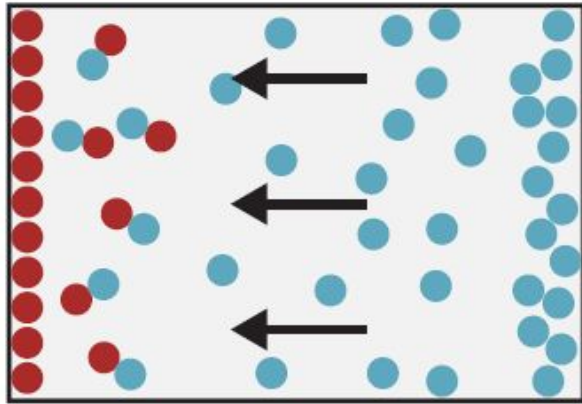


N-acetylgalactosamine (GalNAc)

Chemically induced proximity

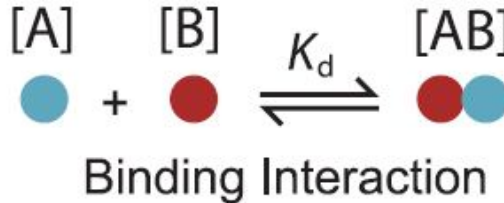
A reaction-diffusion model

Reaction-Diffusion System



Diffusion

$$\frac{\partial u}{\partial t} = D \underbrace{\frac{\partial^2 u}{\partial x^2}}_{\text{Diffusion}} + \underbrace{ku}_{\text{Binding}}$$



x : position

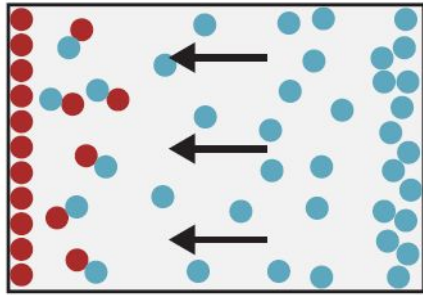
u : product concentration

t : time

The diffusion term follows *Fick's second law of diffusion*; the binding term describes the reaction.

Kinetic and thermodynamic contributions of chemically induced proximity

Reaction-Diffusion System

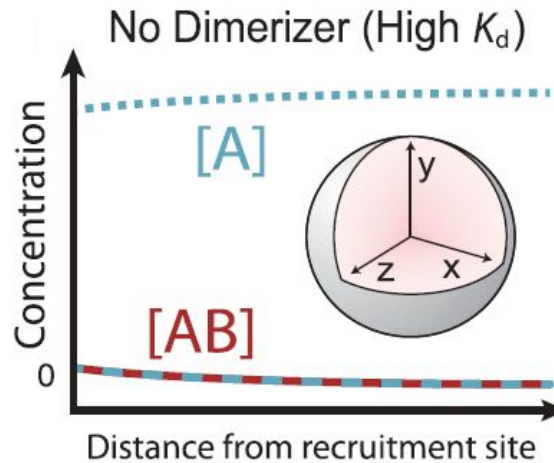


Diffusion

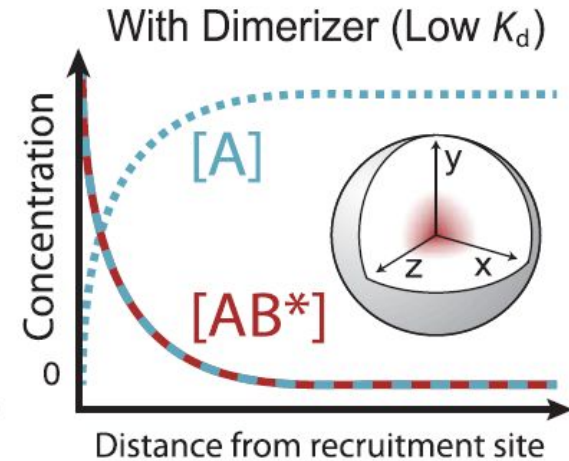
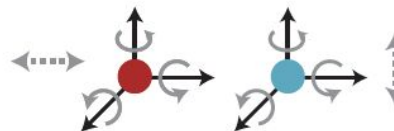
$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + ku$$



Binding Interaction



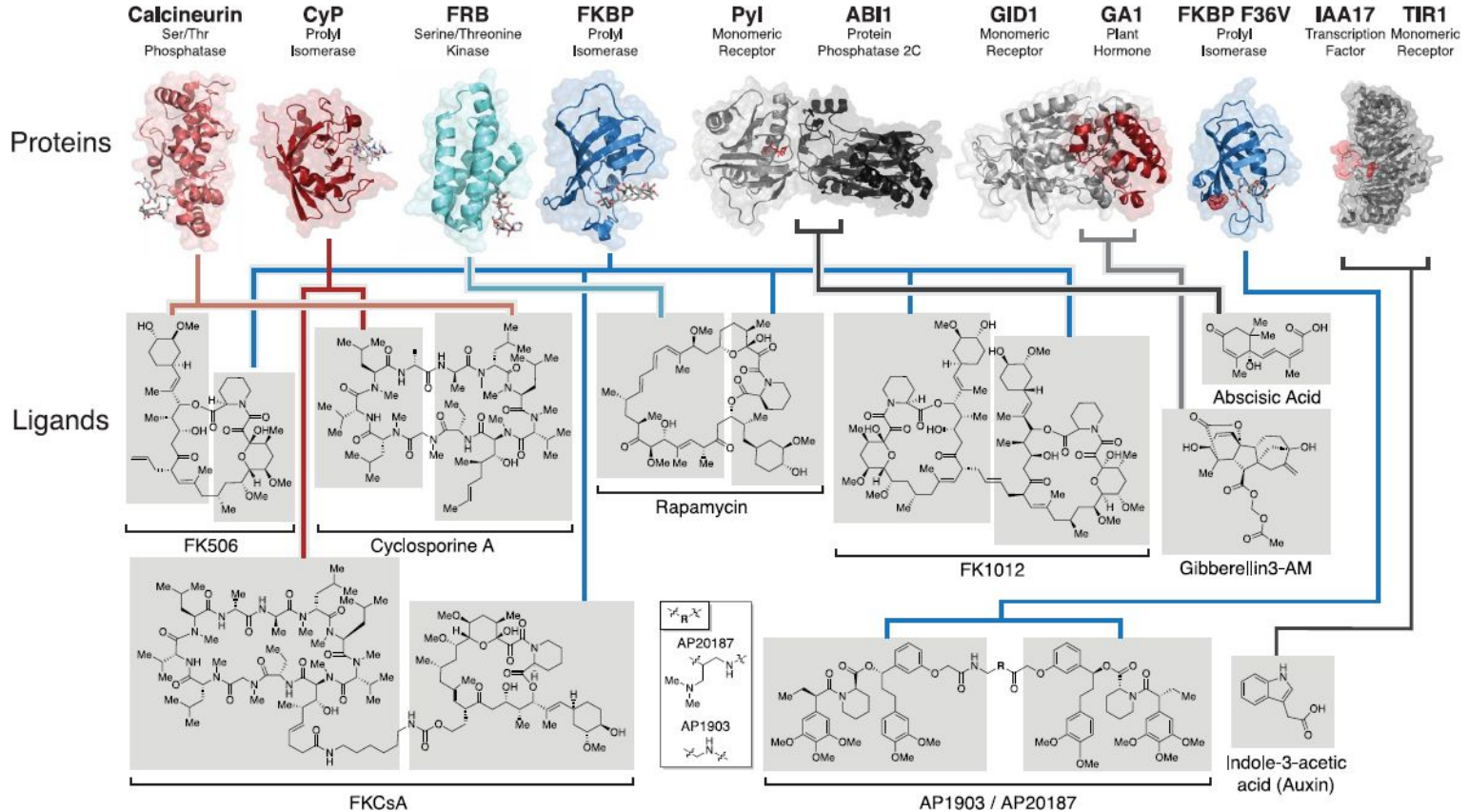
Freely Diffusing



Constrained upon Dimerization

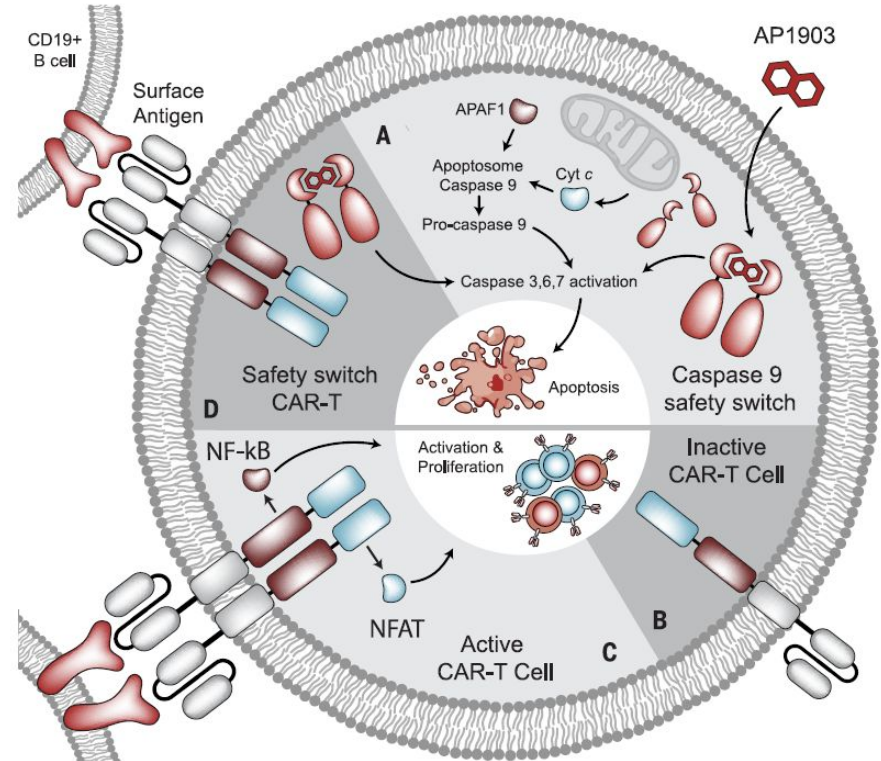


Chemically induced proximity

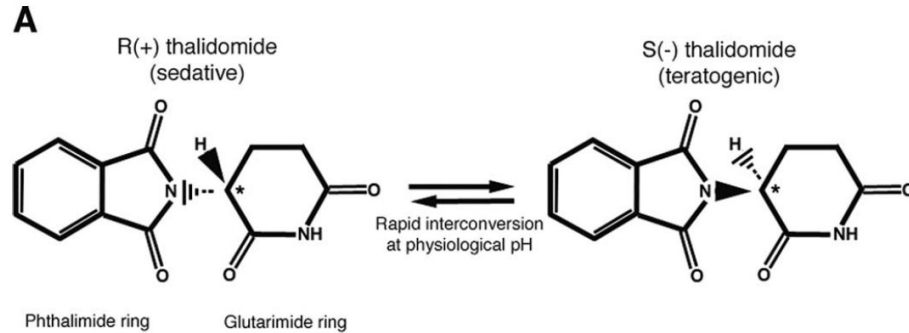


Chemically induced proximity as 'safety switch' for cell therapy

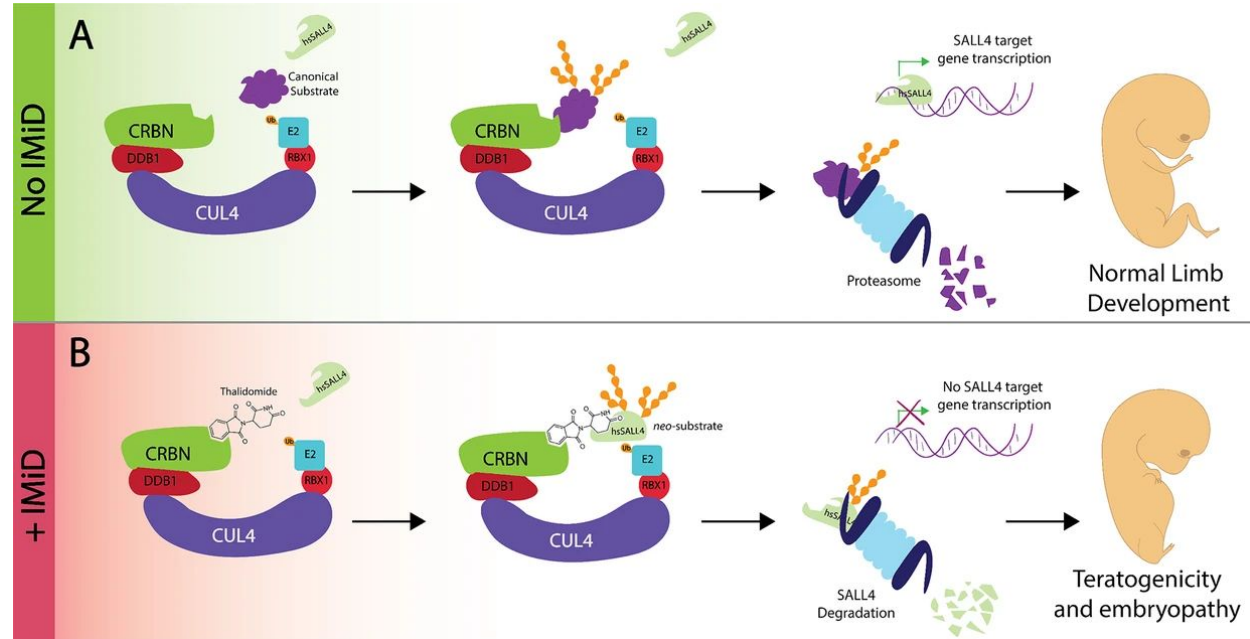
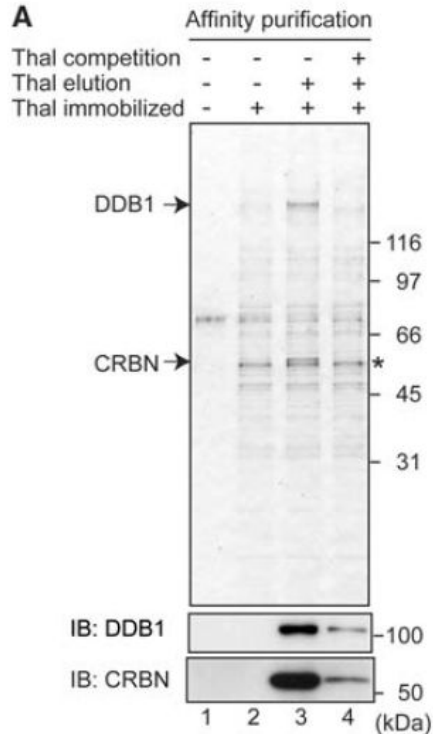
- Too many or too active CAR-T cells may induce serious side effects (cytokine release syndrome, B cell aplasia, *etc.*)
- *Bioinert* small molecules (AP1903 in this case) can be used as 'safety switch' to kill transplanted CAR-T cells.



The Tragedy of teratogenic S(-) thalidomide in 1950s

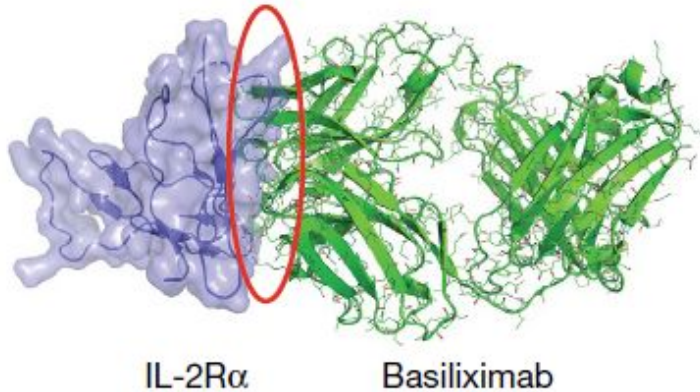


Molecular basis of the teratogenicity of thalidomide reported in 2010



Multispecific Drug Use or Target Interactions

- b** Conventional drug:
- Forms 1 drug–target interface
 - Can act throughout body
 - Only works if its binding to target alters function of target

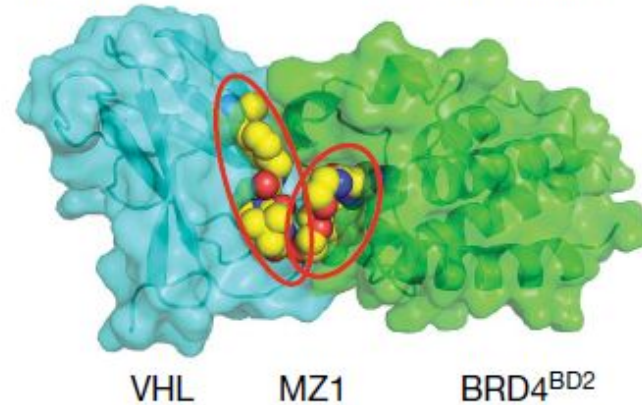


- c** Obligate multispecific drug:
- Forms 2 or more drug–target interfaces
- Class 1 ‘tetherbodies’

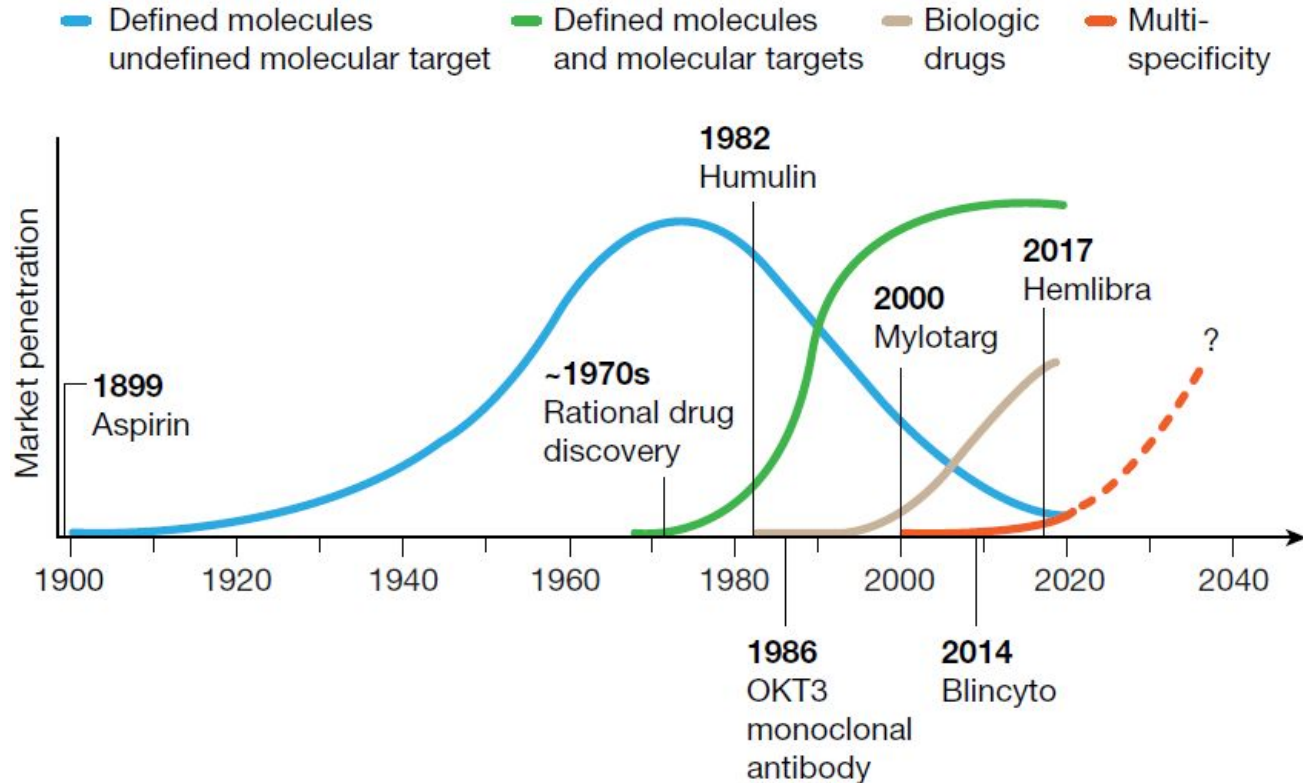
 - Enrich drug at relevant site of action

Class 2 ‘matchmakers’

 - Link drug to a biological effector



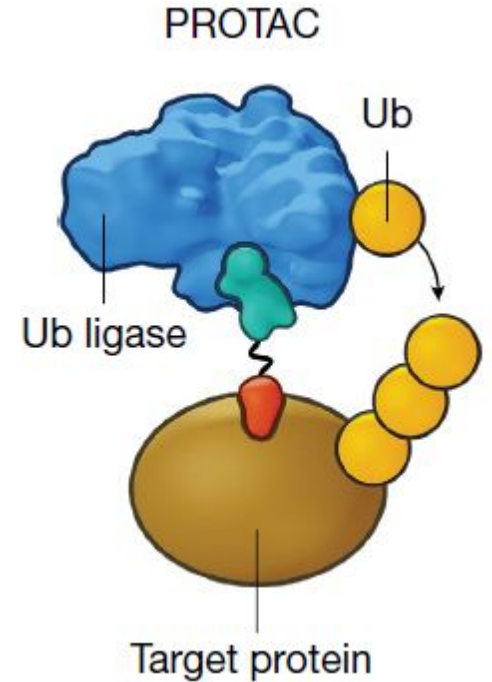
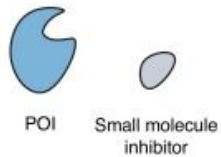
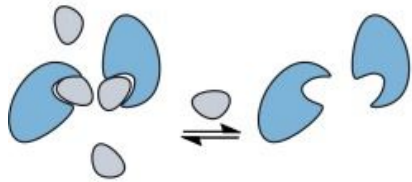
Paradigm shifts and paradigm expansion



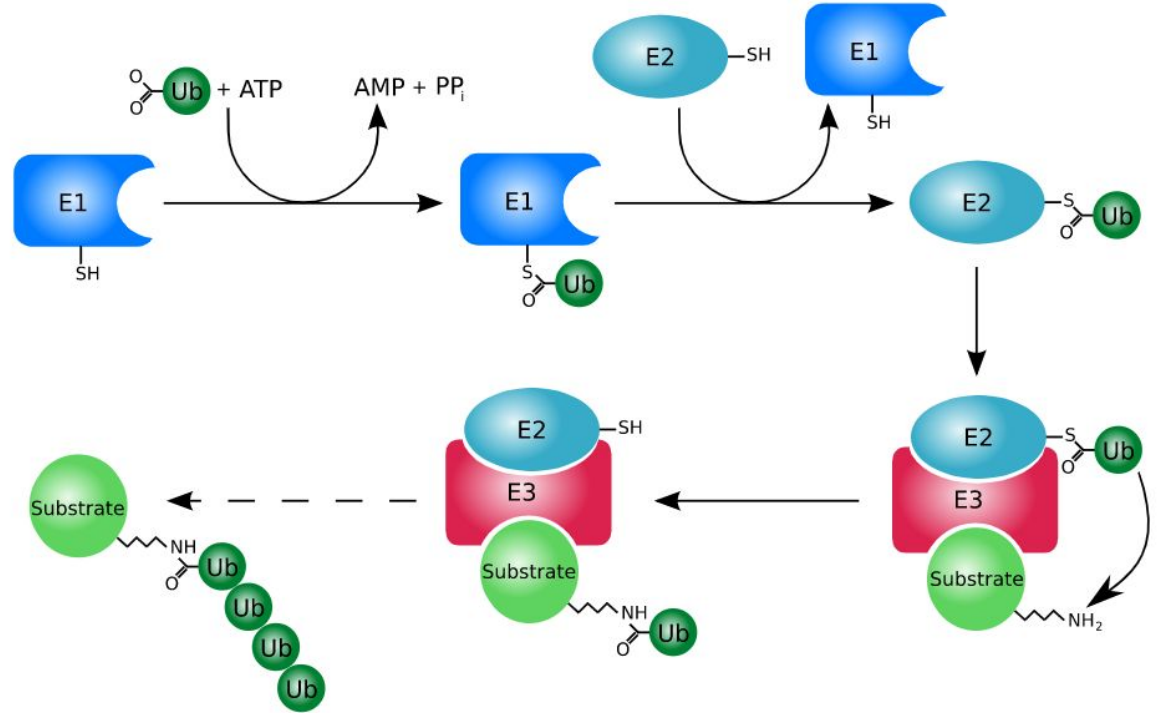
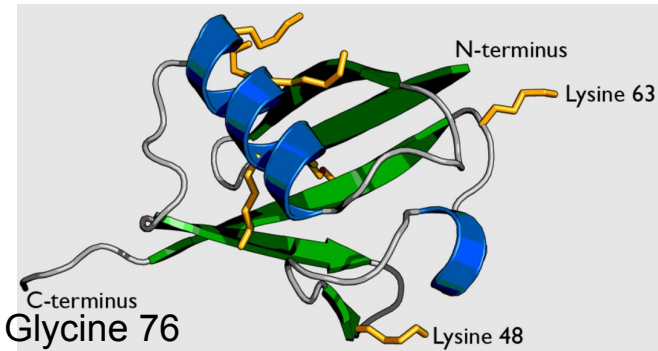
PROteolysis TArgeting Chimera (PROTAC)

(a) Occupancy-driven pharmacology

Protein function is modulated *via* inhibition

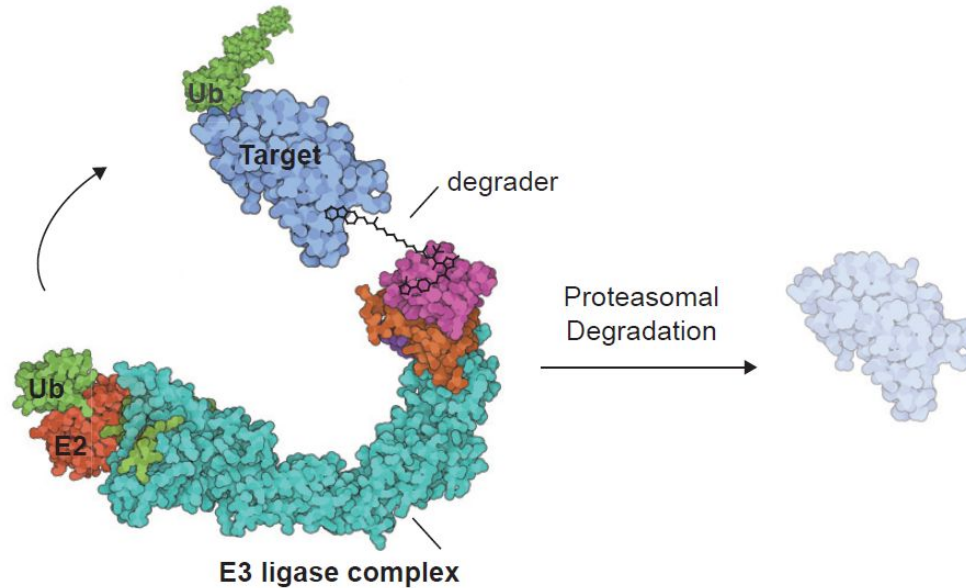


Ubiquitination marks proteins to be degraded

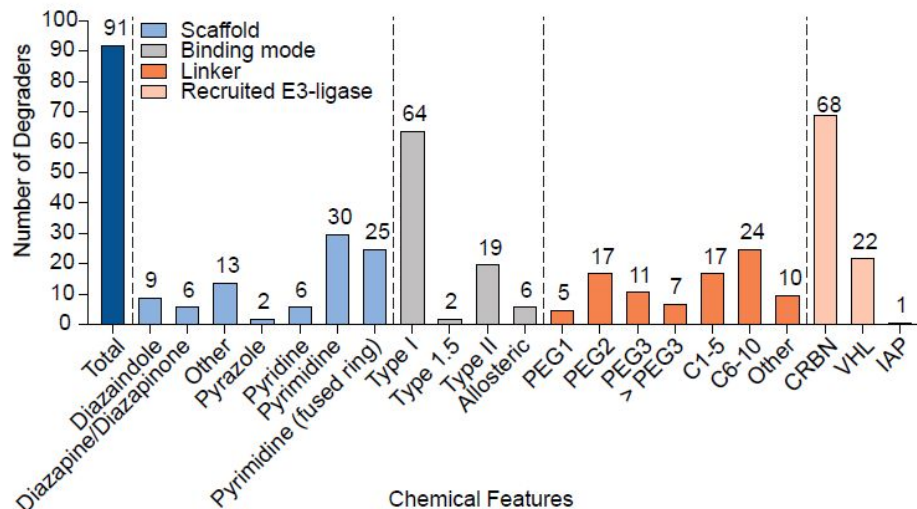
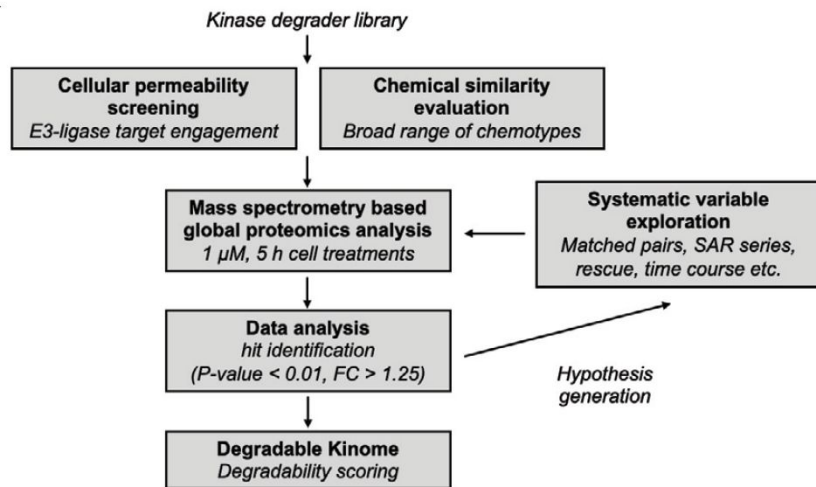


work by [Roger Dodd](#), used under the CC-BY-SA 3.0 licence.

Donovan *et al.* (2020) reports screening results with 91 kinase degraders



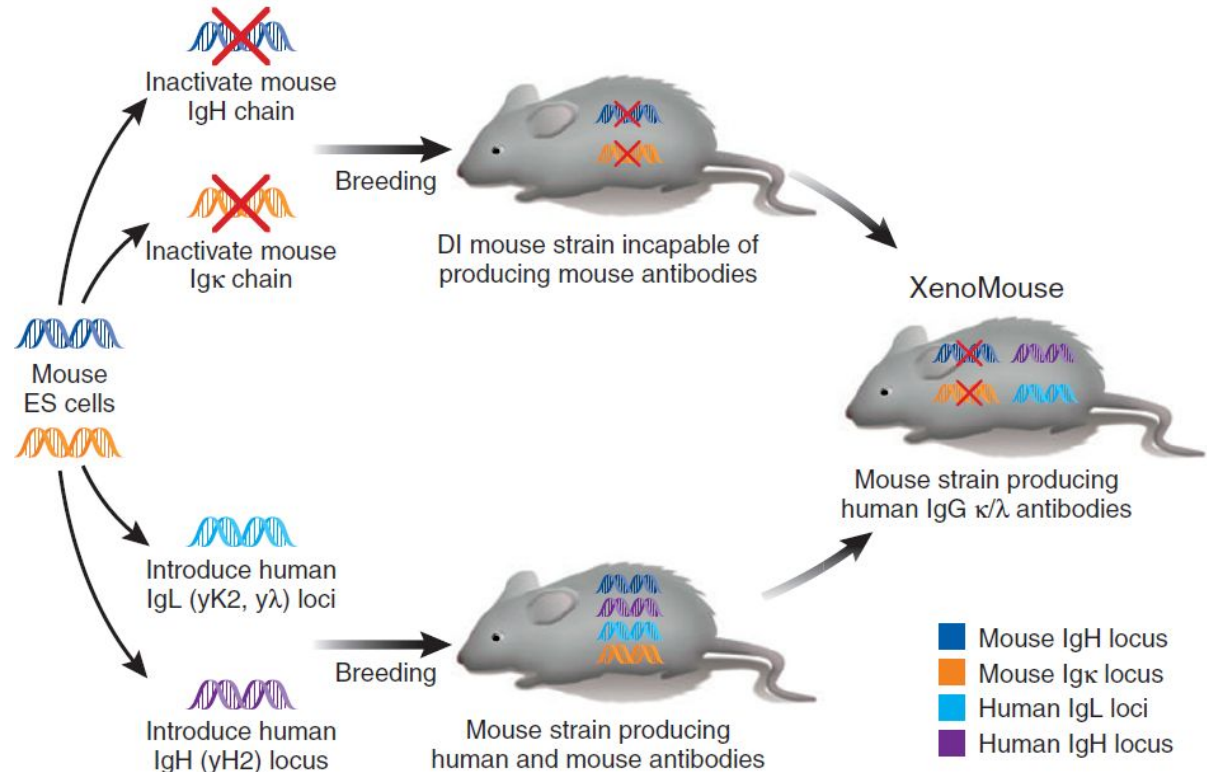
Donovan *et al.* (2020) reports screening results with 91 kinase degraders



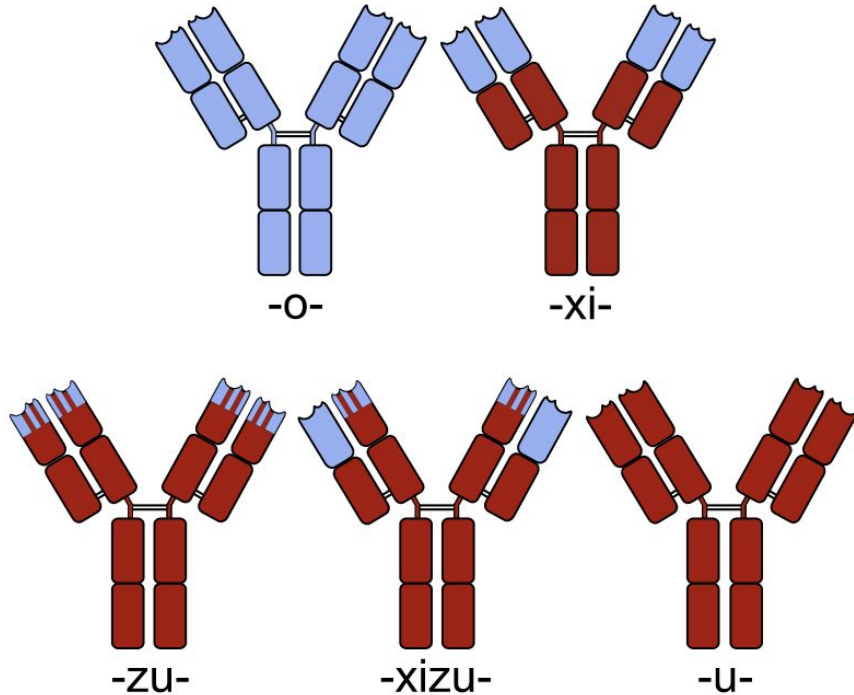
Antibodies and immunogenicity

Therapeutic antibody discovery with transgenic animals

The XenoMouse model, which led to the discovery of panitumumab (Vectibix). Panitumumab targets EGFR for advanced colorectal cancer.

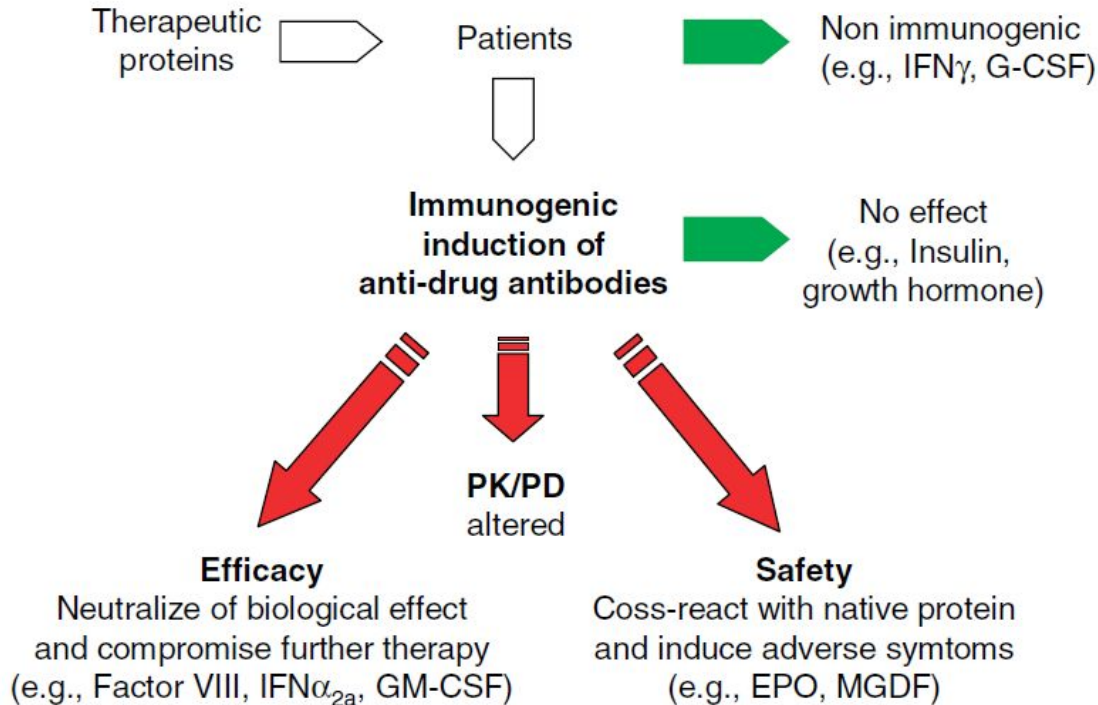


Antibody names suggest their types



- **Chimeric:** Abiciximab (Ab against platelet aggregation inhibitor)
- **Humanized:** Trastuzumab (HER2)
- **Chimeric/Humanized:** Otelixizumab (CD3, a T lymphocyte receptor)
- **Human:** Adalimumab (TNF-alpha)

Immunogenicity affects both efficacy and safety



Immune response underlies immunogenicity

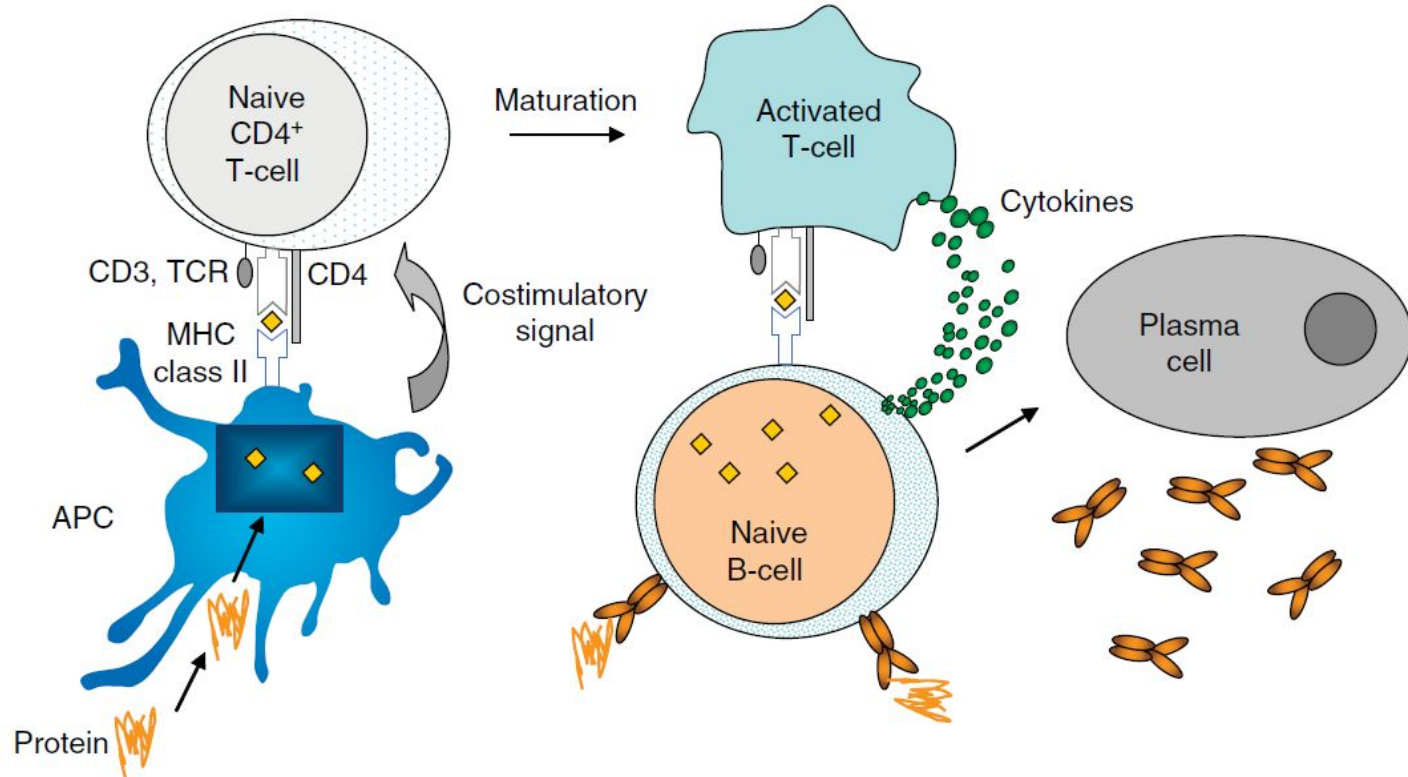


Table 2. The factors contributing to immunogenicity are divided into three groups.

	Immunogenicity potential
<i>Drug product</i>	
Host	Non-human: ▲
Immunomodulatory properties	▲?
Glycosylation	▲
Aggregation	▲
Size	Molecular mass < 10 kDa: ▼
Formulation	Polymers ▼
Excipients/stabilizers	To be characterized: silicone oil ▲
Impurities	Product/process related: ▲
Post-translational mod.	Oxidation, deamidation, etc.: ▲
aa Composition	Charged aa: ▲; Aromatic aa: ▼
Conjugates	▲
<i>Patient</i>	
Age	▼?
Disease state	Different indication/different response
Immune status	Immune compromised: ▼
	Infective disease: ▲
Patient to patient variability	Not predictable
Concomitant therapy	Earlier exposure to similar protein - crossreacting antibodies to similar proteins
Genetic factors	Defective gene
	Polymorphisms for cytokines
<i>Administration</i>	
Dose	Higher dose: ▲?
Route	Intravenous administration less immunogenic than subcutaneous or intramuscular
	Short-term administration less immunogenic than long-term treatment
	Continuous administration less immunogenic than intermittent
Frequency	More frequent: ▲
Duration of therapy	Short term: ▼

A mechanistic, multiscale model of immunogenicity: subcellular model

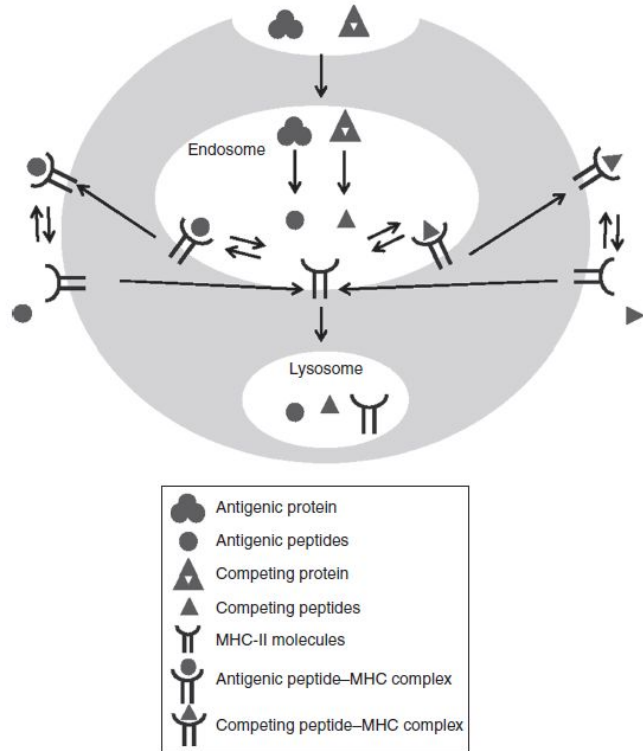


Figure 1. Model structure for the subcellular level, including processes for antigen presentation in mature dendritic cells. The symbols in the figure legends are described below, with corresponding equation number in **Supplementary Materials** shown between parentheses. : Antigenic protein, including antigenic protein in plasma (Ag , Eq. 27 in **Supplementary Material**) and antigenic protein in the endosome: (Ag^E , Eq. 4 in **Supplementary Material**); : antigenic peptide in endosome (p_i^E , Eq. 5 in **Supplementary Material**); : competing protein in the endosome (cp^E , Eq. 9 in **Supplementary Material**); : competing peptide in the endosome (cpt^E , Eq. 10 in **Supplementary Material**); : MHC-II molecules, including those in the endosome (M_k^E , Eq. 6 in **Supplementary Material**) and those on dendritic cell membrane (M_k , Eq. 13 in **Supplementary Material**); : antigenic peptide-MHC complex, including those in the endosome ($p_i M_k^E$, Eq. 7 in **Supplementary Material**) and those on cell membrane ($p_i M_k$, Eq. 8 in **Supplementary Material**); : competing peptide-MHC complex, including those in the endosome ($cpt M_k^E$, Eq. 11 in **Supplementary Material**) and those on cell membrane ($cpt M_k$, Eq. 12 in **Supplementary Material**).

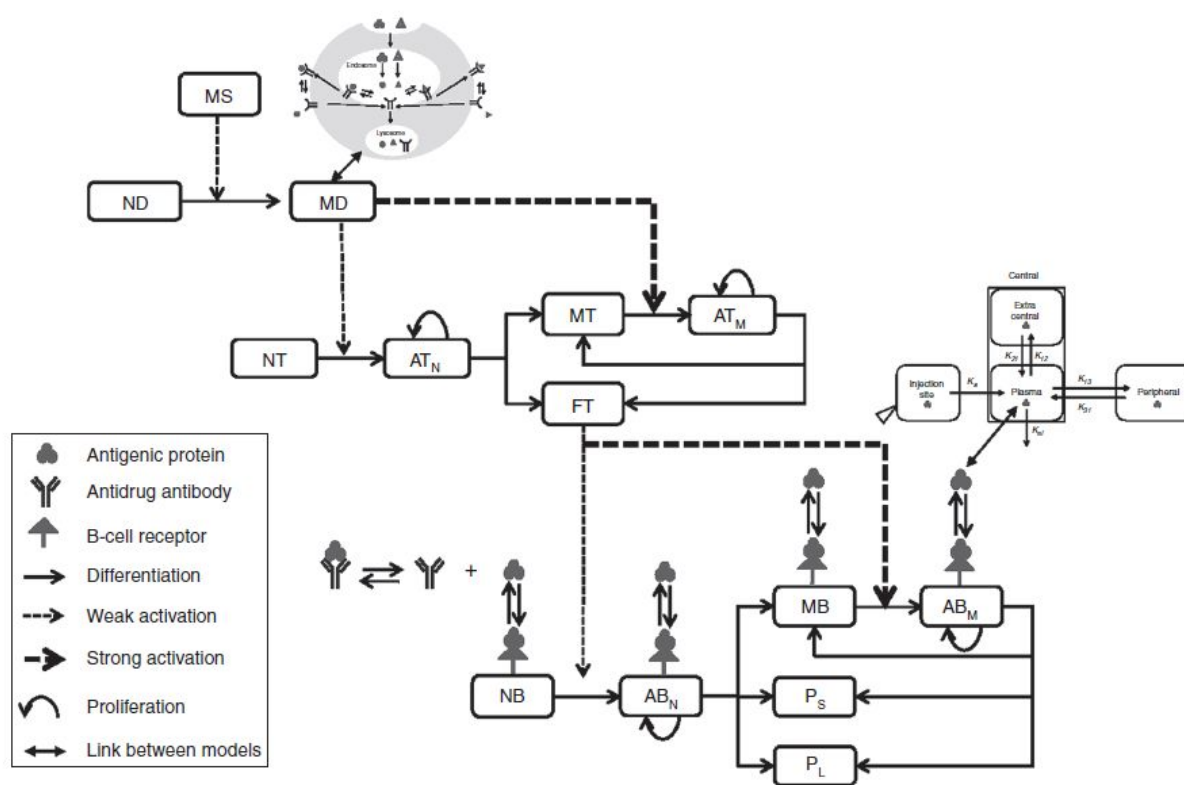


Figure 2. Model structure for the cellular level, including cells, antigen, antidrug antibody, and B-cell receptor. The links between the three levels of the multiscale model are also illustrated to help interpretation. The acronyms are explained below, along with the corresponding equation number in the **Supplementary Material** shown between parentheses. MS: maturation signal (Eq. 1 in **Supplementary Material**); ID: immature dendritic (Eq. 2 in **Supplementary Material**); MD: mature dendritic (Eq. 3 in **Supplementary Material**); NT: naïve T (Eq. 14 in **Supplementary Material**); AT_N : activated T from naïve T (Eq. 15 in **Supplementary Material**); AT_M : activated T from memory T (Eq. 16 in **Supplementary Material**); MT: memory T (Eq. 17 in **Supplementary Material**); FT: functional T (Eq. 18 in **Supplementary Material**); NB: naïve B (Eq. 19 in **Supplementary Material**); AB_N : activated B from naïve B (Eq. 20 in **Supplementary Material**); AB_M : activated B from memory B (Eq. 21 in **Supplementary Material**); MB: memory B (Eq. 22 in **Supplementary Material**); P_S : short-lived plasma (Eq. 23 in **Supplementary Material**); P_L : long-lived plasma cell (Eq. 24 in **Supplementary Material**).

The whole-body model

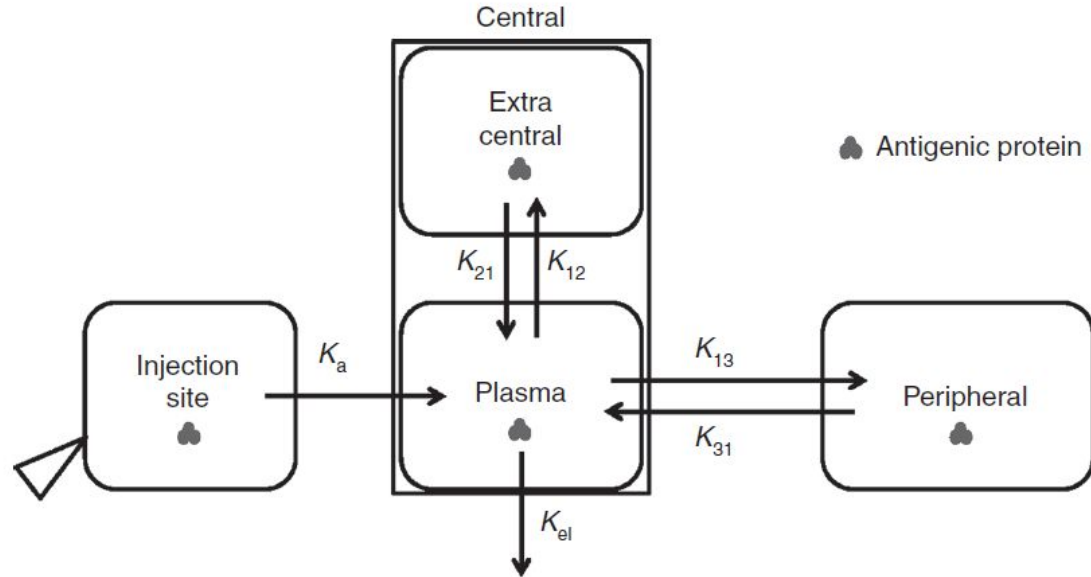


Figure 3. Model structure for the whole-body level, accounting for the *in vivo* disposition of antigenic protein. Details are described in the Results section and also by **Eqs. 26–29 in the Supplementary Materials**.

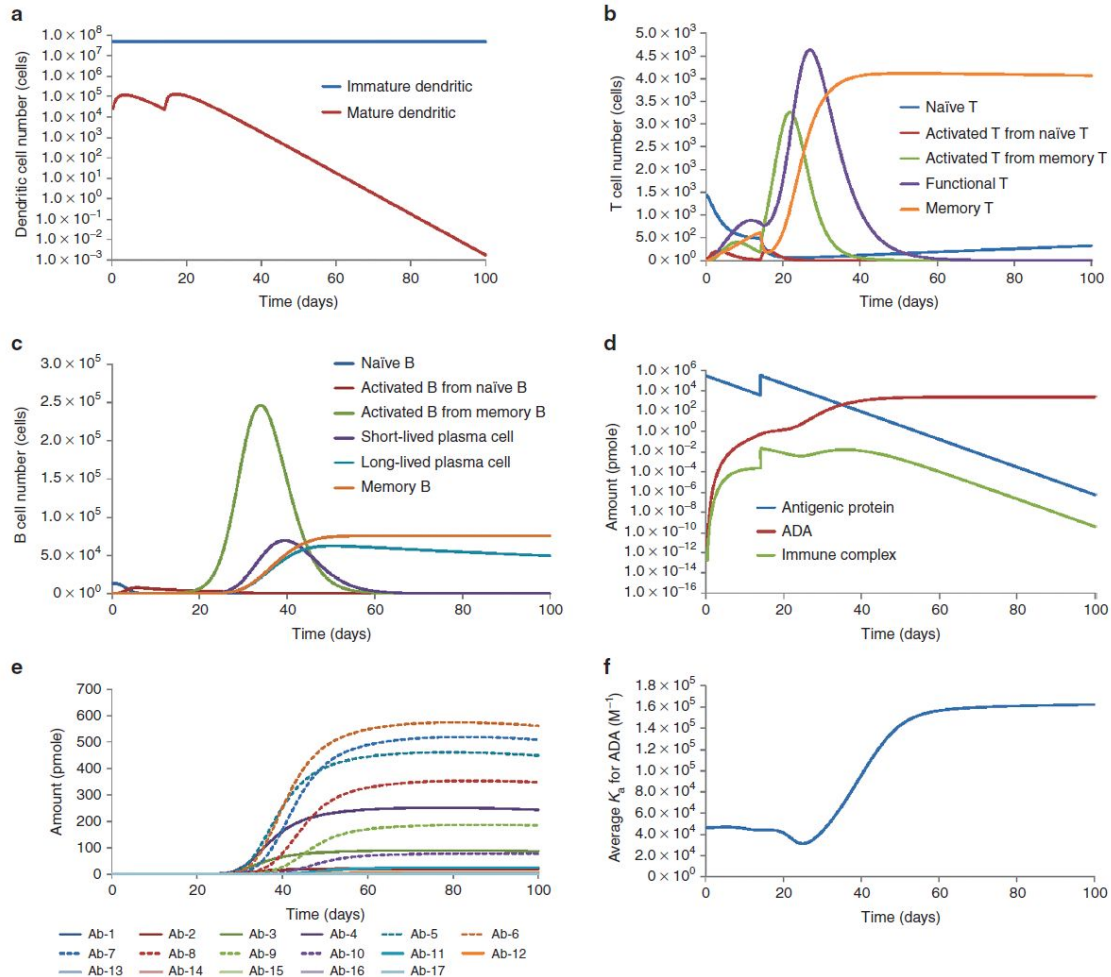
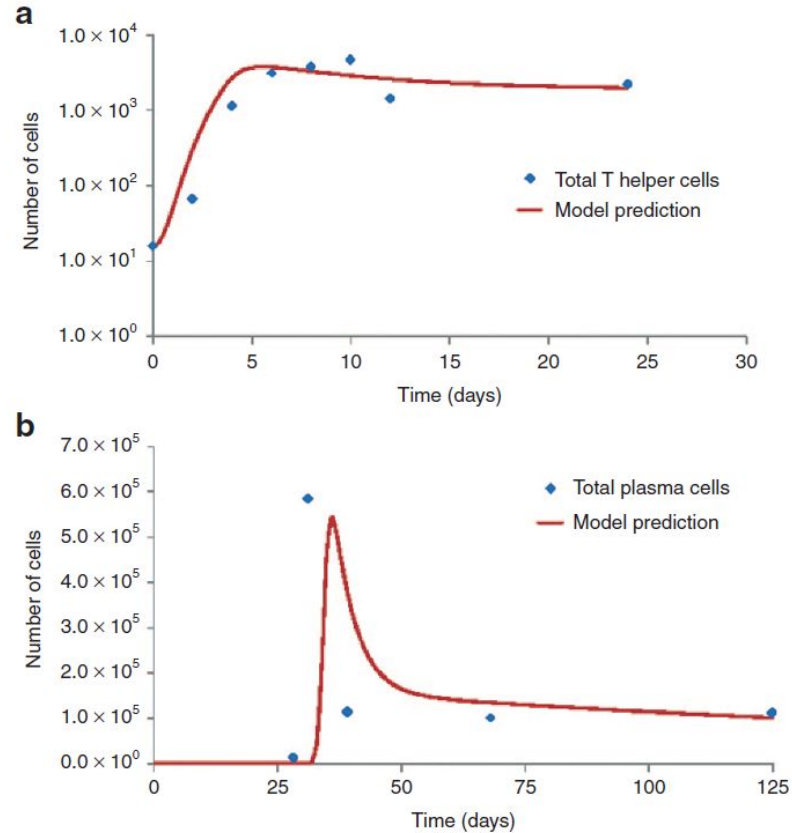


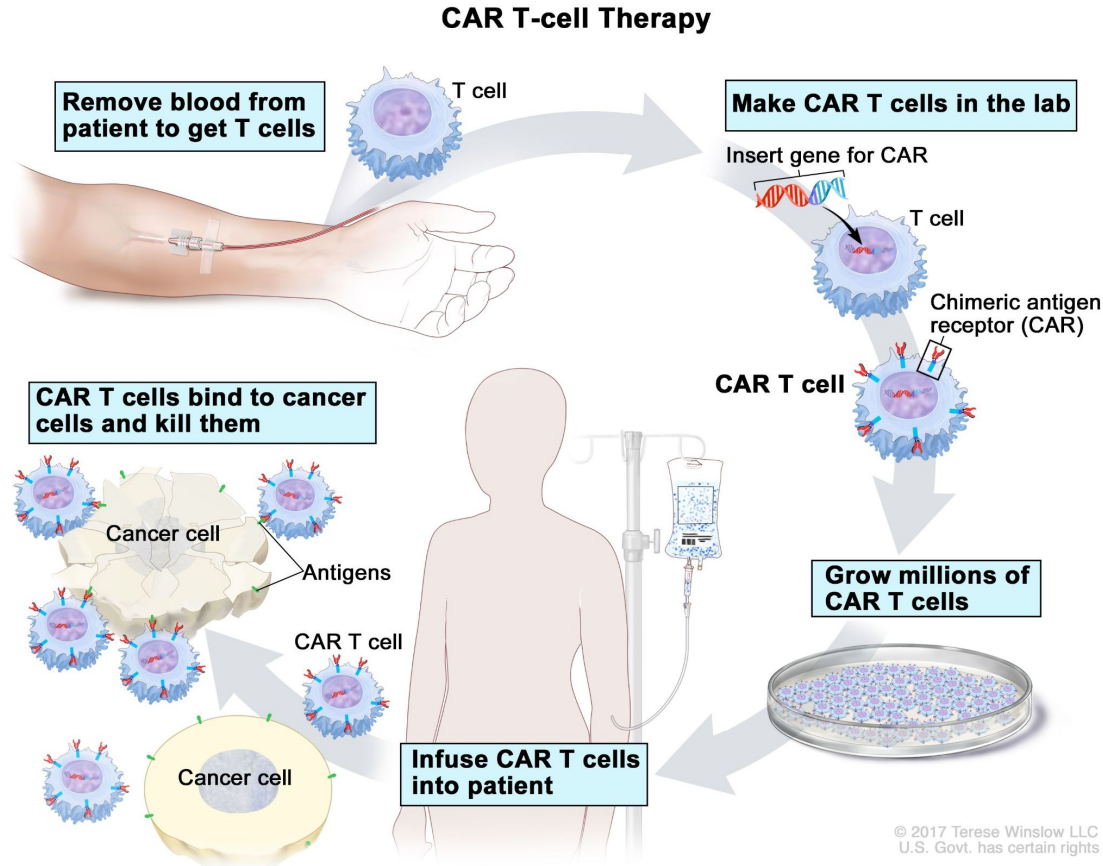
Figure 4. Simulation results of immune responses in human against a theoretical antigenic protein. The results include kinetic profiles for (a) dendritic cells; (b) helper T cells; (c) B cells; (d) antigenic protein, ADA, and immune complex; (e) polyclonal ADA (total 17 clones, whose antigen-binding affinity increases by twofold between clones, from clone 1 to clone 17); (f) average antigen-binding affinity of ADA. ADA, antidrug antibody.

Observation and model prediction

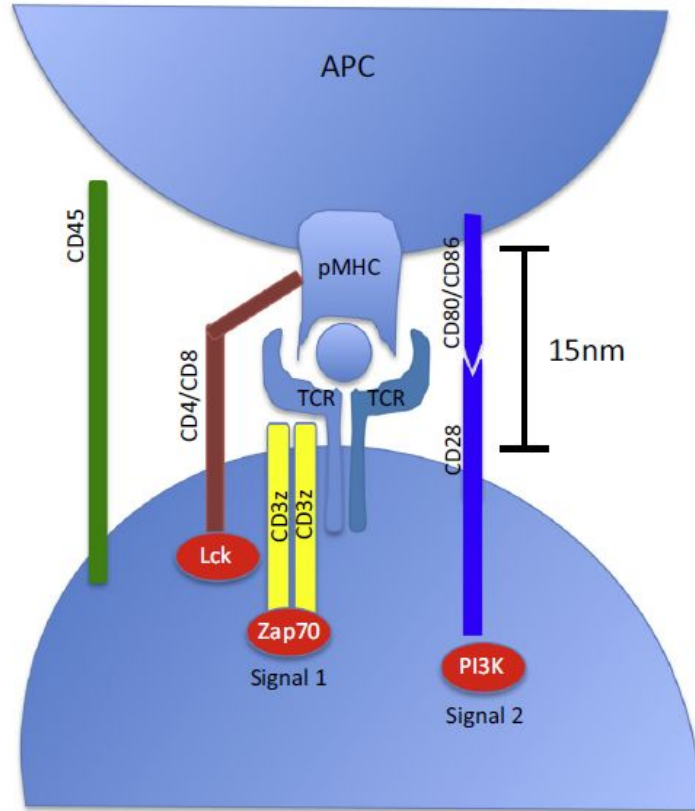


CAR-T and individualized vaccines

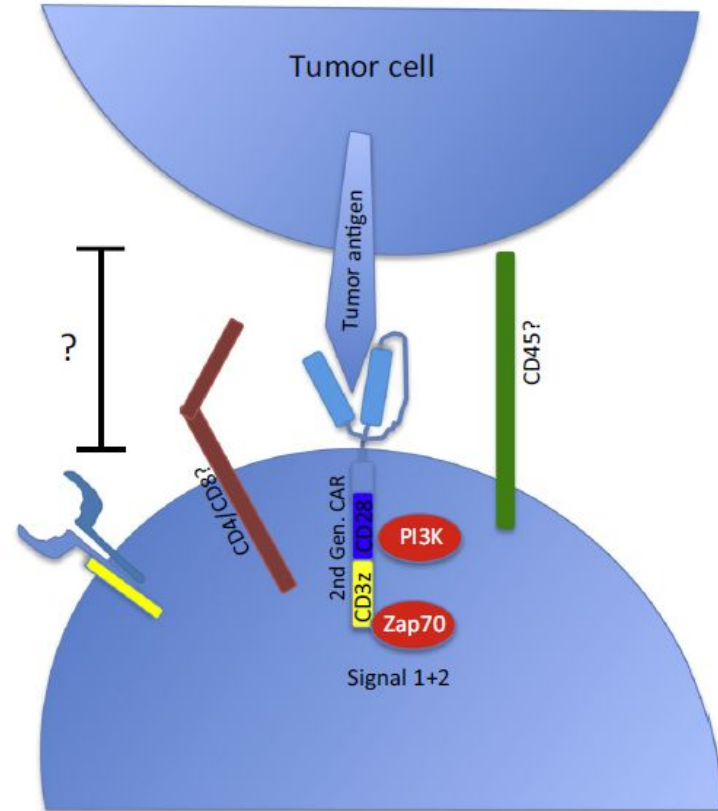
CAR (Chimeric Antigen Receptor) T-Cell Therapy



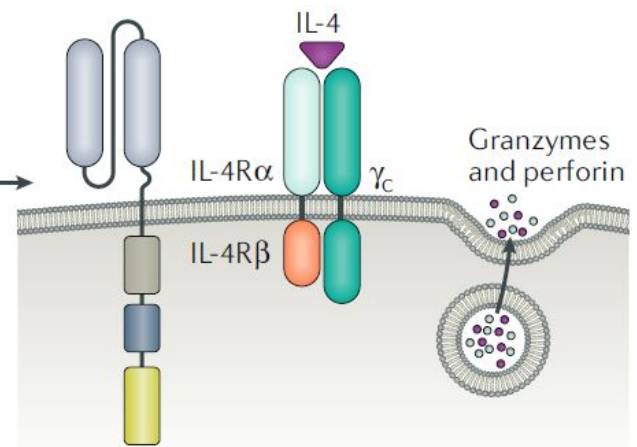
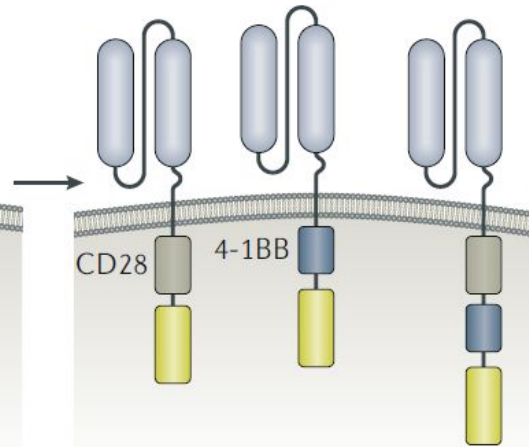
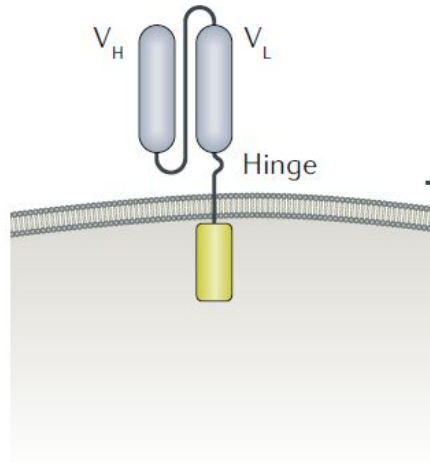
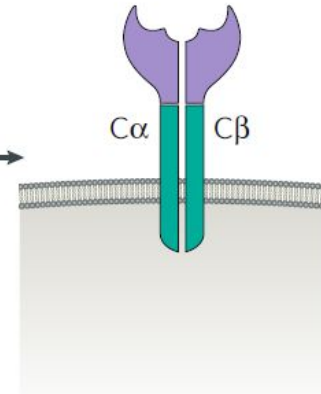
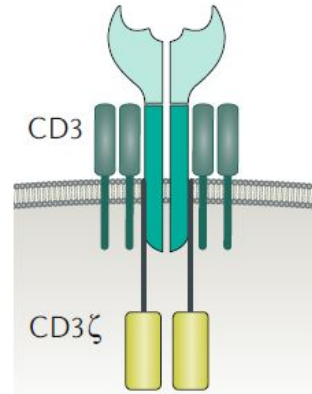
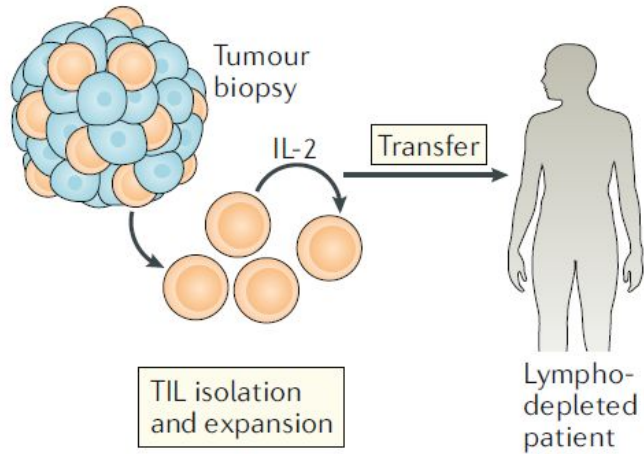
Signaling of conventional and CAR T cells



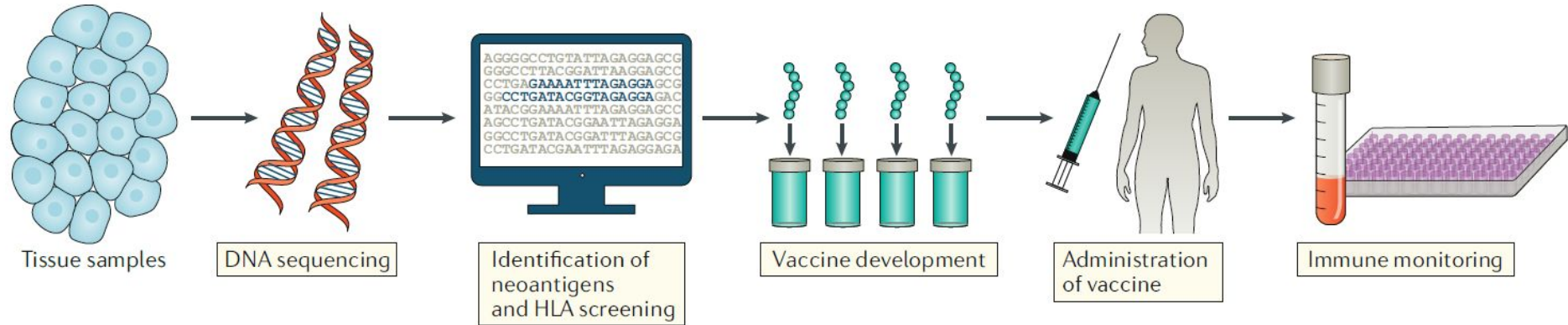
Conventional T cell



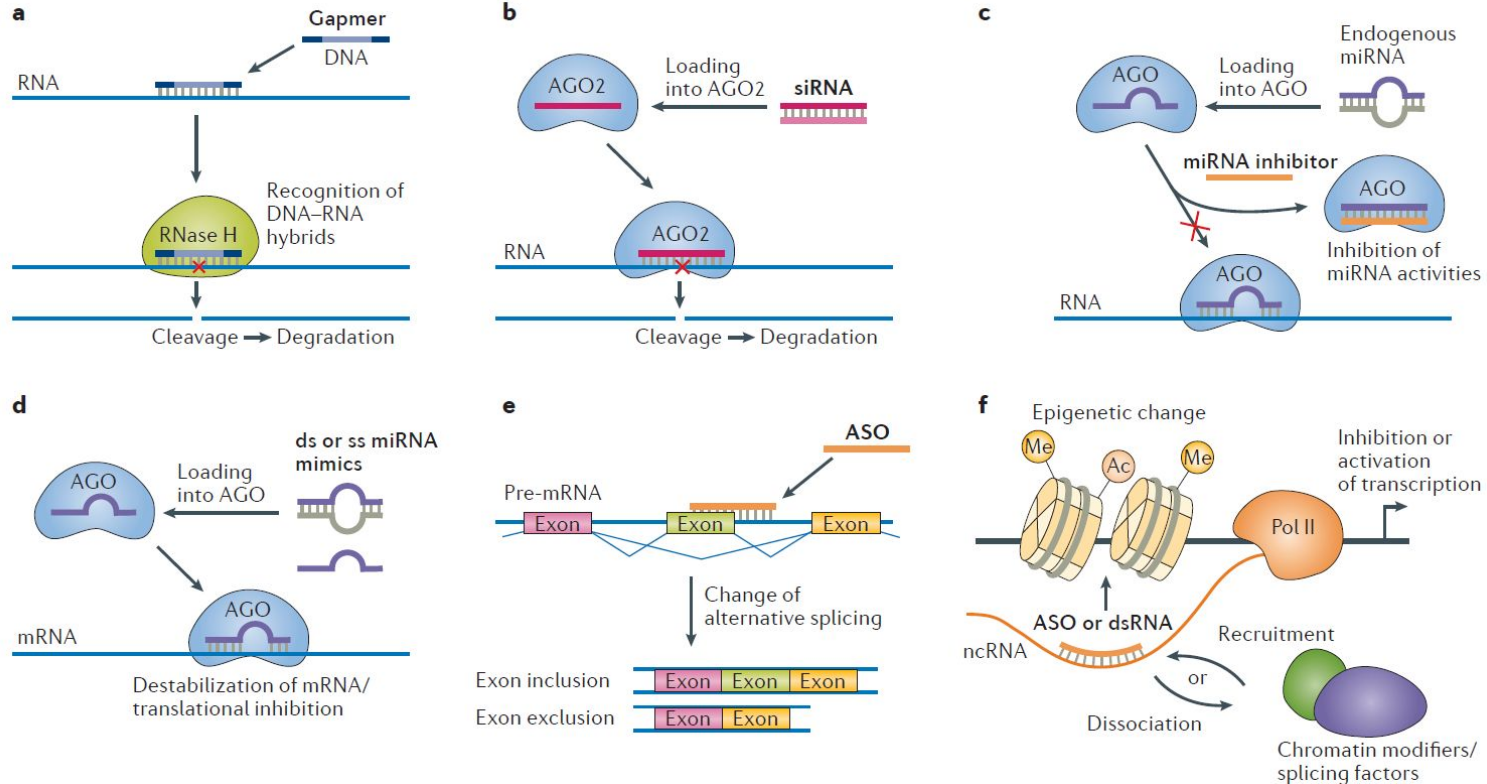
CAR T cell



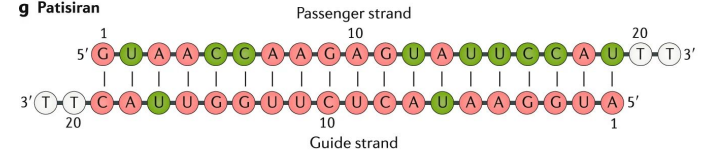
Towards personalized vaccine development



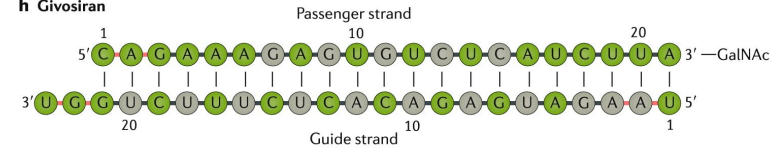
Regulating RNA levels or splicing with ASOs and duplex RNAs



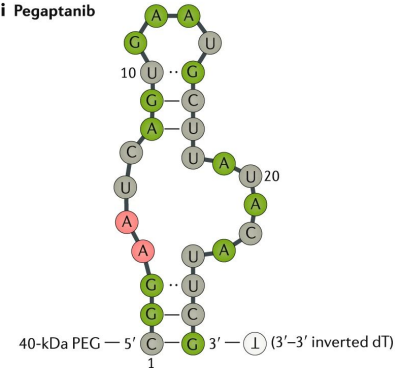
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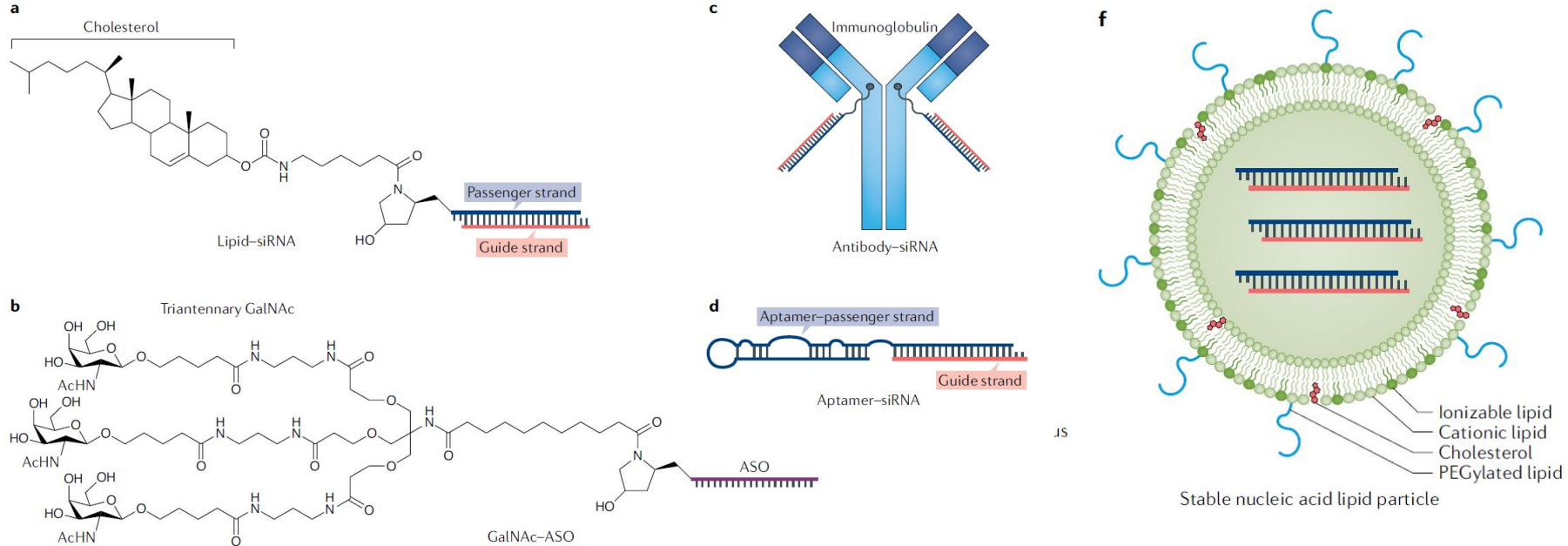


i Pegaptanib



(N)	DNA
(N)	RNA
(N)	PMO
(N)	2'-O-methoxyethyl
(N)	2'-O-methyl
(N)	2'-Fluoro
Y	5-Methyl pyrimidine
—	Phosphorothioate
—	Phosphodiester

Delivery systems of antisense oligonucleotides



Thanks to 2'-MOE modification, Nurinersen can be delivered directly via intrathecal injection.

Other chemistries may need additional delivery systems, for instance mRNA vaccine may require lipid nanoparticles (LNP) shown in diagram *f*.